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Anti-PSMA Antibodies, Bispecific Antigen-Binding Molecules that Bind PSMA and CD3, and Uses Thereof

Abstract

The present invention provides antibodies that bind to prostate-specific membrane antigen (PSMA), bispecific antibodies that bind to PSMA and CD3, and methods of using the same. According to certain embodiments, the antibodies of the invention bind human PSMA with high affinity and bind CD3 to induce human T cell proliferation. The invention includes antibodies that bind PSMA and CD3 and induce T cell-mediated killing of PSMA-expressing tumor cells. According to certain embodiments, the present invention provides bispecific antigen-binding molecules comprising a first antigen-binding domain that specifically binds human CD3, and a second antigen-binding molecule that specifically binds human PSMA. In certain embodiments, the bispecific antigen-binding molecules of the present invention are capable of inhibiting the growth of prostate tumors expressing PSMA. The antibodies and bispecific antigen-binding molecules of the invention are useful for the treatment of diseases and disorders in which an upregulated or induced targeted immune response is desired and/or therapeutically beneficial. For example, the antibodies of the invention are useful for the treatment of various cancers.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 17/476,856, filed Sep. 16, 2021, which is a division of U.S. application Ser. No. 16/205,917, filed Nov. 30, 2018, now U.S. Pat. No. 11,155,633, which is a division of U.S. application Ser. No. 15/223,434, filed Jul. 29, 2016, now U.S. Pat. No. 10,179,819, which claims the benefit under 35 USC § 119(e) of U.S. Provisional Application Nos. 61/199,823, filed Jul. 31, 2015, 62/222,590, filed Sep. 23, 2015, and 62/351,823, filed Jun. 17, 2016, each of which is herein incorporated by reference.

REFERENCE TO A SEQUENCE LISTING

[0002] This application incorporates by reference a computer readable Sequence Listing in ST.26 XML format, titled 10173US04_Sequence, created on May 23, 2025 and containing 2,054,953 bytes.

FIELD OF THE INVENTION

[0003] The present invention relates to antibodies, and antigen-binding fragments thereof, which are specific for prostate-specific membrane antigen (PSMA), and methods of use thereof. The present invention also relates to bispecific antigen-binding molecules that bind PSMA and CD3, and methods of use thereof.

BACKGROUND

[0004] Prostate-specific membrane antigen (PSMA), also known as folate hydrolase 1 (FOLH1), is an integral, non-shed membrane glycoprotein that is highly expressed in prostate epithelial cells and is a cell-surface marker for prostate cancer. Its expression is maintained in castrate-resistant prostate cancer, a condition with poor outcome and limited treatment options. Methods for treating prostate cancer by targeting PSMA have been investigated. For example, Yttrium-90 capromab is a radiotherapeutic comprising a monoclonal antibody to an intracellular epitope of PSMA. In another example, J591, a monoclonal antibody to an extracellular epitope of PSMA, is part of the radiotherapeutic Lutetium-177 J591 and in MLN2704, in which maytansinoid 1 (DM1, an antimicrotubule agent) is conjugated to J591. These therapies have been associated with toxicity.

PSMA is also expressed within the neovasculature of other tumors such as bladder, renal, gastric, and colorectal carcinomas.

[0005] CD3 is a homodimeric or heterodimeric antigen expressed on T cells in association with the T cell receptor complex (TCR) and is required for T cell activation. Functional CD3 is formed from the dimeric association of two of four different chains: epsilon, zeta, delta and gamma. The CD3 dimeric arrangements include gamma/epsilon, delta/epsilon and zeta/zeta. Antibodies against CD3 have been shown to cluster CD3 on T cells, thereby causing T cell activation in a manner similar to the engagement of the TCR by peptide-loaded MHC molecules. Thus, anti-CD3 antibodies have been proposed for therapeutic purposes involving the activation of T cells. In addition, bispecific antibodies that are capable of binding CD3 and a target antigen have been proposed for therapeutic uses involving targeting T cell immune responses to tissues and cells expressing the target antigen. [0006] Antigen-binding molecules that target PSMA, as well as bispecific antigen-binding molecules that bind both PSMA and CD3 would be useful in therapeutic settings in which specific targeting and T cell-mediated killing of cells that express PSMA is desired.

BRIEF SUMMARY OF THE INVENTION

[0007] In a first aspect, the present invention provides antibodies and antigen-binding fragments thereof that bind to human PSMA. The antibodies according to this aspect of the invention are useful, inter alia, for targeting cells expressing PSMA. The present invention also provides bispecific antibodies and antigen-binding fragments thereof that bind human PSMA and human CD3. The bispecific antibodies according to this aspect of the invention are useful, inter alia, for targeting T cells expressing CD3, and for stimulating T cell activation, e.g., under circumstances where T cell-mediated killing of cells expressing PSMA is beneficial or desirable. For example, the bispecific antibodies can direct CD3-mediated T cell activation to specific PSMA-expressing cells, such as prostate tumor cells.

[0008] Exemplary anti-PSMA antibodies of the present invention are listed in Tables 1 and 2 herein. Table 1 sets forth the amino acid sequence identifiers of the heavy chain variable regions (HCVRs) and light chain variable regions (LCVRs), as well as heavy chain complementarity determining regions (HCDR1, HCDR2 and HCDR3), and light chain complementarity determining regions (LCDR1, LCDR2 and LCDR3) of the exemplary anti-PSMA antibodies. Table 2 sets forth the sequence identifiers of the nucleic acid molecules encoding the HCVRs, LCVRs, HCDR1, HCDR2 HCDR3, LCDR1, LCDR2 and LCDR3 of the exemplary anti-PSMA antibodies.

[0009] The present invention provides antibodies, or antigen-binding fragments thereof, comprising an HCVR comprising an amino acid sequence selected from any of the HCVR amino acid sequences listed in Table 1, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity thereto.

[0010] The present invention also provides antibodies, or antigen-binding fragments thereof, comprising an LCVR comprising an amino acid sequence selected from any of the LCVR amino acid sequences listed in Table 1, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity thereto.

[0011] The present invention also provides antibodies, or antigen-binding fragments thereof, comprising an HCVR and an LCVR amino acid sequence pair (HCVR/LCVR) comprising any of the HCVR amino acid sequences listed in Table 1 paired with any of the LCVR amino acid sequences listed in Table 1. According to certain embodiments, the present invention provides antibodies, or antigen-binding fragments thereof, comprising an HCVR/LCVR amino acid sequence pair contained within any of the exemplary anti-PSMA antibodies listed in Table 1. In certain embodiments, the HCVR/LCVR amino acid sequence pair is selected from the group consisting of SEQ ID NOs: 66/1642 (e.g., H1H11810P2); and 122/130 (e.g., H1H3465P).

[0012] The present invention also provides antibodies, or antigen-binding fragments thereof, comprising a heavy chain CDR1 (HCDR1) comprising an amino acid sequence selected from any of the HCDR1 amino acid sequences listed in Table 1 or a substantially similar sequence thereof

having at least 90%, at least 95%, at least 98% or at least 99% sequence identity.

[0013] The present invention also provides antibodies, or antigen-binding fragments thereof, comprising a heavy chain CDR2 (HCDR2) comprising an amino acid sequence selected from any of the HCDR2 amino acid sequences listed in Table 1 or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity.

[0014] The present invention also provides antibodies, or antigen-binding fragments thereof, comprising a heavy chain CDR3 (HCDR3) comprising an amino acid sequence selected from any of the HCDR3 amino acid sequences listed in Table 1 or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity.

[0015] The present invention also provides antibodies, or antigen-binding fragments thereof, comprising a light chain CDR1 (LCDR1) comprising an amino acid sequence selected from any of the LCDR1 amino acid sequences listed in Table 1 or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity.

[0016] The present invention also provides antibodies, or antigen-binding fragments thereof, comprising a light chain CDR2 (LCDR2) comprising an amino acid sequence selected from any of the LCDR2 amino acid sequences listed in Table 1 or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity.

[0017] The present invention also provides antibodies, or antigen-binding fragments thereof, comprising a light chain CDR3 (LCDR3) comprising an amino acid sequence selected from any of the LCDR3 amino acid sequences listed in Table 1 or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity.

[0018] The present invention also provides antibodies, or antigen-binding fragments thereof, comprising an HCDR3 and an LCDR3 amino acid sequence pair (HCDR3/LCDR3) comprising any of the HCDR3 amino acid sequences listed in Table 1 paired with any of the LCDR3 amino acid sequences listed in Table 1. According to certain embodiments, the present invention provides antibodies, or antigen-binding fragments thereof, comprising an HCDR3/LCDR3 amino acid sequence pair contained within any of the exemplary anti-PSMA antibodies listed in Table 1. In certain embodiments, the HCDR3/LCDR3 amino acid sequence pair is selected from the group consisting of SEQ ID NOs: 72/1648 (e.g., H1H11810P2) and 128/136 (e.g., H1H3465P).

[0019] The present invention also provides antibodies, or antigen-binding fragments thereof, comprising a set of six CDRs (i.e., HCDR1-HCDR2-HCDR3-LCDR1-LCDR2-LCDR3) contained within any of the exemplary anti-PSMA antibodies listed in Table 1. In certain embodiments, the HCDR1-HCDR2-HCDR3-LCDR1-LCDR2-LCDR3 amino acid sequences set is selected from the group consisting of SEQ ID NOs: 68-70-72-1644-1646-1648 (e.g., H1H11810P2); and 124-126-128-132-134-136 (e.g., H1H3465P).

[0020] In a related embodiment, the present invention provides antibodies, or antigen-binding fragments thereof, comprising a set of six CDRs (i.e., HCDR1-HCDR2-HCDR3-LCDR1-LCDR2-LCDR3) contained within an HCVR/LCVR amino acid sequence pair as defined by any of the exemplary anti-PSMA antibodies listed in Table 1. For example, the present invention includes antibodies, or antigen-binding fragments thereof, comprising the HCDR1-HCDR2-HCDR3-LCDR1-LCDR2-LCDR3 amino acid sequences set contained within an HCVR/LCVR amino acid sequence pair selected from the group consisting of SEQ ID NOs: 66/146 (e.g., H1H11810P2); and 122/130 (e.g., H1H3465P). Methods and techniques for identifying CDRs within HCVR and LCVR amino acid sequences are well known in the art and can be used to identify CDRs within the specified HCVR and/or LCVR amino acid sequences disclosed herein. Exemplary conventions that can be used to identify the boundaries of CDRs include, e.g., the Kabat definition, the Chothia definition, and the AbM definition. In general terms, the Kabat definition is based on sequence variability, the Chothia definition is based on the location of the structural loop regions, and the AbM definition is a compromise between the Kabat and Chothia approaches. See, e.g., Kabat, "Sequences of Proteins of Immunological Interest," National Institutes of Health, Bethesda, Md.

(1991); Al-Lazikani et al., *J. Mol. Biol.* 273:927-948 (1997); and Martin et al., *Proc. Natl. Acad. Sci. USA* 86:9268-9272 (1989). Public databases are also available for identifying CDR sequences within an antibody.

[0021] The present invention also provides nucleic acid molecules encoding anti-PSMA antibodies or portions thereof. For example, the present invention provides nucleic acid molecules encoding any of the HCVR amino acid sequences listed in Table 1; in certain embodiments the nucleic acid molecule comprises a polynucleotide sequence selected from any of the HCVR nucleic acid sequences listed in Table 2, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity thereto.

[0022] The present invention also provides nucleic acid molecules encoding any of the LCVR amino acid sequences listed in Table 1; in certain embodiments the nucleic acid molecule comprises a polynucleotide sequence selected from any of the LCVR nucleic acid sequences listed in Table 2, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity thereto.

[0023] The present invention also provides nucleic acid molecules encoding any of the HCDR1 amino acid sequences listed in Table 1; in certain embodiments the nucleic acid molecule comprises a polynucleotide sequence selected from any of the HCDR1 nucleic acid sequences listed in Table 2, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity thereto.

[0024] The present invention also provides nucleic acid molecules encoding any of the HCDR2 amino acid sequences listed in Table 1; in certain embodiments the nucleic acid molecule comprises a polynucleotide sequence selected from any of the HCDR2 nucleic acid sequences listed in Table 2, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity thereto.

[0025] The present invention also provides nucleic acid molecules encoding any of the HCDR3 amino acid sequences listed in Table 1; in certain embodiments the nucleic acid molecule comprises a polynucleotide sequence selected from any of the HCDR3 nucleic acid sequences listed in Table 2, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity thereto.

[0026] The present invention also provides nucleic acid molecules encoding any of the LCDR1 amino acid sequences listed in Table 1; in certain embodiments the nucleic acid molecule comprises a polynucleotide sequence selected from any of the LCDR1 nucleic acid sequences listed in Table 2, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity thereto.

[0027] The present invention also provides nucleic acid molecules encoding any of the LCDR2 amino acid sequences listed in Table 1; in certain embodiments the nucleic acid molecule comprises a polynucleotide sequence selected from any of the LCDR2 nucleic acid sequences listed in Table 2, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity thereto.

[0028] The present invention also provides nucleic acid molecules encoding any of the LCDR3 amino acid sequences listed in Table 1; in certain embodiments the nucleic acid molecule comprises a polynucleotide sequence selected from any of the LCDR3 nucleic acid sequences listed in Table 2, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity thereto.

[0029] The present invention also provides nucleic acid molecules encoding an HCVR, wherein the HCVR comprises a set of three CDRs (i.e., HCDR1-HCDR2-HCDR3), wherein the HCDR1-HCDR2-HCDR3 amino acid sequence set is as defined by any of the exemplary anti-PSMA antibodies listed in Table 1.

[0030] The present invention also provides nucleic acid molecules encoding an LCVR, wherein the LCVR comprises a set of three CDRs (i.e., LCDR1-LCDR2-LCDR3), wherein the LCDR1-

LCDR2-LCDR3 amino acid sequence set is as defined by any of the exemplary anti-PSMA antibodies listed in Table 1.

[0031] The present invention also provides nucleic acid molecules encoding both an HCVR and an LCVR, wherein the HCVR comprises an amino acid sequence of any of the HCVR amino acid sequences listed in Table 1, and wherein the LCVR comprises an amino acid sequence of any of the LCVR amino acid sequences listed in Table 1. In certain embodiments, the nucleic acid molecule comprises a polynucleotide sequence selected from any of the HCVR nucleic acid sequences listed in Table 2, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity thereto, and a polynucleotide sequence selected from any of the LCVR nucleic acid sequences listed in Table 2, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity thereto. In certain embodiments according to this aspect of the invention, the nucleic acid molecule encodes an HCVR and LCVR, wherein the HCVR and LCVR are both derived from the same anti-PSMA antibody listed in Table 1.

[0032] The present invention also provides recombinant expression vectors capable of expressing a polypeptide comprising a heavy or light chain variable region of an anti-PSMA antibody. For example, the present invention includes recombinant expression vectors comprising any of the nucleic acid molecules mentioned above, i.e., nucleic acid molecules encoding any of the HCVR, LCVR, and/or CDR sequences as set forth in Table 1. Also included within the scope of the present invention are host cells into which such vectors have been introduced, as well as methods of producing the antibodies or portions thereof by culturing the host cells under conditions permitting production of the antibodies or antibody fragments, and recovering the antibodies and antibody fragments so produced.

[0033] The present invention includes anti-PSMA antibodies having a modified glycosylation pattern. In some embodiments, modification to remove undesirable glycosylation sites may be useful, or an antibody lacking a fucose moiety present on the oligosaccharide chain, for example, to increase antibody dependent cellular cytotoxicity (ADCC) function (see Shield et al. (2002) JBC 277:26733). In other applications, modification of galactosylation can be made in order to modify complement dependent cytotoxicity (CDC).

[0034] In another aspect, the invention provides a pharmaceutical composition comprising a recombinant human antibody or fragment thereof which specifically binds PSMA and a pharmaceutically acceptable carrier. In a related aspect, the invention features a composition which is a combination of an anti-PSMA antibody and a second therapeutic agent. In one embodiment, the second therapeutic agent is any agent that is advantageously combined with an anti-PSMA antibody. Additional combination therapies and co-formulations involving the anti-PSMA antibodies of the present invention are disclosed elsewhere herein.

[0035] In another aspect, the invention provides therapeutic methods for targeting/killing tumor cells expressing PSMA using an anti-PSMA antibody of the invention, wherein the therapeutic methods comprise administering a therapeutically effective amount of a pharmaceutical composition comprising an anti-PSMA antibody of the invention to a subject in need thereof. In some cases, the anti-PSMA antibodies (or antigen-binding fragments thereof) can be used for treating prostate cancer, or may be modified to be more cytotoxic by methods, including but not limited to, modified Fc domains to increase ADCC (see e.g. Shield et al. (2002) JBC 277:26733), radioimmunotherapy (Akhtar, et al., 2012, Prostate-Specific Membrane Antigen-Based Therapeutics; Adv Urol. 2012:973820), antibody-drug conjugates (Olson, W C and Israel, R J, 2014, Front Biosci (Landmark Ed). 19:12-33; DiPippo, et al. Feb. 15, 2015, The Prostate, 75(3):303-313, first published on line Oct. 18, 2014), or other methods for increasing the efficiency of tumor ablation.

[0036] The present invention also includes the use of an anti-PSMA antibody of the invention in the manufacture of a medicament for the treatment of a disease or disorder related to or caused by

PSMA-expressing cells.

[0037] In yet another aspect, the invention provides monospecific anti-PSMA antibodies for diagnostic applications, such as, e.g., imaging reagents.

[0038] In yet another aspect, the invention provides therapeutic methods for stimulating T cell activation using an anti-CD3 antibody or antigen-binding portion of an antibody of the invention, wherein the therapeutic methods comprise administering a therapeutically effective amount of a pharmaceutical composition comprising an antibody

[0039] In another aspect, the present invention provides an isolated antibody or antigen-binding fragment thereof that binds human prostate-specific membrane antigen (PSMA) with a binding dissociation equilibrium constant ($K_{sub.D}$) of less than about 80 nM as measured in a surface plasmon resonance assay at 37° C. In yet another aspect, the present invention provides an isolated antibody or antigen-binding fragment thereof that binds human PSMA with a dissociative half-life ($t_{1/2}$) of greater than about 10 minutes as measured in a surface plasmon resonance assay at 37° C.

[0040] The invention further provides an antibody or antigen-binding fragment that competes for binding to human PSMA with a reference antibody comprising an HCVR/LCVR amino acid sequence pair as set forth in Table 1. In another aspect, the invention provides an antibody or antigen-binding fragment that competes for binding to human PSMA with a reference antibody comprising an HCVR/LCVR amino acid sequence pair selected from the group consisting of SEQ ID NOs: 2/1642; 10/1642; 18/1642; 26/1642; 34/1642; 42/1642; 50/1642; 58/1642; 66/1642; 74/1642; 82/1642; 90/1642; 98/1642; 106/1642; 114/1642; 122/130; and 138/146.

[0041] The invention furthermore provides an antibody or antigen-binding fragment, wherein the antibody or antigen-binding fragment thereof binds to the same epitope on human PSMA as a reference antibody comprising an HCVR/LCVR amino acid sequence pair as set forth in Table 1. In another aspect, the antibody or antigen-binding fragment binds to the same epitope on human PSMA as a reference antibody comprising an HCVR/LCVR amino acid sequence pair selected from the group consisting of SEQ ID NOs: 2/1642; 10/1642; 18/1642; 26/1642; 34/1642; 42/1642; 50/1642; 58/1642; 66/1642; 74/1642; 82/1642; 90/1642; 98/1642; 106/1642; 114/1642; 122/130; and 138/146.

[0042] The invention further provides an isolated antibody or antigen-binding fragment thereof that binds human PSMA, wherein the antibody or antigen-binding fragment comprises: the complementarity determining regions (CDRs) of a heavy chain variable region (HCVR) having an amino acid sequence as set forth in Table 1; and the CDRs of a light chain variable region (LCVR) having an amino acid sequence as set forth in Table 1. In another aspect, the isolated antibody or antigen-binding fragment comprises the heavy and light chain CDRs of a HCVR/LCVR amino acid sequence pair selected from the group consisting of: SEQ ID NOs: 2/1642; 10/1642; 18/1642; 26/1642; 34/1642; 42/1642; 50/1642; 58/1642; 66/1642; 74/1642; 82/1642; 90/1642; 98/1642; 106/1642; 114/1642; 122/130; and 138/146. In yet another aspect, the isolated antibody or antigen-binding fragment comprises HCDR1-HCDR2-HCDR3-LCDR1-LCDR2-LCDR3 domains, respectively, selected from the group consisting of: SEQ ID NOs: 4-6-8-1644-1646-1648; Dec. 14, 2016-1644-1646-1648; 20-22-24-1644-1646-1648; 28-30-32-1644-1646-1648; 36-38-40-1644-1646-1648; 44-46-48-1644-1646-1648; 52-54-56-1644-1646-1648; 60-62-64-1644-1646-1648; 68-70-72-1644-1646-1648; 76-78-80-1644-1646-1648; 84-86-88-1644-1646-1648; 92-94-96-1644-1646-1648; 100-102-104-1644-1646-1648; 108-110-112-1644-1646-1648; 116-118-120-1644-1646-1648; 124-126-128-132-134-136; and 140-142-144-148-150-152.

[0043] In another aspect, the invention provides an isolated antibody or antigen-binding fragment thereof that binds human PSMA, wherein the antibody or antigen-binding fragment comprises: (a) a heavy chain variable region (HCVR) having an amino acid sequence selected from the group consisting of SEQ ID NOs: 2, 10, 18, 26, 34, 42, 50, 58, 66, 74, 82, 90, 98, 106, 114, 122, and 138; and (b) a light chain variable region (LCVR) having an amino acid sequence selected from the group consisting of SEQ ID NOs: 130 and 146. In a further aspect, the isolated antibody or antigen-

binding fragment of claim 10, wherein the antibody or antigen-binding fragment comprises a HCVR/LCVR amino acid sequence pair selected from the group consisting of: SEQ ID NOs: 2/1642; 10/1642; 18/1642; 26/1642; 34/1642; 42/1642; 50/1642; 58/1642; 66/1642; 74/1642; 82/1642; 90/1642; 98/1642; 106/1642; 114/1642; 122/130; and 138/146.

[0044] According to another aspect, the present invention provides bispecific antigen-binding molecules (e.g., antibodies) that bind PSMA and CD3. Such bispecific antigen-binding molecules are also referred to herein as “anti-PSMA/anti-CD3 bispecific molecules,” “anti-CD3/anti-PSMA bispecific molecules,” or “PSMAxCD3 bsAbs.” The anti-PSMA portion of the anti-PSMA/anti-CD3 bispecific molecule is useful for targeting cells (e.g., tumor cells) that express PSMA (e.g., prostate tumors), and the anti-CD3 portion of the bispecific molecule is useful for activating T-cells. The simultaneous binding of PSMA on a tumor cell and CD3 on a T-cell facilitates directed killing (cell lysis) of the targeted tumor cell by the activated T-cell. The anti-PSMA/anti-CD3 bispecific molecules of the invention are therefore useful, inter alia, for treating diseases and disorders related to or caused by PSMA-expressing tumors (e.g., prostate cancers).

[0045] The bispecific antigen-binding molecules according to this aspect of the present invention comprise a first antigen-binding domain that specifically binds human CD3, and a second antigen-binding domain that specifically binds PSMA. The present invention includes anti-PSMA/anti-CD3 bispecific molecules (e.g., bispecific antibodies) wherein each antigen-binding domain comprises a heavy chain variable region (HCVR) paired with a light chain variable region (LCVR). In certain exemplary embodiments of the invention, the anti-CD3 antigen-binding domain and the anti-PSMA antigen binding domain each comprise different, distinct HCVRs paired with a common LCVR. For example, as illustrated in Example 4 herein, bispecific antibodies were constructed comprising a first antigen-binding domain that specifically binds CD3, wherein the first antigen-binding domain comprises an HCVR/LCVR pair derived from an anti-CD3 antibody; and a second antigen-binding domain that specifically binds PSMA, wherein the second antigen-binding domain comprises an HCVR derived from an anti-PSMA antibody paired with an LCVR derived from an anti-CD3 antibody (e.g., the same LCVR that is included in the anti-CD3 antigen-binding domain). In other words, in the exemplary molecules disclosed herein, the pairing of an HCVR from an anti-PSMA antibody with an LCVR from an anti-CD3 antibody creates an antigen-binding domain that specifically binds PSMA (but does not bind CD3). In such embodiments, the first and second antigen-binding domains comprise distinct anti-CD3 and anti-PSMA HCVRs but share a common anti-CD3 LCVR. In other embodiments, the bispecific antigen-binding molecules comprise distinct anti-CD3 and anti-PSMA HCVRs, but share a common LCVR. The amino acid sequence of this LCVR is shown, e.g., in SEQ ID NO:1386, and the amino acid sequences of the corresponding CDRs (i.e., LCDR1-LCDR2-LCDR3) are shown in SEQ ID NOs: 1388, 1390 and 1392, respectively. Genetically modified mice can be used to produce fully human bispecific antigen-binding molecules comprising two different heavy chains that associate with an identical light chain that comprises a variable domain derived from one of two different human light chain variable region gene segments. Alternatively, variable heavy chains may be paired with one common light chain and expressed recombinantly in host cells. As such, the antibodies of the invention can comprise immunoglobulin heavy chains associated with a single rearranged light chain. In some embodiments, the light chain comprises a variable domain derived from a human V κ 1-39 gene segment or a V κ 3-20 gene segment. In other embodiments, the light chain comprises a variable domain derived from a human V κ 1-39 gene segment rearranged with a human JK5 or a human JK1 gene segment.

[0046] The present invention provides anti-CD3/anti-PSMA bispecific molecules, wherein the first antigen-binding domain that specifically binds CD3 comprises any of the HCVR amino acid sequences, any of the LCVR amino acid sequences, any of the HCVR/LCVR amino acid sequence pairs, any of the heavy chain CDR1-CDR2-CDR3 amino acid sequences, or any of the light chain CDR1-CDR2-CDR3 amino acid sequences as set forth in US publication 2014/0088295.

[0047] In addition, the present invention provides anti-CD3/anti-PSMA bispecific molecules, wherein the first antigen-binding domain that specifically binds CD3 comprises any of the HCVR amino acid sequences as set forth in Tables 12, 14, and 18 herein. The first antigen-binding domain that specifically binds CD3 may also comprise any of the LCVR amino acid sequences as set forth in Tables 12, 15, and 20 herein. According to certain embodiments, the first antigen-binding domain that specifically binds CD3 comprises any of the HCVR/LCVR amino acid sequence pairs as set forth in Tables 12, 14, 15, 18, and 20 herein. The present invention also provides anti-CD3/anti-PSMA bispecific molecules, wherein the first antigen-binding domain that specifically binds CD3 comprises any of the heavy chain CDR1-CDR2-CDR3 amino acid sequences as set forth in Tables 12, 14, and 18 herein, and/or any of the light chain CDR1-CDR2-CDR3 amino acid sequences as set forth in Tables 12, 15, and 20 herein.

[0048] According to certain embodiments, the present invention provides anti-CD3/anti-PSMA bispecific molecules, wherein the first antigen-binding domain that specifically binds CD3 comprises a heavy chain variable region (HCVR) having an amino acid sequence as set forth in Tables 12, 14, and 18 herein or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity.

[0049] The present invention also provides anti-CD3/anti-PSMA bispecific molecules, wherein the first antigen-binding domain that specifically binds CD3 comprises a light chain variable region (LCVR) having an amino acid sequence as set forth in Tables 12, 15, and 20 herein, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity.

[0050] The present invention also provides anti-CD3/anti-PSMA bispecific molecules, wherein the first antigen-binding domain that specifically binds CD3 comprises a HCVR and LCVR (HCVR/LCVR) amino acid sequence pair as set forth in Tables 12, 14, 15, 18, and 20 herein.

[0051] The present invention also provides anti-CD3/anti-PSMA bispecific molecules, wherein the first antigen-binding domain that specifically binds CD3 comprises a heavy chain CDR3 (HCDR3) domain having an amino acid sequence as set forth in Tables 12, 14, and 18 herein, or a substantially similar sequence thereto having at least 90%, at least 95%, at least 98% or at least 99% sequence identity; and a light chain CDR3 (LCDR3) domain having an amino acid sequence as set forth in Tables 12, 15, and 20 herein, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity.

[0052] In certain embodiments, the first antigen-binding domain that specifically binds CD3 comprises a HCDR3/LCDR3 amino acid sequence pair as set forth in Tables 12, 14, 15, 18, and 20 herein.

[0053] The present invention also provides anti-CD3/anti-PSMA bispecific antigen-binding molecules, wherein the first antigen-binding domain that specifically binds CD3 comprises a heavy chain CDR1 (HCDR1) domain having an amino acid as set forth in Tables 12, 14, and 18 herein, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity; a heavy chain CDR2 (HCDR2) domain having an amino acid as set forth in Tables 12, 14, and 18, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity; a heavy chain CDR3 (HCDR3) domain having an amino acid as set forth in Tables 12, 14, and 18, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity; a light chain CDR1 (LCDR1) domain having an amino acid sequence as set forth in Tables 12, 15, and 20 herein, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity; a light chain CDR2 (LCDR2) domain having an amino acid sequence as set forth in Tables 12, 15, and 20 herein, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity, and a light chain CDR3 (LCDR3) domain having an amino acid sequence as set forth in Tables 12, 15, and 20 herein, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least

99% sequence identity.

[0054] Certain non-limiting, exemplary anti-CD3/anti-PSMA bispecific antigen-binding molecules of the invention include a first antigen-binding domain that specifically binds CD3 comprising HCDR1-HCDR2-HCDR3-LCDR1-LCDR2-LCDR3 domains, respectively, having the amino acid sequences as set forth in Tables 12, 14, 15, 18, and 20 herein.

[0055] The present invention further provides a bispecific antigen-binding molecule, wherein the first antigen-binding domain that specifically binds human CD3 comprises heavy chain complementarity determining regions (HCDR1, HCDR2 and HCDR3) from a heavy chain variable region (HCVR) comprising an amino acid sequence as set forth in Table 12, Table 14, or Table 18 and light chain complementarity determining regions (LCDR1, LCDR2 and LCDR3) from a light chain variable region (LCVR) comprising an amino acid sequence as set forth in Table 12, Table 15, or Table 20.

[0056] In another aspect, the invention provides a bispecific antigen-binding molecule wherein the first antigen-binding domain that specifically binds human CD3 comprises heavy chain complementarity determining regions (HCDR1, HCDR2 and HCDR3) from a heavy chain variable region (HCVR) selected from the group consisting of SEQ ID NOs: 922, 154, 1482, 1490, 1498, 1506, 1514, 1522, 1530, 1538, 1546, 1554, 1562, 1570, 1578, 1586, 1594, 1602, 1610, 1618, and 1626, and light chain complementarity determining regions (LCDR1, LCDR2 and LCDR3) from a light chain variable region (LCVR) comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 162, 930 and 1642.

[0057] The invention further provides a bispecific antigen-binding molecule, wherein the first antigen-binding domain that specifically binds human CD3 comprises three heavy chain complementarity determining regions (A1-HCDR1, A1-HCDR2 and A1-HCDR3) and three light chain complementarity determining regions (A1-LCDR1, A1-LCDR2 and A1-LCDR3), wherein A1-HCDR1 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 924, 156; 1484, 1492, 1500, 1508, 1516, 1524, 1532, 1540, 1548, 1556, 1564, 1572, 1580, 1588, 1596, 1604, 1612, 1620, and 1628; A1-HCDR2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 926, 158, 1486, 1494, 1502, 1510, 1518, 1526, 1534, 1542, 1550, 1558, 1566, 1574, 1582, 1590, 1598, 1606, 1614, 1622, and 1630; A1-HCDR3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 928, 160, 1488, 1496, 1504, 1512, 1520, 1528, 1536, 1544, 1552, 1560, 1568, 1576, 1584, 1592, 1600, 1608, 1616, 1624, and 1632; A1-LCDR1 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 932, 164, and 1644; A1-LCDR2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 166, 934 and 1646; and A1-LCDR3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 168, 936 and 1648.

[0058] In a further aspect, the invention provides a bispecific antigen-binding molecule, wherein the first antigen-binding domain that specifically binds human CD3 comprises the heavy and light chain CDRs of a HCVR/LCVR amino acid sequence pair selected from the group consisting of: SEQ ID NOs: 922/930, 154/162, 1482/1642, 1490/1642, 1498/1642, 1506/1642, 1514/1642, 1522/1642, 1530/1642, 1538/1642, 1546/1642, 1554/1642, 1562/1642, 1570/1642, 1578/1642, 1586/1642, 1594/1642, 1602/1642, 1610/1642, 1618/1642, and 1626/1642

[0059] In another aspect, the invention provides a bispecific antigen-binding molecule, wherein the first antigen-binding domain that specifically binds human CD3 comprises three heavy chain complementarity determining regions (A1-HCDR1, A1-HCDR2 and A1-HCDR3) and three light chain complementarity determining regions (A1-LCDR1, A1-LCDR2 and A1-LCDR3), and wherein the second antigen-binding domain that specifically binds human PSMA comprises three heavy chain complementarity determining regions (A2-HCDR1, A2-HCDR2 and A2-HCDR3) and three light chain complementarity determining regions (A2-LCDR1, A2-LCDR2 and A2-LCDR3); wherein A1-HCDR1 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 924, 156, 1484, 1492, 1500, 1508, 1516, 1524, 1532, 1540, 1548, 1556, 1564, 1572, 1580,

1588, 1596, 1604, 1612, 1620, and 1628; A1-HCDR2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 926, 158, 1486, 1494, 1502, 1510, 1518, 1526, 1534, 1542, 1550, 1558, 1566, 1574, 1582, 1590, 1598, 1606, 1614, 1622, and 1630; A1-HCDR3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 928, 160, 1488, 1496, 1504, 1512, 1520, 1528, 1536, 1544, 1552, 1560, 1568, 1576, 1584, 1592, 1600, 1608, 1616, 1624, and 1632; A1-LCDR1 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 164, 932, and 1644; A1-LCDR2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 166, 934, and 1646; and A1-LCDR3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 168, 936, and 1648; and wherein A2-HCDR1 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 124 and 68; A2-HCDR2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 126 and 70; A2-HCDR3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 128 and 72; A2-LCDR1 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 932, 164, and 1644; A2-LCDR2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 934, 166, and 1646; and A2-LCDR3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 936, 168, and 1648.

[0060] Certain non-limiting, exemplary anti-CD3/anti-PSMA bispecific antigen-binding molecules of the invention include a first antigen-binding domain that specifically binds CD3 comprising a heavy chain comprising variable domain framework regions having an amino acid sequence selected from FR1 (SEQ ID NO: 1654), FR2 (SEQ ID NO: 1656), FR3 (SEQ ID NO: 1657), and FR4 (SEQ ID NO: 1658).

[0061] In more embodiments, exemplary anti-CD3/anti-PSMA bispecific antigen-binding molecules of the invention include a bispecific antigen-binding molecule wherein the first antigen-binding domain that specifically binds human CD3 comprises a HCVR comprising HCDR1-HCDR2-HCDR3 having the amino acid sequences of SEQ ID NOs: 1659-1660-1661.

[0062] The present invention also provides anti-CD3/anti-PSMA bispecific molecules, wherein the second antigen-binding domain that specifically binds PSMA comprises a heavy chain variable region (HCVR) having the amino acid sequence selected from the group consisting of SEQ ID NOs: 2, 10, 18, 26, 34, 42, 50, 58, 66, 74, 82, 90, 98, 106, 114, 122, and 138, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity.

[0063] The present invention also provides anti-CD3/anti-PSMA bispecific molecules, wherein the second antigen-binding domain that specifically binds PSMA comprises a light chain variable region (LCVR) having the amino acid sequence selected from the group consisting of SEQ ID NOs: 930, 162, and 1642, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity.

[0064] The present invention also provides anti-CD3/anti-PSMA bispecific molecules, wherein the second antigen-binding domain that specifically binds PSMA comprises a HCVR and LCVR (HCVR/LCVR) amino acid sequence pair selected from the group consisting of SEQ ID NOs: 122/930, 122/162, and 66/1642.

[0065] The present invention also provides anti-CD3/anti-PSMA bispecific molecules, wherein the second antigen-binding domain that specifically binds PSMA comprises a heavy chain CDR3 (HCDR3) domain having an amino acid sequence selected from the group consisting of SEQ ID NOs: 8, 16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104, 112, 120, 128, and 144, or a substantially similar sequence thereto having at least 90%, at least 95%, at least 98% or at least 99% sequence identity; and a light chain CDR3 (LCDR3) domain having an amino acid sequence selected from the group consisting of SEQ ID NOs: 936, 168, and 1648, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity.

[0066] In certain embodiments, the second antigen-binding domain that specifically binds PSMA

comprises a HCDR3/LCDR3 amino acid sequence pair selected from the group consisting of SEQ ID NOs: 128/936, 128/168, and 72/1648.

[0067] The present invention also provides anti-CD3/anti-PSMA bispecific antigen-binding molecules, wherein the second antigen-binding domain that specifically binds PSMA comprises a heavy chain CDR1 (HCDR1) domain having an amino acid sequence selected from the group consisting of SEQ ID NOs: 4, 12, 20, 28, 36, 44, 52, 60, 68, 76, 84, 92, 100, 108, 116, 124, and 140, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity; a heavy chain CDR2 (HCDR2) domain having an amino acid sequence selected from the group consisting of SEQ ID NOs: 6, 14, 22, 30, 38, 46, 54, 62, 70, 78, 86, 94, 102, 110, 118, 126 and 142, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity; a heavy chain CDR3 (HCDR3) domain having an amino acid sequence selected from the group consisting of SEQ ID NOs: 8, 16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104, 112, 120, 128, and 144, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity; a light chain CDR1 (LCDR1) domain having an amino acid sequence selected from the group consisting of SEQ ID NOs: 932, 164, and 1644, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity; and a light chain CDR2 (LCDR2) domain having an amino acid sequence selected from the group consisting of SEQ ID NOs: 934, 166, and 1646, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity; and a light chain CDR3 (LCDR3) domain having an amino acid sequence selected from the group consisting of SEQ ID NOs: 936, 168, and 1648, or a substantially similar sequence thereof having at least 90%, at least 95%, at least 98% or at least 99% sequence identity.

[0068] Certain non-limiting, exemplary anti-CD3/anti-PSMA bispecific antigen-binding molecules of the invention include a second antigen-binding domain that specifically binds PSMA comprising HCDR1-HCDR2-HCDR3-LCDR1-LCDR2-LCDR3 domains, respectively, having the amino acid sequences selected from the group consisting of: SEQ ID NOs: 124-126-128-932-934-936, 124-126-128-164-166-168, and 68-70-72-1644-1646-1648.

[0069] In a related embodiment, the invention includes anti-CD3/anti-PSMA bispecific antigen-binding molecules wherein the second antigen-binding domain that specifically binds PSMA comprises the heavy and light chain CDR domains contained within heavy and light chain variable region (HCVR/LCVR) sequences selected from the group consisting of SEQ ID NOs: 122/930, 122/162, and 66/1642.

[0070] In another aspect, the invention provides a bispecific antigen-binding molecule comprising a first antigen-binding domain that binds human CD3 and a second antigen-binding domain that binds human PSMA, wherein the second antigen-binding domain is derived from the antibody or antigen-binding fragment of any one of the anti-PSMA antibodies of the invention. In a further aspect, the invention provides a bispecific antigen-binding molecule comprising a first antigen-binding domain that specifically binds human CD3, and a second antigen-binding domain that specifically binds human PSMA.

[0071] The invention further provides a bispecific antigen-binding molecule which binds human cells expressing human CD3 and cynomolgus monkey cells expressing cynomolgus CD3. In another aspect, the bispecific antigen-binding molecule binds human cells expressing human PSMA and cynomolgus monkey cells expressing cynomolgus PSMA.

[0072] In another aspect the invention provides a bispecific antigen-binding molecule which inhibits tumor growth in immunocompromised mice bearing human prostate cancer xenografts. The invention further provides a bispecific antigen-binding molecule which inhibits tumor growth in immunocompetent mice bearing human prostate cancer xenografts. The invention further provides a bispecific antigen-binding molecule which suppresses tumor growth of established tumors in immunocompromised mice bearing human prostate cancer xenografts. The invention

further provides a bispecific antigen-binding molecule which reduces tumor growth of established tumors in immunocompetent mice bearing human prostate cancer xenografts.

[0073] In another aspect the invention provides a bispecific antigen-binding molecule comprising i) a first antigen-binding domain that specifically binds an effector cell with an EC.sub.50 value of greater than about 40 nM and, and ii) a second antigen-binding domain that specifically binds a target human prostate tumor cell with an EC.sub.50 value of less than 40 nM, wherein such EC.sub.50 binding affinity value is measured in an in vitro FACS binding assay.

[0074] For example, the bispecific antigen-binding molecule can include a first antigen-binding domain that specifically binds human CD3 with an EC.sub.50 value of greater than about 40 nM, or greater than about 100 nM, greater than about 200 nM, or greater than about 1 μ M. In one embodiment, the bispecific antigen-binding molecule can include a second antigen-binding domain that specifically binds the target prostate tumor cell with an EC.sub.50 value of less than about 6 nM. In some cases, the first antigen-binding domain specifically binds each of human CD3 and cynomolgus CD3 with an EC.sub.50 value of greater than about 40 nM, greater than about 100 nM, greater than about 200 nM, or greater than about 1 μ M. In some cases, the first antigen-binding domain specifically binds each of human CD3 and cynomolgus CD3 with weak or no measurable affinity.

[0075] In some embodiments, the antigen-binding molecule induces T cell-mediated tumor cell killing with an EC.sub.50 value of less than about 1.3 nM, as measured in an in vitro T cell-mediated tumor cell killing assay, for example, where the tumor cells are C4-2, 22Rv1, and TRAMPC2_P SMA cells.

[0076] In some applications, the first antigen-binding domain binds human CD3 with an K.sub.D value of greater than about 11 nM, as measured in an in vitro surface plasmon resonance binding assay. In some instances, the first antigen-binding domain binds each of human CD3 and cynomolgus CD3 with an K.sub.D value of greater than about 15 nM, greater than about 30 nM, greater than about 60 nM, greater than about 120 nM, or greater than about 300 nM, as measured in an in vitro surface plasmon resonance binding assay.

[0077] In certain embodiments, anti-CD3 antibodies of the invention, antigen-binding fragments and bispecific antibodies thereof were made by replacing amino acid residues of a parental in a stepwise manner based on differences between the germline sequence and the parental antibody sequence.

[0078] In some embodiments, the invention provides a bispecific antigen-binding molecule, wherein the second antigen-binding domain competes for binding to human PSMA with a reference antigen-binding protein comprising three heavy chain complementarity determining regions (A2-HCDR1, A2-HCDR2 and A2-HCDR3) and three light chain complementarity determining regions (A2-LCDR1, A2-LCDR2 and A2-LCDR3), wherein A2-HCDR1 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 124 and 68; A2-HCDR2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 126 and 70; A2-HCDR3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 128 and 72; A2-LCDR1 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 164, 932 and 1644; A2-LCDR2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 166, 934 and 1646; and A2-LCDR3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 168, 936, and 1648. In some embodiments, the invention provides a bispecific antigen-binding molecule, wherein the second antigen-binding domain competes for binding to human PSMA with a reference antigen-binding protein comprising a heavy chain variable region (HCVR) comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 122 and 66, and a light chain variable region (LCVR) comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 162, 930 and 1642.

[0079] In some embodiments, the invention provides a bispecific antigen-binding molecule,

wherein the first antigen-binding domain competes for binding to human CD3 with a reference antigen-binding protein comprising three heavy chain complementarity determining regions (A1-HCDR1, A1-HCDR2 and A1-HCDR3) and three light chain complementarity determining regions (A1-LCDR1, A1-LCDR2 and A1-LCDR3), wherein A1-HCDR1 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 924, 156, 1484, 1492, 1500, 1508, 1516, 1524, 1532, 1540, 1548, 1556, 1564, 1572, 1580, 1588, 1596, 1604, 1612, 1620, and 1628; A1-HCDR2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 926, 158, 1486, 1494, 1502, 1510, 1518, 1526, 1534, 1542, 1550, 1558, 1566, 1574, 1582, 1590, 1598, 1606, 1614, 1622, and 1630; A1-HCDR3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 928, 160, 1488, 1496, 1504, 1512, 1520, 1528, 1536, 1544, 1552, 1560, 1568, 1576, 1584, 1592, 1600, 1608, 1616, 1624, and 1632; A1-LCDR1 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 164, 932, and 1644; A1-LCDR2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 166, 934, and 1646; and A1-LCDR3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 168, 936, and 1648. In some embodiments, the invention provides a bispecific antigen-binding molecule, wherein the first antigen-binding domain competes for binding to human CD3 with a reference antigen-binding protein comprising a heavy chain variable region (HCVR) comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 922, 154, 1482, 1490, 1498, 1506, 1514, 1522, 1530, 1538, 1546, 1554, 1562, 1570, 1578, 1586, 1594, 1602, 1610, 1618, and 1626, and a light chain variable region (LCVR) comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 930, 162, and 1642.

[0080] In some embodiments, the invention provides a bispecific antigen-binding molecule, wherein the first antigen-binding domain competes for binding to human CD3 with a reference antigen-binding protein comprising a heavy chain variable region (HCVR) comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 922, 154, 1482, 1490, 1498, 1506, 1514, 1522, 1530, 1538, 1546, 1554, 1562, 1570, 1578, 1586, 1594, 1602, 1610, 1618, and 1626, and a light chain variable region (LCVR) comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 930, 162, and 1642; and wherein the second antigen-binding domain competes for binding to human PSMA with a reference antigen-binding protein comprising a heavy chain variable region (HCVR) comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 122 and 66, and a light chain variable region (LCVR) comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 930, 162, and 1642.

[0081] In one aspect, the invention provides a pharmaceutical composition comprising an anti-PSMA antigen-binding molecule or anti-PSMA/anti-CD3 bispecific antigen-binding molecule and a pharmaceutically acceptable carrier or diluent. The invention further provides a method for treating a cancer in a subject, the method comprising administering to the subject the pharmaceutical composition comprising an anti-PSMA antigen-binding molecule or anti-PSMA/anti-CD3 bispecific antigen-binding molecule and a pharmaceutically acceptable carrier or diluent. In some embodiments, the cancer is selected from the group consisting of prostate cancer, kidney cancer, bladder cancer, colorectal cancer, and gastric cancer. In some cases, the cancer is prostate cancer. In some cases, the prostate cancer is castrate-resistant prostate cancer.

[0082] In another aspect, the present invention provides nucleic acid molecules encoding any of the HCVR, LCVR or CDR sequences of the anti-CD3/anti-PSMA bispecific antigen-binding molecules disclosed herein, including nucleic acid molecules comprising the polynucleotide sequences as set forth in Tables 2, 13, 15, 17, 19, and 21 herein, as well as nucleic acid molecules comprising two or more of the polynucleotide sequences as set forth in Tables 2, 13, 15, 17, 19, and 21 in any functional combination or arrangement thereof. Recombinant expression vectors carrying the nucleic acids of the invention, and host cells into which such vectors have been introduced, are also encompassed by the invention, as are methods of producing the antibodies by culturing the host cells under conditions permitting production of the antibodies, and recovering the antibodies

produced.

[0083] The present invention includes anti-CD3/anti-PSMA bispecific antigen-binding molecules wherein any of the aforementioned antigen-binding domains that specifically bind CD3 are combined, connected or otherwise associated with any of the aforementioned antigen-binding domains that specifically bind PSMA to form a bispecific antigen-binding molecule that binds CD3 and PSMA.

[0084] The present invention includes anti-CD3/anti-PSMA bispecific antigen-binding molecules having a modified glycosylation pattern. In some applications, modification to remove undesirable glycosylation sites may be useful, or an antibody lacking a fucose moiety present on the oligosaccharide chain, for example, to increase antibody dependent cellular cytotoxicity (ADCC) function (see Shield et al. (2002) JBC 277:26733). In other applications, modification of galactosylation can be made in order to modify complement dependent cytotoxicity (CDC).

[0085] In another aspect, the invention provides a pharmaceutical composition comprising an anti-CD3/anti-PSMA bispecific antigen-binding molecule as disclosed herein and a pharmaceutically acceptable carrier. In a related aspect, the invention features a composition which is a combination of an anti-CD3/anti-PSMA bispecific antigen-binding molecule and a second therapeutic agent. In one embodiment, the second therapeutic agent is any agent that is advantageously combined with an anti-CD3/anti-PSMA bispecific antigen-binding molecule. Exemplary agents that may be advantageously combined with an anti-CD3/anti-PSMA bispecific antigen-binding molecule are discussed in detail elsewhere herein.

[0086] In yet another aspect, the invention provides therapeutic methods for targeting/killing tumor cells expressing PSMA using an anti-CD3/anti-PSMA bispecific antigen-binding molecule of the invention, wherein the therapeutic methods comprise administering a therapeutically effective amount of a pharmaceutical composition comprising an anti-CD3/anti-PSMA bispecific antigen-binding molecule of the invention to a subject in need thereof.

[0087] The present invention also includes the use of an anti-CD3/anti-PSMA bispecific antigen-binding molecule of the invention in the manufacture of a medicament for the treatment of a disease or disorder related to or caused by PSMA-expressing cells.

[0088] Other embodiments will become apparent from a review of the ensuing detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0089] FIG. 1 illustrates that PSMAxCD3 bispecific antibody inhibits growth of a human prostate cell line in vivo. NSG mice were co-implanted with 22Rv1 cells and human PBMCs subcutaneously. The animals were dosed three times total on Days 0, 3 and 7 with 1 ug of PSMAxCD3 bispecific i.p. Data are expressed as mean (SEM) and were analyzed using analysis of variance (ANOVA).

[0090] FIGS. 2A-2D show that treatment with a PSMAxCD3 bispecific antibody induces transient dose-dependent increase in circulating cytokines, where the cytokine levels (interferon-gamma, IFN-g; tumor necrosis factor, TNF; interleukin-2, IL-2; and interleukin-6, IL-6) are tested at 4 hrs after treatment.

[0091] FIGS. 3A-3B illustrate that treatment with PSMAxCD3 bispecific antibody in a humanized T cell mouse (100 µg/mouse) induces acute increase in cytokines (e.g. IFNγ) (FIG. 3A) as well as transient decrease in circulating T cells (FIG. 3B).

[0092] FIGS. 4A-4C illustrate the effect of PSMAxCD3 bispecific antibodies on effector T cells in the spleen of the immunocompetent mice.

DETAILED DESCRIPTION

[0093] Before the present invention is described, it is to be understood that this invention is not

limited to particular methods and experimental conditions described, as such methods and conditions may vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting, since the scope of the present invention will be limited only by the appended claims.

[0094] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. As used herein, the term “about,” when used in reference to a particular recited numerical value, means that the value may vary from the recited value by no more than 1%. For example, as used herein, the expression “about 100” includes 99 and 101 and all values in between (e.g., 99.1, 99.2, 99.3, 99.4, etc.).

[0095] Although any methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present invention, the preferred methods and materials are now described. All patents, applications and non-patent publications mentioned in this specification are incorporated herein by reference in their entireties.

Definitions

[0096] The expression “CD3,” as used herein, refers to an antigen which is expressed on T cells as part of the multimolecular T cell receptor (TCR) and which consists of a homodimer or heterodimer formed from the association of two of four receptor chains: CD3-epsilon, CD3-delta, CD3-zeta, and CD3-gamma. Human CD3-epsilon comprises the amino acid sequence as set forth in SEQ ID NO:1649; human CD3-delta comprises the amino acid sequence as set forth in SEQ ID NO: 1650. All references to proteins, polypeptides and protein fragments herein are intended to refer to the human version of the respective protein, polypeptide or protein fragment unless explicitly specified as being from a non-human species. Thus, the expression “CD3” means human CD3 unless specified as being from a non-human species, e.g., “mouse CD3,” “monkey CD3,” etc.

[0097] As used herein, “an antibody that binds CD3” or an “anti-CD3 antibody” includes antibodies and antigen-binding fragments thereof that specifically recognize a single CD3 subunit (e.g., epsilon, delta, gamma or zeta), as well as antibodies and antigen-binding fragments thereof that specifically recognize a dimeric complex of two CD3 subunits (e.g., gamma/epsilon, delta/epsilon, and zeta/zeta CD3 dimers). The antibodies and antigen-binding fragments of the present invention may bind soluble CD3 and/or cell surface expressed CD3. Soluble CD3 includes natural CD3 proteins as well as recombinant CD3 protein variants such as, e.g., monomeric and dimeric CD3 constructs, that lack a transmembrane domain or are otherwise unassociated with a cell membrane.

[0098] As used herein, the expression “cell surface-expressed CD3” means one or more CD3 protein(s) that is/are expressed on the surface of a cell in vitro or in vivo, such that at least a portion of a CD3 protein is exposed to the extracellular side of the cell membrane and is accessible to an antigen-binding portion of an antibody. “Cell surface-expressed CD3” includes CD3 proteins contained within the context of a functional T cell receptor in the membrane of a cell. The expression “cell surface-expressed CD3” includes CD3 protein expressed as part of a homodimer or heterodimer on the surface of a cell (e.g., gamma/epsilon, delta/epsilon, and zeta/zeta CD3 dimers). The expression, “cell surface-expressed CD3” also includes a CD3 chain (e.g., CD3-epsilon, CD3-delta or CD3-gamma) that is expressed by itself, without other CD3 chain types, on the surface of a cell. A “cell surface-expressed CD3” can comprise or consist of a CD3 protein expressed on the surface of a cell which normally expresses CD3 protein. Alternatively, “cell surface-expressed CD3” can comprise or consist of CD3 protein expressed on the surface of a cell that normally does not express human CD3 on its surface but has been artificially engineered to express CD3 on its surface.

[0099] The expression “PSMA,” as used herein, refers to prostate-specific membrane antigen, also known as folate hydrolase 1 (FOLH1). PSMA is an integral, non-shed membrane glycoprotein that is highly expressed in prostate epithelial cells and is a cell-surface marker for prostate cancer. The

amino acid sequence of human PSMA is set forth in SEQ ID NO:1651.

[0100] As used herein, “an antibody that binds PSMA” or an “anti-PSMA antibody” includes antibodies and antigen-binding fragments thereof that specifically recognize PSMA.

[0101] The term “antigen-binding molecule” includes antibodies and antigen-binding fragments of antibodies, including, e.g., bispecific antibodies.

[0102] The term “antibody”, as used herein, means any antigen-binding molecule or molecular complex comprising at least one complementarity determining region (CDR) that specifically binds to or interacts with a particular antigen (e.g., PSMA or CD3). The term “antibody” includes immunoglobulin molecules comprising four polypeptide chains, two heavy (H) chains and two light (L) chains inter-connected by disulfide bonds, as well as multimers thereof (e.g., IgM). Each heavy chain comprises a heavy chain variable region (abbreviated herein as HCVR or V.sub.H) and a heavy chain constant region. The heavy chain constant region comprises three domains, C.sub.H1, C.sub.H2 and C.sub.H3. Each light chain comprises a light chain variable region (abbreviated herein as LCVR or V.sub.L) and a light chain constant region. The light chain constant region comprises one domain (C.sub.L1). The V.sub.H and V.sub.L regions can be further subdivided into regions of hypervariability, termed complementarity determining regions (CDRs), interspersed with regions that are more conserved, termed framework regions (FR). Each V.sub.H and V.sub.L is composed of three CDRs and four FRs, arranged from amino-terminus to carboxy-terminus in the following order: FR1, CDR1, FR2, CDR2, FR3, CDR3, FR4. In different embodiments of the invention, the FRs of the anti-PSMA antibody or anti-CD3 antibody (or antigen-binding portion thereof) may be identical to the human germline sequences, or may be naturally or artificially modified. An amino acid consensus sequence may be defined based on a side-by-side analysis of two or more CDRs.

[0103] The term “antibody”, as used herein, also includes antigen-binding fragments of full antibody molecules. The terms “antigen-binding portion” of an antibody, “antigen-binding fragment” of an antibody, and the like, as used herein, include any naturally occurring, enzymatically obtainable, synthetic, or genetically engineered polypeptide or glycoprotein that specifically binds an antigen to form a complex. Antigen-binding fragments of an antibody may be derived, e.g., from full antibody molecules using any suitable standard techniques such as proteolytic digestion or recombinant genetic engineering techniques involving the manipulation and expression of DNA encoding antibody variable and optionally constant domains. Such DNA is known and/or is readily available from, e.g., commercial sources, DNA libraries (including, e.g., phage-antibody libraries), or can be synthesized. The DNA may be sequenced and manipulated chemically or by using molecular biology techniques, for example, to arrange one or more variable and/or constant domains into a suitable configuration, or to introduce codons, create cysteine residues, modify, add or delete amino acids, etc.

[0104] Non-limiting examples of antigen-binding fragments include: (i) Fab fragments; (ii) F(ab')₂ fragments; (iii) Fd fragments; (iv) Fv fragments; (v) single-chain Fv (scFv) molecules; (vi) dAb fragments; and (vii) minimal recognition units consisting of the amino acid residues that mimic the hypervariable region of an antibody (e.g., an isolated complementarity determining region (CDR) such as a CDR3 peptide), or a constrained FR3-CDR3-FR4 peptide. Other engineered molecules, such as domain-specific antibodies, single domain antibodies, domain-deleted antibodies, chimeric antibodies, CDR-grafted antibodies, diabodies, triabodies, tetrabodies, minibodies, nanobodies (e.g. monovalent nanobodies, bivalent nanobodies, etc.), small modular immunopharmaceuticals (SMIPs), and shark variable IgNAR domains, are also encompassed within the expression “antigen-binding fragment,” as used herein.

[0105] An antigen-binding fragment of an antibody will typically comprise at least one variable domain. The variable domain may be of any size or amino acid composition and will generally comprise at least one CDR which is adjacent to or in frame with one or more framework sequences. In antigen-binding fragments having a V.sub.H domain associated with a V.sub.L domain, the

V.sub.H and V.sub.L domains may be situated relative to one another in any suitable arrangement. For example, the variable region may be dimeric and contain V.sub.H-V.sub.H, V.sub.H-V.sub.L or V.sub.L-V.sub.L dimers. Alternatively, the antigen-binding fragment of an antibody may contain a monomeric V.sub.H or V.sub.L domain.

[0106] In certain embodiments, an antigen-binding fragment of an antibody may contain at least one variable domain covalently linked to at least one constant domain. Non-limiting, exemplary configurations of variable and constant domains that may be found within an antigen-binding fragment of an antibody of the present invention include: (i) V.sub.H-C.sub.H1; (ii) V.sub.H-C.sub.H2; (iii) V.sub.H-C.sub.H3; (iv) V.sub.H-C.sub.H1-C.sub.H2; (v) V.sub.H-C.sub.H1-C.sub.H2-C.sub.H3; (vi) V.sub.H-C.sub.H2-C.sub.H3; (vii) V.sub.H-C.sub.L; (viii) V.sub.L-C.sub.H1; (ix) V.sub.L-C.sub.H2; (x) V.sub.L-C.sub.H3; (xi) V.sub.L-C.sub.H1-C.sub.H2; (xii) V.sub.L-C.sub.H1-C.sub.H2-C.sub.H3; (xiii) V.sub.L-C.sub.H2-C.sub.H3; and (xiv) V.sub.L-C.sub.L. In any configuration of variable and constant domains, including any of the exemplary configurations listed above, the variable and constant domains may be either directly linked to one another or may be linked by a full or partial hinge or linker region. A hinge region may consist of at least 2 (e.g., 5, 10, 15, 20, 40, 60 or more) amino acids which result in a flexible or semi-flexible linkage between adjacent variable and/or constant domains in a single polypeptide molecule.

Moreover, an antigen-binding fragment of an antibody of the present invention may comprise a homo-dimer or hetero-dimer (or other multimer) of any of the variable and constant domain configurations listed above in non-covalent association with one another and/or with one or more monomeric V.sub.H or V.sub.L domain (e.g., by disulfide bond(s)).

[0107] As with full antibody molecules, antigen-binding fragments may be monospecific or multispecific (e.g., bispecific). A multispecific antigen-binding fragment of an antibody will typically comprise at least two different variable domains, wherein each variable domain is capable of specifically binding to a separate antigen or to a different epitope on the same antigen. Any multispecific antibody format, including the exemplary bispecific antibody formats disclosed herein, may be adapted for use in the context of an antigen-binding fragment of an antibody of the present invention using routine techniques available in the art.

[0108] The antibodies of the present invention may function through complement-dependent cytotoxicity (CDC) or antibody-dependent cell-mediated cytotoxicity (ADCC). "Complement-dependent cytotoxicity" (CDC) refers to lysis of antigen-expressing cells by an antibody of the invention in the presence of complement. "Antibody-dependent cell-mediated cytotoxicity" (ADCC) refers to a cell-mediated reaction in which nonspecific cytotoxic cells that express Fc receptors (FcRs) (e.g., Natural Killer (NK) cells, neutrophils, and macrophages) recognize bound antibody on a target cell and thereby lead to lysis of the target cell. CDC and ADCC can be measured using assays that are well known and available in the art. (See, e.g., U.S. Pat. Nos. 5,500,362 and 5,821,337, and Clynes et al. (1998) Proc. Natl. Acad. Sci. (USA) 95:652-656). The constant region of an antibody is important in the ability of an antibody to fix complement and mediate cell-dependent cytotoxicity. Thus, the isotype of an antibody may be selected on the basis of whether it is desirable for the antibody to mediate cytotoxicity.

[0109] In certain embodiments of the invention, the anti-PSMA monospecific antibodies or anti-PSMA/anti-CD3 bispecific antibodies of the invention are human antibodies. The term "human antibody", as used herein, is intended to include antibodies having variable and constant regions derived from human germline immunoglobulin sequences. The human antibodies of the invention may include amino acid residues not encoded by human germline immunoglobulin sequences (e.g., mutations introduced by random or site-specific mutagenesis in vitro or by somatic mutation in vivo), for example in the CDRs and in particular CDR3. However, the term "human antibody", as used herein, is not intended to include antibodies in which CDR sequences derived from the germline of another mammalian species, such as a mouse, have been grafted onto human framework sequences.

[0110] The antibodies of the invention may, in some embodiments, be recombinant human antibodies. The term “recombinant human antibody”, as used herein, is intended to include all human antibodies that are prepared, expressed, created or isolated by recombinant means, such as antibodies expressed using a recombinant expression vector transfected into a host cell (described further below), antibodies isolated from a recombinant, combinatorial human antibody library (described further below), antibodies isolated from an animal (e.g., a mouse) that is transgenic for human immunoglobulin genes (see e.g., Taylor et al. (1992) Nucl. Acids Res. 20:6287-6295) or antibodies prepared, expressed, created or isolated by any other means that involves splicing of human immunoglobulin gene sequences to other DNA sequences. Such recombinant human antibodies have variable and constant regions derived from human germline immunoglobulin sequences. In certain embodiments, however, such recombinant human antibodies are subjected to in vitro mutagenesis (or, when an animal transgenic for human Ig sequences is used, in vivo somatic mutagenesis) and thus the amino acid sequences of the V.sub.H and V.sub.L regions of the recombinant antibodies are sequences that, while derived from and related to human germline V.sub.H and V.sub.L sequences, may not naturally exist within the human antibody germline repertoire in vivo.

[0111] Human antibodies can exist in two forms that are associated with hinge heterogeneity. In one form, an immunoglobulin molecule comprises a stable four chain construct of approximately 150-160 kDa in which the dimers are held together by an interchain heavy chain disulfide bond. In a second form, the dimers are not linked via inter-chain disulfide bonds and a molecule of about 75-80 kDa is formed composed of a covalently coupled light and heavy chain (half-antibody). These forms have been extremely difficult to separate, even after affinity purification.

[0112] The frequency of appearance of the second form in various intact IgG isotypes is due to, but not limited to, structural differences associated with the hinge region isotype of the antibody. A single amino acid substitution in the hinge region of the human IgG4 hinge can significantly reduce the appearance of the second form (Angal et al. (1993) Molecular Immunology 30:105) to levels typically observed using a human IgG1 hinge. The instant invention encompasses antibodies having one or more mutations in the hinge, C.sub.H2 or C.sub.H3 region which may be desirable, for example, in production, to improve the yield of the desired antibody form.

[0113] The antibodies of the invention may be isolated antibodies. An “isolated antibody,” as used herein, means an antibody that has been identified and separated and/or recovered from at least one component of its natural environment. For example, an antibody that has been separated or removed from at least one component of an organism, or from a tissue or cell in which the antibody naturally exists or is naturally produced, is an “isolated antibody” for purposes of the present invention. An isolated antibody also includes an antibody in situ within a recombinant cell. Isolated antibodies are antibodies that have been subjected to at least one purification or isolation step. According to certain embodiments, an isolated antibody may be substantially free of other cellular material and/or chemicals.

[0114] The present invention also includes one-arm antibodies that bind PSMA. As used herein, a “one-arm antibody” means an antigen-binding molecule comprising a single antibody heavy chain and a single antibody light chain. The one-arm antibodies of the present invention may comprise any of the HCVR/LCVR or CDR amino acid sequences as set forth in Table 1.

[0115] The anti-PSMA or anti-PSMA/anti-CD3 antibodies disclosed herein may comprise one or more amino acid substitutions, insertions and/or deletions in the framework and/or CDR regions of the heavy and light chain variable domains as compared to the corresponding germline sequences from which the antibodies were derived. Such mutations can be readily ascertained by comparing the amino acid sequences disclosed herein to germline sequences available from, for example, public antibody sequence databases. The present invention includes antibodies, and antigen-binding fragments thereof, which are derived from any of the amino acid sequences disclosed herein, wherein one or more amino acids within one or more framework and/or CDR regions are mutated

to the corresponding residue(s) of the germline sequence from which the antibody was derived, or to the corresponding residue(s) of another human germline sequence, or to a conservative amino acid substitution of the corresponding germline residue(s) (such sequence changes are referred to herein collectively as “germline mutations”). A person of ordinary skill in the art, starting with the heavy and light chain variable region sequences disclosed herein, can easily produce numerous antibodies and antigen-binding fragments which comprise one or more individual germline mutations or combinations thereof. In certain embodiments, all of the framework and/or CDR residues within the V.sub.H and/or V.sub.L domains are mutated back to the residues found in the original germline sequence from which the antibody was derived. In other embodiments, only certain residues are mutated back to the original germline sequence, e.g., only the mutated residues found within the first 8 amino acids of FR1 or within the last 8 amino acids of FR4, or only the mutated residues found within CDR1, CDR2 or CDR3. In other embodiments, one or more of the framework and/or CDR residue(s) are mutated to the corresponding residue(s) of a different germline sequence (i.e., a germline sequence that is different from the germline sequence from which the antibody was originally derived). Furthermore, the antibodies of the present invention may contain any combination of two or more germline mutations within the framework and/or CDR regions, e.g., wherein certain individual residues are mutated to the corresponding residue of a particular germline sequence while certain other residues that differ from the original germline sequence are maintained or are mutated to the corresponding residue of a different germline sequence. Once obtained, antibodies and antigen-binding fragments that contain one or more germline mutations can be easily tested for one or more desired property such as, improved binding specificity, increased binding affinity, improved or enhanced antagonistic or agonistic biological properties (as the case may be), reduced immunogenicity, etc. Antibodies and antigen-binding fragments obtained in this general manner are encompassed within the present invention.

[0116] The present invention also includes anti-PSMA or anti-PSMA/anti-CD3 antibodies comprising variants of any of the HCVR, LCVR, and/or CDR amino acid sequences disclosed herein having one or more conservative substitutions. For example, the present invention includes anti-PSMA or anti-PSMA/anti-CD3 antibodies having HCVR, LCVR, and/or CDR amino acid sequences with, e.g., 10 or fewer, 8 or fewer, 6 or fewer, 4 or fewer, etc. conservative amino acid substitutions relative to any of the HCVR, LCVR, and/or CDR amino acid sequences set forth in Table 1 herein or as described in Tables 12, 14, 15, 18, and 20 herein.

[0117] The term “epitope” refers to an antigenic determinant that interacts with a specific antigen binding site in the variable region of an antibody molecule known as a paratope. A single antigen may have more than one epitope. Thus, different antibodies may bind to different areas on an antigen and may have different biological effects. Epitopes may be either conformational or linear. A conformational epitope is produced by spatially juxtaposed amino acids from different segments of the linear polypeptide chain. A linear epitope is one produced by adjacent amino acid residues in a polypeptide chain. In certain circumstance, an epitope may include moieties of saccharides, phosphoryl groups, or sulfonyl groups on the antigen.

[0118] The term “substantial identity” or “substantially identical,” when referring to a nucleic acid or fragment thereof, indicates that, when optimally aligned with appropriate nucleotide insertions or deletions with another nucleic acid (or its complementary strand), there is nucleotide sequence identity in at least about 95%, and more preferably at least about 96%, 97%, 98% or 99% of the nucleotide bases, as measured by any well-known algorithm of sequence identity, such as FASTA, BLAST or Gap, as discussed below. A nucleic acid molecule having substantial identity to a reference nucleic acid molecule may, in certain instances, encode a polypeptide having the same or substantially similar amino acid sequence as the polypeptide encoded by the reference nucleic acid molecule.

[0119] As applied to polypeptides, the term “substantial similarity” or “substantially similar” means that two peptide sequences, when optimally aligned, such as by the programs GAP or BESTFIT

using default gap weights, share at least 95% sequence identity, even more preferably at least 98% or 99% sequence identity. Preferably, residue positions which are not identical differ by conservative amino acid substitutions. A “conservative amino acid substitution” is one in which an amino acid residue is substituted by another amino acid residue having a side chain (R group) with similar chemical properties (e.g., charge or hydrophobicity). In general, a conservative amino acid substitution will not substantially change the functional properties of a protein. In cases where two or more amino acid sequences differ from each other by conservative substitutions, the percent sequence identity or degree of similarity may be adjusted upwards to correct for the conservative nature of the substitution. Means for making this adjustment are well-known to those of skill in the art. See, e.g., Pearson (1994) *Methods Mol. Biol.* 24:307-331, herein incorporated by reference. Examples of groups of amino acids that have side chains with similar chemical properties include (1) aliphatic side chains: glycine, alanine, valine, leucine and isoleucine; (2) aliphatic-hydroxyl side chains: serine and threonine; (3) amide-containing side chains: asparagine and glutamine; (4) aromatic side chains: phenylalanine, tyrosine, and tryptophan; (5) basic side chains: lysine, arginine, and histidine; (6) acidic side chains: aspartate and glutamate, and (7) sulfur-containing side chains are cysteine and methionine. Preferred conservative amino acids substitution groups are: valine-leucine-isoleucine, phenylalanine-tyrosine, lysine-arginine, alanine-valine, glutamate-aspartate, and asparagine-glutamine. Alternatively, a conservative replacement is any change having a positive value in the PAM250 log-likelihood matrix disclosed in Gonnet et al. (1992) *Science* 256:1443-1445, herein incorporated by reference. A “moderately conservative” replacement is any change having a nonnegative value in the PAM250 log-likelihood matrix.

[0120] Sequence similarity for polypeptides, which is also referred to as sequence identity, is typically measured using sequence analysis software. Protein analysis software matches similar sequences using measures of similarity assigned to various substitutions, deletions and other modifications, including conservative amino acid substitutions. For instance, GCG software contains programs such as Gap and Bestfit which can be used with default parameters to determine sequence homology or sequence identity between closely related polypeptides, such as homologous polypeptides from different species of organisms or between a wild type protein and a mutein thereof. See, e.g., GCG Version 6.1. Polypeptide sequences also can be compared using FASTA using default or recommended parameters, a program in GCG Version 6.1. FASTA (e.g., FASTA2 and FASTA3) provides alignments and percent sequence identity of the regions of the best overlap between the query and search sequences (Pearson (2000) *supra*). Another preferred algorithm when comparing a sequence of the invention to a database containing a large number of sequences from different organisms is the computer program BLAST, especially BLASTP or TBLASTN, using default parameters. See, e.g., Altschul et al. (1990) *J. Mol. Biol.* 215:403-410 and Altschul et al. (1997) *Nucleic Acids Res.* 25:3389-402, each herein incorporated by reference.

Germline Mutations

[0121] The anti-CD3 antibodies disclosed herein comprise one or more amino acid substitutions, insertions and/or deletions in the framework and/or CDR regions of the heavy chain variable domains as compared to the corresponding germline sequences from which the antibodies were derived.

[0122] The present invention also includes antibodies, and antigen-binding fragments thereof, which are derived from any of the amino acid sequences disclosed herein, wherein one or more amino acids within one or more framework and/or CDR regions are mutated to the corresponding residue(s) of the germline sequence from which the antibody was derived, or to the corresponding residue(s) of another human germline sequence, or to a conservative amino acid substitution of the corresponding germline residue(s) (such sequence changes are referred to herein collectively as “germline mutations”), and having weak or no detectable binding to a CD3 antigen. Several such exemplary antibodies that recognize CD3 are described in Tables 12 and 18 herein.

[0123] Furthermore, the antibodies of the present invention may contain any combination of two or

more germline mutations within the framework and/or CDR regions, e.g., wherein certain individual residues are mutated to the corresponding residue of a particular germline sequence while certain other residues that differ from the original germline sequence are maintained or are mutated to the corresponding residue of a different germline sequence. Once obtained, antibodies and antigen-binding fragments that contain one or more germline mutations can be tested for one or more desired properties such as, improved binding specificity, weak or reduced binding affinity, improved or enhanced pharmacokinetic properties, reduced immunogenicity, etc. Antibodies and antigen-binding fragments obtained in this general manner given the guidance of the present disclosure are encompassed within the present invention.

[0124] The present invention also includes anti-CD3 antibodies comprising variants of any of the HCVR, LCVR, and/or CDR amino acid sequences disclosed herein having one or more conservative substitutions. For example, the present invention includes anti-CD3 antibodies having HCVR, LCVR, and/or CDR amino acid sequences with, e.g., 10 or fewer, 8 or fewer, 6 or fewer, 4 or fewer, etc. conservative amino acid substitutions relative to any of the HCVR, LCVR, and/or CDR amino acid sequences set forth in Tables 12, 14, 15, 18, and 20 herein. The antibodies and bispecific antigen-binding molecules of the present invention comprise one or more amino acid substitutions, insertions and/or deletions in the framework and/or CDR regions of the heavy and light chain variable domains as compared to the corresponding germline sequences from which the individual antigen-binding domains were derived, while maintaining or improving the desired weak-to-no detectable binding to CD3 antigen. A “conservative amino acid substitution” is one in which an amino acid residue is substituted by another amino acid residue having a side chain (R group) with similar chemical properties (e.g., charge or hydrophobicity). In general, a conservative amino acid substitution will not substantially change the functional properties of a protein, i.e. the amino acid substitution maintains or improves the desired weak to no detectable binding affinity in the case of anti-CD3 binding molecules. Examples of groups of amino acids that have side chains with similar chemical properties include (1) aliphatic side chains: glycine, alanine, valine, leucine and isoleucine; (2) aliphatic-hydroxyl side chains: serine and threonine; (3) amide-containing side chains: asparagine and glutamine; (4) aromatic side chains: phenylalanine, tyrosine, and tryptophan; (5) basic side chains: lysine, arginine, and histidine; (6) acidic side chains: aspartate and glutamate, and (7) sulfur-containing side chains are cysteine and methionine. Preferred conservative amino acids substitution groups are: valine-leucine-isoleucine, phenylalanine-tyrosine, lysine-arginine, alanine-valine, glutamate-aspartate, and asparagine-glutamine. Alternatively, a conservative replacement is any change having a positive value in the PAM250 log-likelihood matrix disclosed in Gonnet et al. (1992) Science 256:1443-1445. A “moderately conservative” replacement is any change having a nonnegative value in the PAM250 log-likelihood matrix.

[0125] The present invention also includes antigen-binding molecules comprising an antigen-binding domain with an HCVR and/or CDR amino acid sequence that is substantially identical to any of the HCVR and/or CDR amino acid sequences disclosed herein, while maintaining or improving the desired weak affinity to CD3 antigen. The term “substantial identity” or “substantially identical,” when referring to an amino acid sequence means that two amino acid sequences, when optimally aligned, such as by the programs GAP or BESTFIT using default gap weights, share at least 95% sequence identity, even more preferably at least 98% or 99% sequence identity. Preferably, residue positions which are not identical differ by conservative amino acid substitutions. In cases where two or more amino acid sequences differ from each other by conservative substitutions, the percent sequence identity or degree of similarity may be adjusted upwards to correct for the conservative nature of the substitution. Means for making this adjustment are well-known to those of skill in the art. See, e.g., Pearson (1994) Methods Mol. Biol. 24:307-331.

[0126] Sequence similarity for polypeptides, which is also referred to as sequence identity, is

typically measured using sequence analysis software. Protein analysis software matches similar sequences using measures of similarity assigned to various substitutions, deletions and other modifications, including conservative amino acid substitutions. For instance, GCG software contains programs such as Gap and Bestfit which can be used with default parameters to determine sequence homology or sequence identity between closely related polypeptides, such as homologous polypeptides from different species of organisms or between a wild type protein and a mutein thereof. See, e.g., GCG Version 6.1. Polypeptide sequences also can be compared using FASTA using default or recommended parameters, a program in GCG Version 6.1. FASTA (e.g., FASTA2 and FASTA3) provides alignments and percent sequence identity of the regions of the best overlap between the query and search sequences (Pearson (2000) supra). Another preferred algorithm when comparing a sequence of the invention to a database containing a large number of sequences from different organisms is the computer program BLAST, especially BLASTP or TBLASTN, using default parameters. See, e.g., Altschul et al. (1990) *J. Mol. Biol.* 215:403-410 and Altschul et al. (1997) *Nucleic Acids Res.* 25:3389-402.

[0127] Once obtained, antigen-binding domains that contain one or more germline mutations were tested for decreased binding affinity utilizing one or more in vitro assays. Although antibodies that recognize a particular antigen are typically screened for their purpose by testing for high (i.e. strong) binding affinity to the antigen, the antibodies of the present invention exhibit weak binding or no detectable binding. Bispecific antigen-binding molecules comprising one or more antigen-binding domains obtained in this general manner are also encompassed within the present invention and were found to be advantageous as avidity-driven tumor therapies.

[0128] Unexpected benefits, for example, improved pharmacokinetic properties and low toxicity to the patient may be realized from the methods described herein.

Binding Properties of the Antibodies

[0129] As used herein, the term “binding” in the context of the binding of an antibody, immunoglobulin, antibody-binding fragment, or Fc-containing protein to either, e.g., a predetermined antigen, such as a cell surface protein or fragment thereof, typically refers to an interaction or association between a minimum of two entities or molecular structures, such as an antibody-antigen interaction.

[0130] For instance, binding affinity typically corresponds to a $K_{sub.D}$ value of about 10^{-7} M or less, such as about 10^{-8} M or less, such as about 10^{-9} M or less when determined by, for instance, surface plasmon resonance (SPR) technology in a BIAcore™ 3000 instrument using the antigen as the ligand and the antibody, Ig, antibody-binding fragment, or Fc-containing protein as the analyte (or antiligand). Cell-based binding strategies, such as fluorescent-activated cell sorting (FACS) binding assays, are also routinely used, and FACS data correlates well with other methods such as radioligand competition binding and SPR (Benedict, C A, *J Immunol Methods*. 1997, 201(2):223-31; Geuijen, C A, et al. *J Immunol Methods*. 2005, 302(1-2):68-77).

[0131] Accordingly, the antibody or antigen-binding protein of the invention binds to the predetermined antigen or cell surface molecule (receptor) having an affinity corresponding to a $K_{sub.D}$ value that is at least ten-fold lower than its affinity for binding to a non-specific antigen (e.g., BSA, casein). According to the present invention, the affinity of an antibody corresponding to a $K_{sub.D}$ value that is equal to or less than ten-fold lower than a non-specific antigen may be considered non-detectable binding, however such an antibody may be paired with a second antigen binding arm for the production of a bispecific antibody of the invention.

[0132] The term “ $K_{sub.D}$ ” (M) refers to the dissociation equilibrium constant of a particular antibody-antigen interaction, or the dissociation equilibrium constant of an antibody or antibody-binding fragment binding to an antigen. There is an inverse relationship between $K_{sub.D}$ and binding affinity, therefore the smaller the $K_{sub.D}$ value, the higher, i.e. stronger, the affinity. Thus, the terms “higher affinity” or “stronger affinity” relate to a higher ability to form an interaction and therefore a smaller $K_{sub.D}$ value, and conversely the terms “lower affinity” or “weaker affinity”

relate to a lower ability to form an interaction and therefore a larger $K_{sub.D}$ value. In some circumstances, a higher binding affinity (or $K_{sub.D}$) of a particular molecule (e.g. antibody) to its interactive partner molecule (e.g. antigen X) compared to the binding affinity of the molecule (e.g. antibody) to another interactive partner molecule (e.g. antigen Y) may be expressed as a binding ratio determined by dividing the larger $K_{sub.D}$ value (lower, or weaker, affinity) by the smaller $K_{sub.D}$ (higher, or stronger, affinity), for example expressed as 5-fold or 10-fold greater binding affinity, as the case may be.

[0133] The term “ $k_{sub.d}$ ” (sec^{-1} or $1/s$) refers to the dissociation rate constant of a particular antibody-antigen interaction, or the dissociation rate constant of an antibody or antibody-binding fragment. Said value is also referred to as the $k_{sub.off}$ value.

[0134] The term “ $k_{sub.a}$ ” ($M^{-1} \times sec^{-1}$ or $1/M$) refers to the association rate constant of a particular antibody-antigen interaction, or the association rate constant of an antibody or antibody-binding fragment.

[0135] The term “ $K_{sub.A}$ ” (M^{-1} or $1/M$) refers to the association equilibrium constant of a particular antibody-antigen interaction, or the association equilibrium constant of an antibody or antibody-binding fragment. The association equilibrium constant is obtained by dividing the $k_{sub.a}$ by the $k_{sub.d}$.

[0136] The term “EC₅₀” or “EC_{sub.50}” refers to the half maximal effective concentration, which includes the concentration of an antibody which induces a response halfway between the baseline and maximum after a specified exposure time. The EC_{sub.50} essentially represents the concentration of an antibody where 50% of its maximal effect is observed. In certain embodiments, the EC_{sub.50} value equals the concentration of an antibody of the invention that gives half-maximal binding to cells expressing CD3 or tumor-associated antigen, as determined by e.g. a FACS binding assay. Thus, reduced or weaker binding is observed with an increased EC_{sub.50}, or half maximal effective concentration value.

[0137] In one embodiment, decreased binding can be defined as an increased EC_{sub.50} antibody concentration which enables binding to the half-maximal amount of target cells.

[0138] In another embodiment, the EC_{sub.50} value represents the concentration of an antibody of the invention that elicits half-maximal depletion of target cells by T cell cytotoxic activity. Thus, increased cytotoxic activity (e.g. T cell-mediated tumor cell killing) is observed with a decreased EC_{sub.50}, or half maximal effective concentration value.

Bispecific Antigen-Binding Molecules

[0139] The antibodies of the present invention may be monospecific, bi-specific, or multispecific. Multispecific antibodies may be specific for different epitopes of one target polypeptide or may contain antigen-binding domains specific for more than one target polypeptide. See, e.g., Tutt et al., 1991, J. Immunol. 147:60-69; Kufer et al., 2004, Trends Biotechnol. 22:238-244. The anti-PSMA monospecific antibodies or anti-PSMA/anti-CD3 bispecific antibodies of the present invention can be linked to or co-expressed with another functional molecule, e.g., another peptide or protein. For example, an antibody or fragment thereof can be functionally linked (e.g., by chemical coupling, genetic fusion, noncovalent association or otherwise) to one or more other molecular entities, such as another antibody or antibody fragment to produce a bi-specific or a multispecific antibody with a second or additional binding specificity.

[0140] Use of the expression “anti-CD3 antibody” or “anti-PSMA antibody” herein is intended to include both monospecific anti-CD3 or anti-PSMA antibodies as well as bispecific antibodies comprising a CD3-binding arm and a PSMA-binding arm. Thus, the present invention includes bispecific antibodies wherein one arm of an immunoglobulin binds human CD3, and the other arm of the immunoglobulin is specific for human PSMA. The CD3-binding arm can comprise any of the HCVR/LCVR or CDR amino acid sequences as set forth in Tables 12, 14, 15, 18, and 20 herein.

[0141] In certain embodiments, the CD3-binding arm binds to human CD3 and induces human T

cell activation. In certain embodiments, the CD3-binding arm binds weakly to human CD3 and induces human T cell activation. In other embodiments, the CD3-binding arm binds weakly to human CD3 and induces tumor-associated antigen-expressing cell killing in the context of a bispecific or multispecific antibody. In other embodiments, the CD3-binding arm binds or associated weakly with human and cynomolgus (monkey) CD3, yet the binding interaction is not detectable by in vitro assays known in the art. The PSMA-binding arm can comprise any of the HCVR/LCVR or CDR amino acid sequences as set forth in Table 1 herein.

[0142] According to certain exemplary embodiments, the present invention includes bispecific antigen-binding molecules that specifically bind CD3 and PSMA. Such molecules may be referred to herein as, e.g., “anti-CD3/anti-PSMA,” or “anti-CD3×PSMA” or “CD3×PSMA” bispecific molecules, or other similar terminology (e.g., anti-PSMA/anti-CD3).

[0143] The term “PSMA,” as used herein, refers to the human PSMA protein unless specified as being from a non-human species (e.g., “mouse PSMA,” “monkey PSMA,” etc.). The human PSMA protein has the amino acid sequence shown in SEQ ID NO:1651.

[0144] The aforementioned bispecific antigen-binding molecules that specifically bind CD3 and PSMA may comprise an anti-CD3 antigen-binding molecule which binds to CD3 with a weak binding affinity such as exhibiting a K_{sub.D} of greater than about 40 nM, as measured by an in vitro affinity binding assay.

[0145] As used herein, the expression “antigen-binding molecule” means a protein, polypeptide or molecular complex comprising or consisting of at least one complementarity determining region (CDR) that alone, or in combination with one or more additional CDRs and/or framework regions (FRs), specifically binds to a particular antigen. In certain embodiments, an antigen-binding molecule is an antibody or a fragment of an antibody, as those terms are defined elsewhere herein.

[0146] As used herein, the expression “bispecific antigen-binding molecule” means a protein, polypeptide or molecular complex comprising at least a first antigen-binding domain and a second antigen-binding domain. Each antigen-binding domain within the bispecific antigen-binding molecule comprises at least one CDR that alone, or in combination with one or more additional CDRs and/or FRs, specifically binds to a particular antigen. In the context of the present invention, the first antigen-binding domain specifically binds a first antigen (e.g., CD3), and the second antigen-binding domain specifically binds a second, distinct antigen (e.g., PSMA).

[0147] In certain exemplary embodiments of the present invention, the bispecific antigen-binding molecule is a bispecific antibody. Each antigen-binding domain of a bispecific antibody comprises a heavy chain variable domain (HCVR) and a light chain variable domain (LCVR). In the context of a bispecific antigen-binding molecule comprising a first and a second antigen-binding domain (e.g., a bispecific antibody), the CDRs of the first antigen-binding domain may be designated with the prefix “A1” and the CDRs of the second antigen-binding domain may be designated with the prefix “A2”. Thus, the CDRs of the first antigen-binding domain may be referred to herein as A1-HCDR1, A1-HCDR2, and A1-HCDR3; and the CDRs of the second antigen-binding domain may be referred to herein as A2-HCDR1, A2-HCDR2, and A2-HCDR3.

[0148] The first antigen-binding domain and the second antigen-binding domain may be directly or indirectly connected to one another to form a bispecific antigen-binding molecule of the present invention. Alternatively, the first antigen-binding domain and the second antigen-binding domain may each be connected to a separate multimerizing domain. The association of one multimerizing domain with another multimerizing domain facilitates the association between the two antigen-binding domains, thereby forming a bispecific antigen-binding molecule. As used herein, a “multimerizing domain” is any macromolecule, protein, polypeptide, peptide, or amino acid that has the ability to associate with a second multimerizing domain of the same or similar structure or constitution. For example, a multimerizing domain may be a polypeptide comprising an immunoglobulin C_{sub}.H3 domain. A non-limiting example of a multimerizing component is an Fc portion of an immunoglobulin (comprising a C_{sub}.H2-C_{sub}.H3 domain), e.g., an Fc domain of an

IgG selected from the isotypes IgG1, IgG2, IgG3, and IgG4, as well as any allotype within each isotype group.

[0149] Bispecific antigen-binding molecules of the present invention will typically comprise two multimerizing domains, e.g., two Fc domains that are each individually part of a separate antibody heavy chain. The first and second multimerizing domains may be of the same IgG isotype such as, e.g., IgG1/IgG1, IgG2/IgG2, IgG4/IgG4. Alternatively, the first and second multimerizing domains may be of different IgG isotypes such as, e.g., IgG1/IgG2, IgG1/IgG4, IgG2/IgG4, etc.

[0150] In certain embodiments, the multimerizing domain is an Fc fragment or an amino acid sequence of from 1 to about 200 amino acids in length containing at least one cysteine residue. In other embodiments, the multimerizing domain is a cysteine residue, or a short cysteine-containing peptide. Other multimerizing domains include peptides or polypeptides comprising or consisting of a leucine zipper, a helix-loop motif, or a coiled-coil motif.

[0151] Any bispecific antibody format or technology may be used to make the bispecific antigen-binding molecules of the present invention. For example, an antibody or fragment thereof having a first antigen binding specificity can be functionally linked (e.g., by chemical coupling, genetic fusion, noncovalent association or otherwise) to one or more other molecular entities, such as another antibody or antibody fragment having a second antigen-binding specificity to produce a bispecific antigen-binding molecule. Specific exemplary bispecific formats that can be used in the context of the present invention include, without limitation, e.g., scFv-based or diabody bispecific formats, IgG-scFv fusions, dual variable domain (DVD)-Ig, Quadroma, knobs-into-holes, common light chain (e.g., common light chain with knobs-into-holes, etc.), CrossMab, CrossFab, (SEED) body, leucine zipper, Duobody, IgG1/IgG2, dual acting Fab (DAF)-IgG, and Mab2 bispecific formats (see, e.g., Klein et al. 2012, mAbs 4:6, 1-11, and references cited therein, for a review of the foregoing formats).

[0152] In the context of bispecific antigen-binding molecules of the present invention, the multimerizing domains, e.g., Fc domains, may comprise one or more amino acid changes (e.g., insertions, deletions or substitutions) as compared to the wild-type, naturally occurring version of the Fc domain. For example, the invention includes bispecific antigen-binding molecules comprising one or more modifications in the Fc domain that results in a modified Fc domain having a modified binding interaction (e.g., enhanced or diminished) between Fc and FcRn. In one embodiment, the bispecific antigen-binding molecule comprises a modification in a C.sub.H2 or a C.sub.H3 region, wherein the modification increases the affinity of the Fc domain to FcRn in an acidic environment (e.g., in an endosome where pH ranges from about 5.5 to about 6.0). Non-limiting examples of such Fc modifications include, e.g., a modification at position 250 (e.g., E or Q); 250 and 428 (e.g., L or F); 252 (e.g., L/Y/F/W or T), 254 (e.g., S or T), and 256 (e.g., S/R/Q/E/D or T); or a modification at position 428 and/or 433 (e.g., L/R/S/P/Q or K) and/or 434 (e.g., H/F or Y); or a modification at position 250 and/or 428; or a modification at position 307 or 308 (e.g., 308F, V308F), and 434. In one embodiment, the modification comprises a 428L (e.g., M428L) and 434S (e.g., N434S) modification; a 428L, 259I (e.g., V259I), and 308F (e.g., V308F) modification; a 433K (e.g., H433K) and a 434 (e.g., 434Y) modification; a 252, 254, and 256 (e.g., 252Y, 254T, and 256E) modification; a 250Q and 428L modification (e.g., T250Q and M428L); and a 307 and/or 308 modification (e.g., 308F or 308P).

[0153] The present invention also includes bispecific antigen-binding molecules comprising a first C.sub.H3 domain and a second Ig C.sub.H3 domain, wherein the first and second Ig C.sub.H3 domains differ from one another by at least one amino acid, and wherein at least one amino acid difference reduces binding of the bispecific antibody to Protein A as compared to a bi-specific antibody lacking the amino acid difference. In one embodiment, the first Ig C.sub.H3 domain binds Protein A and the second Ig C.sub.H3 domain contains a mutation that reduces or abolishes Protein A binding such as an H95R modification (by IMGT exon numbering; H435R by EU numbering). The second C.sub.H3 may further comprise a Y96F modification (by IMGT; Y436F by EU). See,

for example, U.S. Pat. No. 8,586,713. Further modifications that may be found within the second C.sub.H3 include: D16E, L18M, N44S, K52N, V57M, and V82I (by IMGT; D356E, L358M, N384S, K392N, V397M, and V422I by EU) in the case of IgG1 antibodies; N44S, K52N, and V82I (IMGT; N384S, K392N, and V422I by EU) in the case of IgG2 antibodies; and Q15R, N44S, K52N, V57M, R69K, E79Q, and V82I (by IMGT; Q355R, N384S, K392N, V397M, R409K, E419Q, and V422I by EU) in the case of IgG4 antibodies.

[0154] In certain embodiments, the Fc domain may be chimeric, combining Fc sequences derived from more than one immunoglobulin isotype. For example, a chimeric Fc domain can comprise part or all of a C.sub.H2 sequence derived from a human IgG1, human IgG2 or human IgG4 C.sub.H2 region, and part or all of a C.sub.H3 sequence derived from a human IgG1, human IgG2 or human IgG4. A chimeric Fc domain can also contain a chimeric hinge region. For example, a chimeric hinge may comprise an “upper hinge” sequence, derived from a human IgG1, a human IgG2 or a human IgG4 hinge region, combined with a “lower hinge” sequence, derived from a human IgG1, a human IgG2 or a human IgG4 hinge region. A particular example of a chimeric Fc domain that can be included in any of the antigen-binding molecules set forth herein comprises, from N- to C-terminus: [IgG4 C.sub.H1]–[IgG4 upper hinge]–[IgG2 lower hinge]–[IgG4 CH2]–[IgG4 CH3]. Another example of a chimeric Fc domain that can be included in any of the antigen-binding molecules set forth herein comprises, from N- to C-terminus: [IgG1 CH1]–[IgG1 upper hinge]–[IgG2 lower hinge]–[IgG4 CH2]–[IgG1 CH3]. These and other examples of chimeric Fc domains that can be included in any of the antigen-binding molecules of the present invention are described in US Publication 2014/0243504, published Aug. 28, 2014, which is herein incorporated in its entirety. Chimeric Fc domains having these general structural arrangements, and variants thereof, can have altered Fc receptor binding, which in turn affects Fc effector function.

[0155] In certain embodiments, the invention provides an antibody heavy chain wherein the heavy chain constant region (CH) region comprises an amino acid sequence at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to any one of SEQ ID NO: 1663, SEQ ID NO: 1664, SEQ ID NO: 1665, SEQ ID NO: 1666, SEQ ID NO: 1667, SEQ ID NO: 1668, SEQ ID NO: 1669, SEQ ID NO: 1670 SEQ ID NO: 1671 or SEQ ID NO: 1672. In some embodiments, the heavy chain constant region (CH) region comprises an amino acid sequence selected from the group consisting of SEQ ID NO: 1663, SEQ ID NO: 1664, SEQ ID NO: 1665, SEQ ID NO: 1666, SEQ ID NO: 1667, SEQ ID NO: 1668, SEQ ID NO: 1669, SEQ ID NO: 1670, SEQ ID NO: 1671 and SEQ ID NO: 1672.

[0156] In other embodiments, the invention provides an antibody heavy chain wherein the Fc domain comprises an amino acid sequence at least 95%, at least 96%, at least 97%, at least 98%, at least 99% identical to any one of SEQ ID NO: 1673, SEQ ID NO: 1674, SEQ ID NO: 1675, SEQ ID NO: 1676, SEQ ID NO: 1677, SEQ ID NO: 1678, SEQ ID NO: 1679, SEQ ID NO: 1680, SEQ ID NO: 1681 or SEQ ID NO: 1682. In some embodiments, the Fc domain comprises an amino acid sequence selected from the group consisting of SEQ ID NO: 1673, SEQ ID NO: 1674, SEQ ID NO: 1675, SEQ ID NO: 1676, SEQ ID NO: 1677, SEQ ID NO: 1678, SEQ ID NO: 1679, SEQ ID NO: 1680, SEQ ID NO: 1681 and SEQ ID NO: 1682.

Sequence Variants

[0157] The antibodies and bispecific antigen-binding molecules of the present invention may comprise one or more amino acid substitutions, insertions and/or deletions in the framework and/or CDR regions of the heavy and light chain variable domains as compared to the corresponding germline sequences from which the individual antigen-binding domains were derived. Such mutations can be readily ascertained by comparing the amino acid sequences disclosed herein to germline sequences available from, for example, public antibody sequence databases. The antigen-binding molecules of the present invention may comprise antigen-binding domains which are derived from any of the exemplary amino acid sequences disclosed herein, wherein one or more amino acids within one or more framework and/or CDR regions are mutated to the corresponding

residue(s) of the germline sequence from which the antibody was derived, or to the corresponding residue(s) of another human germline sequence, or to a conservative amino acid substitution of the corresponding germline residue(s) (such sequence changes are referred to herein collectively as “germline mutations”). A person of ordinary skill in the art, starting with the heavy and light chain variable region sequences disclosed herein, can easily produce numerous antibodies and antigen-binding fragments which comprise one or more individual germline mutations or combinations thereof. In certain embodiments, all of the framework and/or CDR residues within the V.sub.H and/or V.sub.L domains are mutated back to the residues found in the original germline sequence from which the antigen-binding domain was originally derived. In other embodiments, only certain residues are mutated back to the original germline sequence, e.g., only the mutated residues found within the first 8 amino acids of FR1 or within the last 8 amino acids of FR4, or only the mutated residues found within CDR1, CDR2 or CDR3. In other embodiments, one or more of the framework and/or CDR residue(s) are mutated to the corresponding residue(s) of a different germline sequence (i.e., a germline sequence that is different from the germline sequence from which the antigen-binding domain was originally derived). Furthermore, the antigen-binding domains may contain any combination of two or more germline mutations within the framework and/or CDR regions, e.g., wherein certain individual residues are mutated to the corresponding residue of a particular germline sequence while certain other residues that differ from the original germline sequence are maintained or are mutated to the corresponding residue of a different germline sequence. Once obtained, antigen-binding domains that contain one or more germline mutations can be easily tested for one or more desired property such as, improved binding specificity, increased binding affinity, improved or enhanced antagonistic or agonistic biological properties (as the case may be), reduced immunogenicity, etc. Bispecific antigen-binding molecules comprising one or more antigen-binding domains obtained in this general manner are encompassed within the present invention.

[0158] The present invention also includes antigen-binding molecules wherein one or both antigen-binding domains comprise variants of any of the HCVR, LCVR, and/or CDR amino acid sequences disclosed herein having one or more conservative substitutions. For example, the present invention includes antigen-binding molecules comprising an antigen-binding domain having HCVR, LCVR, and/or CDR amino acid sequences with, e.g., 10 or fewer, 8 or fewer, 6 or fewer, 4 or fewer, etc. conservative amino acid substitutions relative to any of the HCVR, LCVR, and/or CDR amino acid sequences disclosed herein. A “conservative amino acid substitution” is one in which an amino acid residue is substituted by another amino acid residue having a side chain (R group) with similar chemical properties (e.g., charge or hydrophobicity). In general, a conservative amino acid substitution will not substantially change the functional properties of a protein. Examples of groups of amino acids that have side chains with similar chemical properties include (1) aliphatic side chains: glycine, alanine, valine, leucine and isoleucine; (2) aliphatic-hydroxyl side chains: serine and threonine; (3) amide-containing side chains: asparagine and glutamine; (4) aromatic side chains: phenylalanine, tyrosine, and tryptophan; (5) basic side chains: lysine, arginine, and histidine; (6) acidic side chains: aspartate and glutamate, and (7) sulfur-containing side chains are cysteine and methionine. Preferred conservative amino acids substitution groups are: valine-leucine-isoleucine, phenylalanine-tyrosine, lysine-arginine, alanine-valine, glutamate-aspartate, and asparagine-glutamine. Alternatively, a conservative replacement is any change having a positive value in the PAM250 log-likelihood matrix disclosed in Gonnet et al. (1992) Science 256:1443-1445, herein incorporated by reference. A “moderately conservative” replacement is any change having a nonnegative value in the PAM250 log-likelihood matrix.

[0159] The present invention also includes antigen-binding molecules comprising an antigen-binding domain with an HCVR, LCVR, and/or CDR amino acid sequence that is substantially identical to any of the HCVR, LCVR, and/or CDR amino acid sequences disclosed herein. The term “substantial identity” or “substantially identical,” when referring to an amino acid sequence

means that two amino acid sequences, when optimally aligned, such as by the programs GAP or BESTFIT using default gap weights, share at least 95% sequence identity, even more preferably at least 98% or 99% sequence identity. Preferably, residue positions which are not identical differ by conservative amino acid substitutions. In cases where two or more amino acid sequences differ from each other by conservative substitutions, the percent sequence identity or degree of similarity may be adjusted upwards to correct for the conservative nature of the substitution. Means for making this adjustment are well-known to those of skill in the art. See, e.g., Pearson (1994) *Methods Mol. Biol.* 24:307-331, herein incorporated by reference.

[0160] Sequence similarity for polypeptides, which is also referred to as sequence identity, is typically measured using sequence analysis software. Protein analysis software matches similar sequences using measures of similarity assigned to various substitutions, deletions and other modifications, including conservative amino acid substitutions. For instance, GCG software contains programs such as Gap and Bestfit which can be used with default parameters to determine sequence homology or sequence identity between closely related polypeptides, such as homologous polypeptides from different species of organisms or between a wild type protein and a mutin thereof. See, e.g., GCG Version 6.1. Polypeptide sequences also can be compared using FASTA using default or recommended parameters, a program in GCG Version 6.1. FASTA (e.g., FASTA2 and FASTA3) provides alignments and percent sequence identity of the regions of the best overlap between the query and search sequences (Pearson (2000) *supra*). Another preferred algorithm when comparing a sequence of the invention to a database containing a large number of sequences from different organisms is the computer program BLAST, especially BLASTP or TBLASTN, using default parameters. See, e.g., Altschul et al. (1990) *J. Mol. Biol.* 215:403-410 and Altschul et al. (1997) *Nucleic Acids Res.* 25:3389-402, each herein incorporated by reference.

pH-Dependent Binding

[0161] The present invention includes anti-PSMA antibodies, and anti-CD3/anti-PSMA bispecific antigen-binding molecules, with pH-dependent binding characteristics. For example, an anti-PSMA antibody of the present invention may exhibit reduced binding to PSMA at acidic pH as compared to neutral pH. Alternatively, anti-PSMA antibodies of the invention may exhibit enhanced binding to PSMA at acidic pH as compared to neutral pH. The expression “acidic pH” includes pH values less than about 6.2, e.g., about 6.0, 5.95, 5.9, 5.85, 5.8, 5.75, 5.7, 5.65, 5.6, 5.55, 5.5, 5.45, 5.4, 5.35, 5.3, 5.25, 5.2, 5.15, 5.1, 5.05, 5.0, or less. As used herein, the expression “neutral pH” means a pH of about 7.0 to about 7.4. The expression “neutral pH” includes pH values of about 7.0, 7.05, 7.1, 7.15, 7.2, 7.25, 7.3, 7.35, and 7.4.

[0162] In certain instances, “reduced binding . . . at acidic pH as compared to neutral pH” is expressed in terms of a ratio of the K_{sub}.D value of the antibody binding to its antigen at acidic pH to the K_{sub}.D value of the antibody binding to its antigen at neutral pH (or vice versa). For example, an antibody or antigen-binding fragment thereof may be regarded as exhibiting “reduced binding to PSMA at acidic pH as compared to neutral pH” for purposes of the present invention if the antibody or antigen-binding fragment thereof exhibits an acidic/neutral K_{sub}.D ratio of about 3.0 or greater. In certain exemplary embodiments, the acidic/neutral K_{sub}.D ratio for an antibody or antigen-binding fragment of the present invention can be about 3.0, 3.5, 4.0, 4.5, 5.0, 5.5, 6.0, 6.5, 7.0, 7.5, 8.0, 8.5, 9.0, 9.5, 10.0, 10.5, 11.0, 11.5, 12.0, 12.5, 13.0, 13.5, 14.0, 14.5, 15.0, 20.0, 25.0, 30.0, 40.0, 50.0, 60.0, 70.0, 100.0 or greater.

[0163] Antibodies with pH-dependent binding characteristics may be obtained, e.g., by screening a population of antibodies for reduced (or enhanced) binding to a particular antigen at acidic pH as compared to neutral pH. Additionally, modifications of the antigen-binding domain at the amino acid level may yield antibodies with pH-dependent characteristics. For example, by substituting one or more amino acids of an antigen-binding domain (e.g., within a CDR) with a histidine residue, an antibody with reduced antigen-binding at acidic pH relative to neutral pH may be obtained.

Antibodies Comprising Fc Variants

[0164] According to certain embodiments of the present invention, anti-PSMA antibodies, and anti-CD3/anti-PSMA bispecific antigen-binding molecules, are provided comprising an Fc domain comprising one or more mutations which enhance or diminish antibody binding to the FcRn receptor, e.g., at acidic pH as compared to neutral pH. For example, the present invention includes antibodies comprising a mutation in the C.sub.H2 or a C.sub.H3 region of the Fc domain, wherein the mutation(s) increases the affinity of the Fc domain to FcRn in an acidic environment (e.g., in an endosome where pH ranges from about 5.5 to about 6.0). Such mutations may result in an increase in serum half-life of the antibody when administered to an animal. Non-limiting examples of such Fc modifications include, e.g., a modification at position 250 (e.g., E or Q); 250 and 428 (e.g., L or F); 252 (e.g., L/Y/F/W or T), 254 (e.g., S or T), and 256 (e.g., S/R/Q/E/D or T); or a modification at position 428 and/or 433 (e.g., H/L/R/S/P/Q or K) and/or 434 (e.g., H/F or Y); or a modification at position 250 and/or 428; or a modification at position 307 or 308 (e.g., 308F, V308F), and 434. In one embodiment, the modification comprises a 428L (e.g., M428L) and 434S (e.g., N434S) modification; a 428L, 259I (e.g., V259I), and 308F (e.g., V308F) modification; a 433K (e.g., H433K) and a 434 (e.g., 434Y) modification; a 252, 254, and 256 (e.g., 252Y, 254T, and 256E) modification; a 250Q and 428L modification (e.g., T250Q and M428L); and a 307 and/or 308 modification (e.g., 308F or 308P).

[0165] For example, the present invention includes anti-PSMA antibodies, and anti-CD3/anti-PSMA bispecific antigen-binding molecules, comprising an Fc domain comprising one or more pairs or groups of mutations selected from the group consisting of: 250Q and 248L (e.g., T250Q and M248L); 252Y, 254T and 256E (e.g., M252Y, S254T and T256E); 428L and 434S (e.g., M428L and N434S); and 433K and 434F (e.g., H433K and N434F). All possible combinations of the foregoing Fc domain mutations, and other mutations within the antibody variable domains disclosed herein, are contemplated within the scope of the present invention.

Biological Characteristics of the Antibodies and Bispecific Antigen-Binding Molecules

[0166] The present invention includes antibodies and antigen-binding fragments thereof that bind human PSMA with high affinity (e.g., sub-nanomolar K.sub.D values).

[0167] According to certain embodiments, the present invention includes antibodies and antigen-binding fragments of antibodies that bind human PSMA (e.g., at 37° C.) with a K.sub.D of less than about 80 nM as measured by surface plasmon resonance, e.g., using an assay format as defined in Example 3 herein. In certain embodiments, the antibodies or antigen-binding fragments of the present invention bind PSMA with a K.sub.D of less than about 5 nM, less than about 2 nM, less than about 1 nM, less than about 800 pM, less than about 600 pM, less than about 500 pM, less than about 400 pM, less than about 300 pM, less than about 200 pM, less than about 180 pM, less than about 160 pM, less than about 140 pM, less than about 120 pM, less than about 100 pM, less than about 80 pM, less than about 60 pM, less than about 40 pM, less than about 20 pM, or less than about 10 pM, as measured by surface plasmon resonance, e.g., using an assay format as defined in Example 3 herein (e.g., mAb-capture or antigen-capture format), or a substantially similar assay.

[0168] The present invention also includes antibodies and antigen-binding fragments thereof that bind PSMA with a dissociative half-life ($t_{1/2}$) of greater than about 1 minute or greater than about 10 minutes as measured by surface plasmon resonance at 37° C., e.g., using an assay format as defined in Example 3 herein, or a substantially similar assay. In certain embodiments, the antibodies or antigen-binding fragments of the present invention bind PSMA with a $t_{1/2}$ of greater than about 20 minutes, greater than about 30 minutes, greater than about 40 minutes, greater than about 50 minutes, greater than about 60 minutes, greater than about 70 minutes, greater than about 80 minutes, greater than about 90 minutes, greater than about 100 minutes, greater than about 200 minutes, greater than about 300 minutes, greater than about 400 minutes, greater than about 500 minutes, greater than about 600 minutes, greater than about 700 minutes, greater than about 800

minutes, greater than about 900 minutes, greater than about 1000 minutes, or greater than about 1100 minutes, as measured by surface plasmon resonance at 25° C. or 37° C., e.g., using an assay format as defined in Example 3 herein (e.g., mAb-capture or antigen-capture format), or a substantially similar assay. The present invention includes bispecific antigen-binding molecules (e.g., bispecific antibodies) which are capable of simultaneously binding to human CD3 and human PSMA. According to certain embodiments, the bispecific antigen-binding molecules of the invention specifically interact with cells that express CD3 and/or PSMA. The extent to which a bispecific antigen-binding molecule binds cells that express CD3 and/or PSMA can be assessed by fluorescence activated cell sorting (FACS), as illustrated in Example 5 herein. For example, the present invention includes bispecific antigen-binding molecules which specifically bind human T-cell lines which express CD3 (such cell lines do not express PSMA, e.g., Jurkat) and/or human lines which express PSMA (such cell lines do not express CD3, e.g., B16F10.9/hPSMA or 22RV1). The present invention includes bispecific antigen-binding molecules which bind any of the aforementioned cells and cell lines with an EC₅₀ value of about 80 nM, or less, as determined using a FACS assay as set forth in Example 5 or a substantially similar assay.

[0169] The present invention also includes anti-CD3/anti-PSMA bispecific antigen-binding molecules which bind to CD3-expressing human T-cells (e.g., Jurkat) and/or PSMA-expressing cells with an EC₅₀ value of between 1.0 pM and 1000 nM. In certain embodiments, the anti-CD3/anti-PSMA bispecific antigen-binding molecules bind to CD3-expressing human T-cells with an EC₅₀ value of between 1 nM and 60 nM. For example, the present invention includes anti-CD3/anti-PSMA bispecific antigen-binding molecules which bind to CD3-expressing human T-cells (e.g., Jurkat) and/or PSMA-expressing cells with an EC₅₀ value of about 1 pM, about 10 pM, about 100 pM, about 500 pM, about 1 nM, about 2 nM, about 5 nM, about 10 nM, about 20 nM, about 30 nM, about 40 nM, about 50 nM, about 60 nM, about 70 nM, about 80 nM, about 90 nM, about 100 nM, about 200 nM, about 300 nM, about 500 nM, about 800 nM, about 1000 nM, or more.

[0170] The present invention also includes anti-CD3/anti-PSMA bispecific antigen-binding molecules which exhibit one or more characteristics selected from the group consisting of: (a) inhibiting tumor growth in immunocompromised mice bearing human prostate cancer xenografts; (b) inhibiting tumor growth in immunocompetent mice bearing human prostate cancer xenografts; (c) suppressing tumor growth of established tumors in immunocompromised mice bearing human prostate cancer xenografts; and (d) reducing tumor growth of established tumors in immunocompetent mice bearing human prostate cancer xenografts (see, e.g., Example 8).

[0171] The present invention includes antibodies and antigen-binding fragments thereof that bind human CD3 with high affinity. The present invention also includes antibodies and antigen-binding fragments thereof that bind human CD3 with medium or low affinity, depending on the therapeutic context and particular targeting properties that are desired. For example, in the context of a bispecific antigen-binding molecule, wherein one arm binds CD3 and another arm binds a target antigen (e.g., PSMA), it may be desirable for the target antigen-binding arm to bind the target antigen with high affinity while the anti-CD3 arm binds CD3 with only moderate or low affinity. In this manner, preferential targeting of the antigen-binding molecule to cells expressing the target antigen may be achieved while avoiding general/untargeted CD3 binding and the consequent adverse side effects associated therewith.

[0172] The present invention includes bispecific antigen-binding molecules (e.g., bispecific antibodies) which are capable of simultaneously binding to human CD3 and a human PSMA. The binding arm that interacts with cells that express CD3 may have weak to no detectable binding as measured in a suitable in vitro binding assay. The extent to which a bispecific antigen-binding molecule binds cells that express CD3 and/or PSMA can be assessed by fluorescence activated cell sorting (FACS), as illustrated in Example 5 herein.

[0173] For example, the present invention includes antibodies, antigen-binding fragments, and

bispecific antibodies thereof which specifically bind human T-cell lines which express CD3 but do not express PSMA (e.g., Jurkat), primate T-cells (e.g., cynomolgus peripheral blood mononuclear cells [PBMCs]), and/or PSMA-expressing cells. The present invention includes bispecific antigen-binding molecules which bind any of the aforementioned T cells and T cell lines with an EC₅₀ value of from about 1.8×10^{-8} (18 nM) to about 2.1×10^{-7} (210 nM), or more (i.e. weaker affinity), or EC₅₀ is undetectable, as determined using a FACS binding assay as set forth in Example 5 or a substantially similar assay. In certain embodiments, the antibodies, antigen-binding fragments, and bispecific antibodies of the present invention bind CD3 with an EC₅₀ of greater than about 30 nM, greater than about 40 nM, greater than about 45 nM, greater than about 50 nM, greater than about 55 nM, greater than about 60 nM, greater than about 65 nM, greater than about 70 nM, greater than about 75 nM, at least 80 nM, greater than about 90 nM, greater than about 100 nM, greater than about 110 nM, at least 120 nM, greater than about 130 nM, greater than about 140 nM, greater than about 150 nM, at least 160 nM, greater than about 170 nM, greater than about 180 nM, greater than about 190 nM, greater than about 200 nM, greater than about 250 nM, greater than about 300 nM, greater than about 1 μ M, greater than about 2 μ M, or greater than about 3 μ M, or no detectable affinity, as measured by FACS binding, e.g., using an assay format as defined in Example 5 herein, or a substantially similar assay.

[0174] The present invention also includes antibodies, antigen-binding fragments, and bispecific antibodies thereof which bind to PSMA-expressing cells and cell lines, with an EC₅₀ value of less than or equal to 5.6 nM (5.6×10^{-9}), as determined using a FACS binding assay as set forth in Example 5 or a substantially similar assay.

[0175] The present invention includes antibodies, antigen-binding fragments, and bispecific antibodies thereof that bind human CD3 with weak (i.e. low) or even no detectable affinity. According to certain embodiments, the present invention includes antibodies and antigen-binding fragments of antibodies that bind human CD3 (e.g., at 37° C.) with a K_D of greater than about 11 nM as measured by surface plasmon resonance, e.g., using an assay format as defined in Example 6 herein. In certain embodiments, the antibodies or antigen-binding fragments of the present invention bind CD3 with a K_D of greater than about 15 nM, greater than about 20 nM, greater than about 25 nM, greater than about 30 nM, greater than about 35 nM, greater than about 40 nM, greater than about 45 nM, greater than about 50 nM, greater than about 55 nM, greater than about 60 nM, greater than about 65 nM, greater than about 70 nM, greater than about 75 nM, at least 80 nM, greater than about 90 nM, greater than about 100 nM, greater than about 110 nM, at least 120 nM, greater than about 130 nM, greater than about 140 nM, greater than about 150 nM, at least 160 nM, greater than about 170 nM, greater than about 180 nM, greater than about 190 nM, greater than about 200 nM, greater than about 250 nM, greater than about 300 nM, greater than about 1 μ M, greater than about 2 μ M, or greater than about 3 μ M, or no detectable affinity, as measured by surface plasmon resonance, e.g., using an assay format as defined in Example 6 herein (e.g., mAb-capture or antigen-capture format), or a substantially similar assay.

[0176] The present invention includes antibodies, antigen-binding fragments, and bispecific antibodies thereof that bind monkey (i.e. cynomolgus) CD3 with weak (i.e. low) or even no detectable affinity. According to certain embodiments, the present invention includes antibodies, antigen-binding fragments, and bispecific antibodies thereof that bind human CD3 (e.g., at 37° C.) with a K_D of greater than about 10 nM as measured by surface plasmon resonance, e.g., using an assay format as defined in Example 6 herein. In certain embodiments, the antibodies or antigen-binding fragments of the present invention bind CD3 with a K_D of greater than about 15 nM, greater than about 20 nM, greater than about 25 nM, greater than about 30 nM, greater than about 35 nM, greater than about 40 nM, greater than about 45 nM, greater than about 50 nM, greater than about 55 nM, greater than about 60 nM, greater than about 65 nM, greater than about 70 nM, greater than about 75 nM, at least 80 nM, greater than about 90 nM, greater than about 100 nM, greater than about 110 nM, at least 120 nM, greater than about 130 nM, greater than about 140 nM,

greater than about 150 nM, at least 160 nM, greater than about 170 nM, greater than about 180 nM, greater than about 190 nM, greater than about 200 nM, greater than about 250 nM, greater than about 300 nM, greater than about 1 μ M, greater than about 2 μ M, or greater than about 3 μ M, or no detectable affinity, as measured by surface plasmon resonance, e.g., using an assay format as defined in Example 6 herein (e.g., mAb-capture or antigen-capture format), or a substantially similar assay.

[0177] The present invention includes antibodies, antigen-binding fragments, and bispecific antibodies thereof that bind human CD3 and induce T cell activation. For example, the present invention includes anti-CD3 antibodies that induce human T cell activation with an EC.sub.50 value of less than about 113 pM, as measured by an in vitro T cell activation assay, e.g., using the assay format as defined in Example 7 herein [e.g., assessing the percent activated (CD69+) cells out of total T cells (CD2+) in the presence of anti-CD3 antibodies], or a substantially similar assay that assesses T cell in their activated state. In certain embodiments, the antibodies or antigen-binding fragments of the present invention induce human T cell activation [e.g., percent activated (CD69+) T cells] with an EC.sub.50 value of less than about 100 pM, less than about 50 pM, less than about 20 pM, less than about 19 pM, less than about 18 pM, less than about 17 pM, less than about 16 pM, less than about 15 pM, less than about 14 pM, less than about 13 pM, less than about 12 pM, less than about 11 pM, less than about 10 pM, less than about 9 pM, less than about 8 pM, less than about 7 pM, less than about 6 pM, less than about 5 pM, less than about 4 pM, less than about 3 pM, less than about 2 pM, or less than about 1 pM, as measured by an in vitro T cell activation assay, e.g., using the assay format as defined in Example 7 herein, or a substantially similar assay. Anti-CD3 antibodies that have weak or no detectable binding to CD3 have the ability to induce T cell activation with high potency (i.e. pM range), despite having weak or no detectable binding affinity to CD3, as exemplified in Example 7 herein.

[0178] The present invention also includes antibodies, antigen-binding fragments, and bispecific antibodies that bind human CD3 and induce T cell-mediated killing of tumor antigen-expressing cells. For example, the present invention includes anti-CD3 antibodies that induce T cell-mediated killing of tumor cells with an EC.sub.50 of less than about 1.3 nM, as measured in an in vitro T cell-mediated tumor cell killing assay, e.g., using the assay format as defined in Example 7 herein (e.g., assessing the extent of PSMA-expressing cell killing by human PBMCs in the presence of anti-CD3 antibodies), or a substantially similar assay. In certain embodiments, the antibodies or antigen-binding fragments of the present invention induce T cell-mediated tumor cell killing (e.g., PBMC-mediated killing of C4-2, 22Rv1 and TRAMPC2_PSMA cells) with an EC.sub.50 value of less than about 1 nM, less than about 400 pM, less than about 250 pM, less than about 100 pM, less than about 50 pM, less than about 40 pM, less than about 30 pM, less than about 20 pM, less than about 10 pM, less than about 9 pM, less than about 8 pM, less than about 7 pM, less than about 6 pM, less than about 5 pM, less than about 4 pM, less than about 3 pM, less than about 2 pM, or less than about 1 pM, as measured by an in vitro T cell-mediated tumor cell killing assay, e.g., using the assay format as defined in Example 7 herein, or a substantially similar assay. The present invention also includes antibodies, antigen-binding fragments, and bispecific antibodies that bind human and/or monkey (i.e. cynomolgus) CD3 with weak (i.e. low) or even no detectable affinity (i.e. do not bind or exhibit no detectable affinity) and induce T cell-mediated killing of tumor antigen-expressing cells.

[0179] The present invention also includes antibodies, antigen-binding fragments, and bispecific antibodies that bind CD3 with a dissociative half-life ($t_{1/2}$) of less than about 10 minutes as measured by surface plasmon resonance at 25° C. or 37° C., e.g., using an assay format as defined in Example 6 herein, or a substantially similar assay. In certain embodiments, the antibodies or antigen-binding fragments of the present invention bind CD3 with a $t_{1/2}$ of less than about 9 minutes, of less than about 8 minutes, of less than about 7 minutes, of less than about 6 minutes, of less than about 5 minutes, of less than about 4 minutes, of less than about 3 minutes, of less than

about 2 minutes, of less than about 1.9 minutes, or less than about 1.8 minutes, or exhibit very weak or no detectable binding as measured by surface plasmon resonance at 25° C. or 37° C., e.g., using an assay format as defined in Example 6 herein (e.g., mAb-capture or antigen-capture format), or a substantially similar assay.

[0180] The anti-CD3/anti-PSMA bispecific antigen-binding molecules of the present invention may additionally exhibit one or more characteristics selected from the group consisting of: (a) inducing PBMC proliferation in vitro; (b) activating T-cells via inducing IFN-gamma release and CD25 up-regulation in human whole blood; and (c) inducing T-cell mediated cytotoxicity on anti-PSMA-resistant cell lines.

[0181] The present invention includes anti-CD3/anti-PSMA bispecific antigen-binding molecules which are capable of depleting tumor antigen-expressing cells in a subject (see, e.g., Example 8). For example, according to certain embodiments, anti-CD3/anti-PSMA bispecific antigen-binding molecules are provided, wherein a single administration of 1 ug, or 10 ug, or 100 µg of the bispecific antigen-binding molecule to a subject (e.g., at a dose of about 0.1 mg/kg, about 0.08 mg/kg, about 0.06 mg/kg about 0.04 mg/kg, about 0.04 mg/kg, about 0.02 mg/kg, about 0.01 mg/kg, or less) causes a reduction in the number of PSMA-expressing cells in the subject (e.g., tumor growth in the subject is suppressed or inhibited) below detectable levels. In certain embodiments, a single administration of the anti-CD3/anti-PSMA bispecific antigen-binding molecule at a dose of about 0.4 mg/kg causes a reduction in tumor growth in the subject below detectable levels by about day 7, about day 6, about day 5, about day 4, about day 3, about day 2, or about day 1 after administration of the bispecific antigen-binding molecule to the subject. According to certain embodiments, a single administration of an anti-CD3/anti-PSMA bispecific antigen-binding molecule of the invention, at a dose of at least about 0.01 mg/kg, causes the number of PSMA-expressing tumor cells to remain below detectable levels until at least about 7 days, 8 days, 9 days, 10 days, 11 days, 12 days, 13 days, 14 days, 15 days, 16 days, 17 days or more, following the administration. As used herein, the expression “below detectable levels” means that no tumor cells can be directly or indirectly detected growing subcutaneously in a subject using standard caliper measurement methods, e.g., as set forth in Example 8, herein.

[0182] The present invention also includes anti-CD3/anti-PSMA bispecific antigen-binding molecules which exhibit one or more characteristics selected from the group consisting of: (a) inhibiting tumor growth in immunocompromised mice bearing human prostate cancer xenografts; (b) inhibiting tumor growth in immunocompetent mice bearing human prostate cancer xenografts; (c) suppressing tumor growth of established tumors in immunocompromised mice bearing human prostate cancer xenografts; and (d) reducing tumor growth of established tumors in immunocompetent mice bearing human prostate cancer xenografts (see, e.g., Example 8). The present invention also includes anti-CD3/anti-PSMA bispecific antigen-binding molecules which exhibit one or more characteristics selected from the group consisting of: (a) induce transient dose-dependent increases in circulating cytokines, (b) induce transient increases in circulating T cells, and (c) do not deplete effector T cell cells (e.g. CD4+ T cells, CD8+ T cells, and regulatory T cells, i.e. Tregs).

Epitope Mapping and Related Technologies

[0183] The epitope on CD3 and/or PSMA to which the antigen-binding molecules of the present invention bind may consist of a single contiguous sequence of 3 or more (e.g., 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 or more) amino acids of a CD3 or PSMA protein.

Alternatively, the epitope may consist of a plurality of non-contiguous amino acids (or amino acid sequences) of CD3 or PSMA. The antibodies of the invention may interact with amino acids contained within a single CD3 chain (e.g., CD3-epsilon, CD3-delta or CD3-gamma), or may interact with amino acids on two or more different CD3 chains. The term “epitope,” as used herein, refers to an antigenic determinant that interacts with a specific antigen binding site in the variable region of an antibody molecule known as a paratope. A single antigen may have more than one

epitope. Thus, different antibodies may bind to different areas on an antigen and may have different biological effects. Epitopes may be either conformational or linear. A conformational epitope is produced by spatially juxtaposed amino acids from different segments of the linear polypeptide chain. A linear epitope is one produced by adjacent amino acid residues in a polypeptide chain. In certain circumstances, an epitope may include moieties of saccharides, phosphoryl groups, or sulfonyl groups on the antigen.

[0184] Various techniques known to persons of ordinary skill in the art can be used to determine whether an antigen-binding domain of an antibody “interacts with one or more amino acids” within a polypeptide or protein. Exemplary techniques include, e.g., routine cross-blocking assay such as that described *Antibodies*, Harlow and Lane (Cold Spring Harbor Press, Cold Spring Harb., NY), alanine scanning mutational analysis, peptide blots analysis (Reineke, 2004, *Methods Mol Biol* 248:443-463), and peptide cleavage analysis. In addition, methods such as epitope excision, epitope extraction and chemical modification of antigens can be employed (Tomer, 2000, *Protein Science* 9:487-496). Another method that can be used to identify the amino acids within a polypeptide with which an antigen-binding domain of an antibody interacts is hydrogen/deuterium exchange detected by mass spectrometry. In general terms, the hydrogen/deuterium exchange method involves deuterium-labeling the protein of interest, followed by binding the antibody to the deuterium-labeled protein. Next, the protein/antibody complex is transferred to water to allow hydrogen-deuterium exchange to occur at all residues except for the residues protected by the antibody (which remain deuterium-labeled). After dissociation of the antibody, the target protein is subjected to protease cleavage and mass spectrometry analysis, thereby revealing the deuterium-labeled residues which correspond to the specific amino acids with which the antibody interacts. See, e.g., Ehring (1999) *Analytical Biochemistry* 267(2):252-259; Engen and Smith (2001) *Anal. Chem.* 73:256A-265A. X-ray crystallography of the antigen/antibody complex may also be used for epitope mapping purposes.

[0185] The present invention further includes anti-PSMA antibodies that bind to the same epitope as any of the specific exemplary antibodies described herein (e.g. antibodies comprising any of the amino acid sequences as set forth in Table 1 herein). Likewise, the present invention also includes anti-PSMA antibodies that compete for binding to PSMA with any of the specific exemplary antibodies described herein (e.g. antibodies comprising any of the amino acid sequences as set forth in Table 1 herein).

[0186] The present invention also includes bispecific antigen-binding molecules comprising a first antigen-binding domain that specifically binds human CD3 and/or cynomolgus CD3 with low or detectable binding affinity, and a second antigen binding domain that specifically binds human PSMA, wherein the first antigen-binding domain binds to the same epitope on CD3 as any of the specific exemplary CD3-specific antigen-binding domains described herein, and/or wherein the second antigen-binding domain binds to the same epitope on PSMA as any of the specific exemplary PSMA-specific antigen-binding domains described herein.

[0187] Likewise, the present invention also includes bispecific antigen-binding molecules comprising a first antigen-binding domain that specifically binds human CD3, and a second antigen binding domain that specifically binds human PSMA, wherein the first antigen-binding domain competes for binding to CD3 with any of the specific exemplary CD3-specific antigen-binding domains described herein, and/or wherein the second antigen-binding domain competes for binding to PSMA with any of the specific exemplary PSMA-specific antigen-binding domains described herein.

[0188] One can easily determine whether a particular antigen-binding molecule (e.g., antibody) or antigen-binding domain thereof binds to the same epitope as, or competes for binding with, a reference antigen-binding molecule of the present invention by using routine methods known in the art. For example, to determine if a test antibody binds to the same epitope on PSMA (or CD3) as a reference bispecific antigen-binding molecule of the present invention, the reference bispecific

molecule is first allowed to bind to a PSMA protein (or CD3 protein). Next, the ability of a test antibody to bind to the PSMA (or CD3) molecule is assessed. If the test antibody is able to bind to PSMA (or CD3) following saturation binding with the reference bispecific antigen-binding molecule, it can be concluded that the test antibody binds to a different epitope of PSMA (or CD3) than the reference bispecific antigen-binding molecule. On the other hand, if the test antibody is not able to bind to the PSMA (or CD3) molecule following saturation binding with the reference bispecific antigen-binding molecule, then the test antibody may bind to the same epitope of PSMA (or CD3) as the epitope bound by the reference bispecific antigen-binding molecule of the invention. Additional routine experimentation (e.g., peptide mutation and binding analyses) can then be carried out to confirm whether the observed lack of binding of the test antibody is in fact due to binding to the same epitope as the reference bispecific antigen-binding molecule or if steric blocking (or another phenomenon) is responsible for the lack of observed binding. Experiments of this sort can be performed using ELISA, RIA, Biacore™, flow cytometry or any other quantitative or qualitative antibody-binding assay available in the art. In accordance with certain embodiments of the present invention, two antigen-binding proteins bind to the same (or overlapping) epitope if, e.g., a 1-, 5-, 10-, 20- or 100-fold excess of one antigen-binding protein inhibits binding of the other by at least 50% but preferably 75%, 90% or even 99% as measured in a competitive binding assay (see, e.g., Junghans et al., Cancer Res. 1990:50:1495-1502). Alternatively, two antigen-binding proteins are deemed to bind to the same epitope if essentially all amino acid mutations in the antigen that reduce or eliminate binding of one antigen-binding protein reduce or eliminate binding of the other. Two antigen-binding proteins are deemed to have “overlapping epitopes” if only a subset of the amino acid mutations that reduce or eliminate binding of one antigen-binding protein reduce or eliminate binding of the other.

[0189] To determine if an antibody or antigen-binding domain thereof competes for binding with a reference antigen-binding molecule, the above-described binding methodology is performed in two orientations: In a first orientation, the reference antigen-binding molecule is allowed to bind to a PSMA protein (or CD3 protein) under saturating conditions followed by assessment of binding of the test antibody to the PSMA (or CD3) molecule. In a second orientation, the test antibody is allowed to bind to a PSMA (or CD3) molecule under saturating conditions followed by assessment of binding of the reference antigen-binding molecule to the PSMA (or CD3) molecule. If, in both orientations, only the first (saturating) antigen-binding molecule is capable of binding to the PSMA (or CD3) molecule, then it is concluded that the test antibody and the reference antigen-binding molecule compete for binding to PSMA (or CD3). As will be appreciated by a person of ordinary skill in the art, an antibody that competes for binding with a reference antigen-binding molecule may not necessarily bind to the same epitope as the reference antibody, but may sterically block binding of the reference antibody by binding an overlapping or adjacent epitope.

Preparation of Antigen-Binding Domains and Construction of Bispecific Molecules

[0190] Antigen-binding domains specific for particular antigens can be prepared by any antibody generating technology known in the art. Once obtained, two different antigen-binding domains, specific for two different antigens (e.g., CD3 and PSMA), can be appropriately arranged relative to one another to produce a bispecific antigen-binding molecule of the present invention using routine methods. (A discussion of exemplary bispecific antibody formats that can be used to construct the bispecific antigen-binding molecules of the present invention is provided elsewhere herein). In certain embodiments, one or more of the individual components (e.g., heavy and light chains) of the multispecific antigen-binding molecules of the invention are derived from chimeric, humanized or fully human antibodies. Methods for making such antibodies are well known in the art. For example, one or more of the heavy and/or light chains of the bispecific antigen-binding molecules of the present invention can be prepared using VELOCIMMUNE™ technology. Using VELOCIMMUNE™ technology (or any other human antibody generating technology), high affinity chimeric antibodies to a particular antigen (e.g., CD3 or PSMA) are initially isolated

having a human variable region and a mouse constant region. The antibodies are characterized and selected for desirable characteristics, including affinity, selectivity, epitope, etc. The mouse constant regions are replaced with a desired human constant region to generate fully human heavy and/or light chains that can be incorporated into the bispecific antigen-binding molecules of the present invention.

[0191] Genetically engineered animals may be used to make human bispecific antigen-binding molecules. For example, a genetically modified mouse can be used which is incapable of rearranging and expressing an endogenous mouse immunoglobulin light chain variable sequence, wherein the mouse expresses only one or two human light chain variable domains encoded by human immunoglobulin sequences operably linked to the mouse kappa constant gene at the endogenous mouse kappa locus. Such genetically modified mice can be used to produce fully human bispecific antigen-binding molecules comprising two different heavy chains that associate with an identical light chain that comprises a variable domain derived from one of two different human light chain variable region gene segments. (See, e.g., US 2011/0195454 for a detailed discussion of such engineered mice and the use thereof to produce bispecific antigen-binding molecules).

Bioequivalents

[0192] The present invention encompasses antigen-binding molecules having amino acid sequences that vary from those of the exemplary molecules disclosed herein but that retain the ability to bind CD3 and/or PSMA. Such variant molecules may comprise one or more additions, deletions, or substitutions of amino acids when compared to parent sequence, but exhibit biological activity that is essentially equivalent to that of the described bispecific antigen-binding molecules.

[0193] The present invention includes antigen-binding molecules that are bioequivalent to any of the exemplary antigen-binding molecules set forth herein. Two antigen-binding proteins, or antibodies, are considered bioequivalent if, for example, they are pharmaceutical equivalents or pharmaceutical alternatives whose rate and extent of absorption do not show a significant difference when administered at the same molar dose under similar experimental conditions, either single dose or multiple dose. Some antigen-binding proteins will be considered equivalents or pharmaceutical alternatives if they are equivalent in the extent of their absorption but not in their rate of absorption and yet may be considered bioequivalent because such differences in the rate of absorption are intentional and are reflected in the labeling, are not essential to the attainment of effective body drug concentrations on, e.g., chronic use, and are considered medically insignificant for the particular drug product studied.

[0194] In one embodiment, two antigen-binding proteins are bioequivalent if there are no clinically meaningful differences in their safety, purity, and potency.

[0195] In one embodiment, two antigen-binding proteins are bioequivalent if a patient can be switched one or more times between the reference product and the biological product without an expected increase in the risk of adverse effects, including a clinically significant change in immunogenicity, or diminished effectiveness, as compared to continued therapy without such switching.

[0196] In one embodiment, two antigen-binding proteins are bioequivalent if they both act by a common mechanism or mechanisms of action for the condition or conditions of use, to the extent that such mechanisms are known.

[0197] Bioequivalence may be demonstrated by in vivo and in vitro methods. Bioequivalence measures include, e.g., (a) an in vivo test in humans or other mammals, in which the concentration of the antibody or its metabolites is measured in blood, plasma, serum, or other biological fluid as a function of time; (b) an in vitro test that has been correlated with and is reasonably predictive of human in vivo bioavailability data; (c) an in vivo test in humans or other mammals in which the appropriate acute pharmacological effect of the antibody (or its target) is measured as a function of time; and (d) in a well-controlled clinical trial that establishes safety, efficacy, or bioavailability or

bioequivalence of an antigen-binding protein.

[0198] Bioequivalent variants of the exemplary bispecific antigen-binding molecules set forth herein may be constructed by, for example, making various substitutions of residues or sequences or deleting terminal or internal residues or sequences not needed for biological activity. For example, cysteine residues not essential for biological activity can be deleted or replaced with other amino acids to prevent formation of unnecessary or incorrect intramolecular disulfide bridges upon renaturation. In other contexts, bioequivalent antigen-binding proteins may include variants of the exemplary bispecific antigen-binding molecules set forth herein comprising amino acid changes which modify the glycosylation characteristics of the molecules, e.g., mutations which eliminate or remove glycosylation.

Species Selectivity and Species Cross-Reactivity

[0199] According to certain embodiments of the invention, antigen-binding molecules are provided which bind to human CD3 but not to CD3 from other species. Also provided are antigen-binding molecules which bind to human PSMA but not to PSMA from other species. The present invention also includes antigen-binding molecules that bind to human CD3 and to CD3 from one or more non-human species; and/or antigen-binding molecules that bind to human PSMA and to PSMA from one or more non-human species.

[0200] According to certain exemplary embodiments of the invention, antigen-binding molecules are provided which bind to human CD3 and/or human PSMA and may bind or not bind, as the case may be, to one or more of mouse, rat, guinea pig, hamster, gerbil, pig, cat, dog, rabbit, goat, sheep, cow, horse, camel, cynomolgus, marmoset, rhesus or chimpanzee CD3 and/or PSMA. For example, in a particular exemplary embodiment of the present invention bispecific antigen-binding molecules are provided comprising a first antigen-binding domain that binds human CD3 and cynomolgus CD3, and a second antigen-binding domain that specifically binds human PSMA.

Immunoconjugates

[0201] The present invention encompasses antigen-binding molecules conjugated to a therapeutic moiety ("immunoconjugate"), such as a cytotoxin, a chemotherapeutic drug, an immunosuppressant or a radioisotope. Cytotoxic agents include any agent that is detrimental to cells. Examples of suitable cytotoxic agents and chemotherapeutic agents for forming immunoconjugates are known in the art, (see for example, WO 05/103081).

Therapeutic Formulation and Administration

[0202] The present invention provides pharmaceutical compositions comprising the antigen-binding molecules of the present invention. The pharmaceutical compositions of the invention are formulated with suitable carriers, excipients, and other agents that provide improved transfer, delivery, tolerance, and the like. A multitude of appropriate formulations can be found in the formulary known to all pharmaceutical chemists: Remington's Pharmaceutical Sciences, Mack Publishing Company, Easton, PA. These formulations include, for example, powders, pastes, ointments, jellies, waxes, oils, lipids, lipid (cationic or anionic) containing vesicles (such as LIPOFECTIN™, Life Technologies, Carlsbad, CA), DNA conjugates, anhydrous absorption pastes, oil-in-water and water-in-oil emulsions, emulsions carbowax (polyethylene glycols of various molecular weights), semi-solid gels, and semi-solid mixtures containing carbowax. See also Powell et al. "Compendium of excipients for parenteral formulations" PDA (1998) J Pharm Sci Technol 52:238-311.

[0203] The dose of antigen-binding molecule administered to a patient may vary depending upon the age and the size of the patient, target disease, conditions, route of administration, and the like. The preferred dose is typically calculated according to body weight or body surface area. When a bispecific antigen-binding molecule of the present invention is used for therapeutic purposes in an adult patient, it may be advantageous to intravenously administer the bispecific antigen-binding molecule of the present invention normally at a single dose of about 0.01 to about 20 mg/kg body weight, more preferably about 0.02 to about 7, about 0.03 to about 5, or about 0.05 to about 3

mg/kg body weight. Depending on the severity of the condition, the frequency and the duration of the treatment can be adjusted. Effective dosages and schedules for administering a bispecific antigen-binding molecule may be determined empirically; for example, patient progress can be monitored by periodic assessment, and the dose adjusted accordingly. Moreover, interspecies scaling of dosages can be performed using well-known methods in the art (e.g., Mordenti et al., 1991, *Pharmaceut. Res.* 8:1351).

[0204] Various delivery systems are known and can be used to administer the pharmaceutical composition of the invention, e.g., encapsulation in liposomes, microparticles, microcapsules, recombinant cells capable of expressing the mutant viruses, receptor mediated endocytosis (see, e.g., Wu et al., 1987, *J. Biol. Chem.* 262:4429-4432). Methods of introduction include, but are not limited to, intradermal, intramuscular, intraperitoneal, intravenous, subcutaneous, intranasal, epidural, and oral routes. The composition may be administered by any convenient route, for example by infusion or bolus injection, by absorption through epithelial or mucocutaneous linings (e.g., oral mucosa, rectal and intestinal mucosa, etc.) and may be administered together with other biologically active agents. Administration can be systemic or local.

[0205] A pharmaceutical composition of the present invention can be delivered subcutaneously or intravenously with a standard needle and syringe. In addition, with respect to subcutaneous delivery, a pen delivery device readily has applications in delivering a pharmaceutical composition of the present invention. Such a pen delivery device can be reusable or disposable. A reusable pen delivery device generally utilizes a replaceable cartridge that contains a pharmaceutical composition. Once all of the pharmaceutical composition within the cartridge has been administered and the cartridge is empty, the empty cartridge can readily be discarded and replaced with a new cartridge that contains the pharmaceutical composition. The pen delivery device can then be reused. In a disposable pen delivery device, there is no replaceable cartridge. Rather, the disposable pen delivery device comes prefilled with the pharmaceutical composition held in a reservoir within the device. Once the reservoir is emptied of the pharmaceutical composition, the entire device is discarded.

[0206] Numerous reusable pen and autoinjector delivery devices have applications in the subcutaneous delivery of a pharmaceutical composition of the present invention. Examples include, but are not limited to AUTOPEN™ (Owen Mumford, Inc., Woodstock, UK), DISETRONIC™ pen (Disetronic Medical Systems, Bergdorf, Switzerland), HUMALOG MIX 75/25™ pen, HUMALOG™ pen, HUMALIN 70/30™ pen (Eli Lilly and Co., Indianapolis, IN), NOVOPEN™ I, II and III (Novo Nordisk, Copenhagen, Denmark), NOVOPEN JUNIOR™ (Novo Nordisk, Copenhagen, Denmark), BD™ pen (Becton Dickinson, Franklin Lakes, NJ), OPTIPEN™, OPTIPEN PRO™, OPTIPEN STARLET™, and OPTICLIK™ (sanofi-aventis, Frankfurt, Germany), to name only a few. Examples of disposable pen delivery devices having applications in subcutaneous delivery of a pharmaceutical composition of the present invention include, but are not limited to the SOLOSTAR™ pen (sanofi-aventis), the FLEXPEN™ (Novo Nordisk), and the KWIKPEN™ (Eli Lilly), the SURECLICK™ Autoinjector (Amgen, Thousand Oaks, CA), the PENLET™ (Haselmeier, Stuttgart, Germany), the EPIPEN (Dey, L.P.), and the HUMIRA™ Pen (Abbott Labs, Abbott Park Ill.), to name only a few.

[0207] In certain situations, the pharmaceutical composition can be delivered in a controlled release system. In one embodiment, a pump may be used (see Langer, supra; Sefton, 1987, *CRC Crit. Rev. Biomed. Eng.* 14:201). In another embodiment, polymeric materials can be used; see, *Medical Applications of Controlled Release*, Langer and Wise (eds.), 1974, CRC Pres., Boca Raton, Florida. In yet another embodiment, a controlled release system can be placed in proximity of the composition's target, thus requiring only a fraction of the systemic dose (see, e.g., Goodson, 1984, in *Medical Applications of Controlled Release*, supra, vol. 2, pp. 115-138). Other controlled release systems are discussed in the review by Langer, 1990, *Science* 249:1527-1533.

[0208] The injectable preparations may include dosage forms for intravenous, subcutaneous,

intracutaneous and intramuscular injections, drip infusions, etc. These injectable preparations may be prepared by methods publicly known. For example, the injectable preparations may be prepared, e.g., by dissolving, suspending or emulsifying the antibody or its salt described above in a sterile aqueous medium or an oily medium conventionally used for injections. As the aqueous medium for injections, there are, for example, physiological saline, an isotonic solution containing glucose and other auxiliary agents, etc., which may be used in combination with an appropriate solubilizing agent such as an alcohol (e.g., ethanol), a polyalcohol (e.g., propylene glycol, polyethylene glycol), a nonionic surfactant [e.g., polysorbate 80, HCO-50 (polyoxyethylene (50 mol) adduct of hydrogenated castor oil)], etc. As the oily medium, there are employed, e.g., sesame oil, soybean oil, etc., which may be used in combination with a solubilizing agent such as benzyl benzoate, benzyl alcohol, etc. The injection thus prepared is preferably filled in an appropriate ampoule. [0209] Advantageously, the pharmaceutical compositions for oral or parenteral use described above are prepared into dosage forms in a unit dose suited to fit a dose of the active ingredients. Such dosage forms in a unit dose include, for example, tablets, pills, capsules, injections (ampoules), suppositories, etc. The amount of the aforesaid antibody contained is generally about 5 to about 500 mg per dosage form in a unit dose; especially in the form of injection, it is preferred that the aforesaid antibody is contained in about 5 to about 100 mg and in about 10 to about 250 mg for the other dosage forms.

Therapeutic Uses of the Antigen-Binding Molecules

[0210] The present invention includes methods comprising administering to a subject in need thereof a therapeutic composition comprising an anti-PSMA antibody or antigen-binding fragment thereof, or a bispecific antigen-binding molecule that specifically binds CD3 and PSMA. The therapeutic composition can comprise any of the antibodies or bispecific antigen-binding molecules as disclosed herein and a pharmaceutically acceptable carrier or diluent. As used herein, the expression “a subject in need thereof” means a human or non-human animal that exhibits one or more symptoms or indicia of cancer (e.g., a subject expressing a tumor or suffering from any of the cancers mentioned herein below), or who otherwise would benefit from an inhibition or reduction in PSMA activity or a depletion of PSMA+ cells (e.g., prostate cancer cells).

[0211] The antibodies and bispecific antigen-binding molecules of the invention (and therapeutic compositions comprising the same) are useful, inter alia, for treating any disease or disorder in which stimulation, activation and/or targeting of an immune response would be beneficial. In particular, the anti-PSMA antibodies or the anti-CD3/anti-PSMA bispecific antigen-binding molecules of the present invention may be used for the treatment, prevention and/or amelioration of any disease or disorder associated with or mediated by PSMA expression or activity or the proliferation of PSMA+ cells. The mechanism of action by which the therapeutic methods of the invention are achieved include killing of the cells expressing PSMA in the presence of effector cells, for example, by CDC, apoptosis, ADCC, phagocytosis, or by a combination of two or more of these mechanisms. Cells expressing PSMA which can be inhibited or killed using the bispecific antigen-binding molecules of the invention include, for example, prostate tumor cells.

[0212] The antigen-binding molecules of the present invention may be used to treat, e.g., primary and/or metastatic tumors arising in the gastrointestinal tract, prostate, kidney, and/or bladder. In certain embodiments, the bispecific antigen-binding molecules of the invention are used to treat one or more of the following cancers: clear cell renal cell carcinoma, chromophobe renal cell carcinoma, (renal) oncocytoma, (renal) transitional cell carcinoma, prostate cancer, colorectal cancer, gastric cancer, urothelial carcinoma, (bladder) adenocarcinoma, or (bladder) small cell carcinoma. According to certain embodiments of the present invention, the anti-PSMA antibodies or anti-PSMA/anti-CD3 bispecific antibodies are useful for treating a patient afflicted with a castrate-resistant prostate cancer. According to other related embodiments of the invention, methods are provided comprising administering an anti-PSMA antibody or an anti-CD3/anti-PSMA bispecific antigen-binding molecule as disclosed herein to a patient who is afflicted with a castrate-

resistant prostate cancer. Analytic/diagnostic methods known in the art, such as tumor scanning, etc., may be used to ascertain whether a patient harbors a tumor that is castrate-resistant.

[0213] The present invention also includes methods for treating residual cancer in a subject. As used herein, the term “residual cancer” means the existence or persistence of one or more cancerous cells in a subject following treatment with an anti-cancer therapy.

[0214] According to certain aspects, the present invention provides methods for treating a disease or disorder associated with PSMA expression (e.g., prostate cancer) comprising administering one or more of the anti-PSMA or bispecific antigen-binding molecules described elsewhere herein to a subject after the subject has been determined to have prostate cancer (e.g., castrate-resistant prostate cancer). For example, the present invention includes methods for treating prostate cancer comprising administering an anti-PSMA antibody or an anti-CD3/anti-PSMA bispecific antigen-binding molecule to a patient 1 day, 2 days, 3 days, 4 days, 5 days, 6 days, 1 week, 2 weeks, 3 weeks or 4 weeks, 2 months, 4 months, 6 months, 8 months, 1 year, or more after the subject has received hormone therapy (e.g., anti-androgen therapy).

Combination Therapies and Formulations

[0215] The present invention provides methods which comprise administering a pharmaceutical composition comprising any of the exemplary antibodies and bispecific antigen-binding molecules described herein in combination with one or more additional therapeutic agents. Exemplary additional therapeutic agents that may be combined with or administered in combination with an antigen-binding molecule of the present invention include, e.g., an EGFR antagonist (e.g., an anti-EGFR antibody [e.g., cetuximab or panitumumab] or small molecule inhibitor of EGFR [e.g., gefitinib or erlotinib]), an antagonist of another EGFR family member such as Her2/ErbB2, ErbB3 or ErbB4 (e.g., anti-ErbB2, anti-ErbB3 or anti-ErbB4 antibody or small molecule inhibitor of ErbB2, ErbB3 or ErbB4 activity), an antagonist of EGFRvIII (e.g., an antibody that specifically binds EGFRvIII), a cMET antagonist (e.g., an anti-cMET antibody), an IGF1R antagonist (e.g., an anti-IGF1R antibody), a B-raf inhibitor (e.g., vemurafenib, sorafenib, GDC-0879, PLX-4720), a PDGFR- α inhibitor (e.g., an anti-PDGFR- α antibody), a PDGFR- β inhibitor (e.g., an anti-PDGFR- β antibody), a VEGF antagonist (e.g., a VEGF-Trap, see, e.g., U.S. Pat. No. 7,087,411 (also referred to herein as a “VEGF-inhibiting fusion protein”), anti-VEGF antibody (e.g., bevacizumab), a small molecule kinase inhibitor of VEGF receptor (e.g., sunitinib, sorafenib or pazopanib)), a DLL4 antagonist (e.g., an anti-DLL4 antibody disclosed in US 2009/0142354 such as REGN421), an Ang2 antagonist (e.g., an anti-Ang2 antibody disclosed in US 2011/0027286 such as H1H685P), a FOLH1 (PSMA) antagonist, a PRLR antagonist (e.g., an anti-PRLR antibody), a STEAP1 or STEAP2 antagonist (e.g., an anti-STEAP1 antibody or an anti-STEAP2 antibody), a TMPRSS2 antagonist (e.g., an anti-TMPRSS2 antibody), a MSLN antagonist (e.g., an anti-MSLN antibody), a CA9 antagonist (e.g., an anti-CA9 antibody), a uroplakin antagonist (e.g., an anti-uroplakin antibody), etc. Other agents that may be beneficially administered in combination with the antigen-binding molecules of the invention include cytokine inhibitors, including small-molecule cytokine inhibitors and antibodies that bind to cytokines such as IL-1, IL-2, IL-3, IL-4, IL-5, IL-6, IL-8, IL-9, IL-11, IL-12, IL-13, IL-17, IL-18, or to their respective receptors. The pharmaceutical compositions of the present invention (e.g., pharmaceutical compositions comprising an anti-CD3/anti-PSMA bispecific antigen-binding molecule as disclosed herein) may also be administered as part of a therapeutic regimen comprising one or more therapeutic combinations selected from “ICE”: ifosfamide (e.g., Ifex®), carboplatin (e.g., Paraplatin®), etoposide (e.g., Etopophos®, Toposar®, VePesid®, VP-16); “DHAP”: dexamethasone (e.g., Decadron®), cytarabine (e.g., Cytosar-U®, cytosine arabinoside, ara-C), cisplatin (e.g., Platinol®-AQ); and “ESHAP”: etoposide (e.g., Etopophos®, Toposar®, VePesid®, VP-16), methylprednisolone (e.g., Medrol®), high-dose cytarabine, cisplatin (e.g., Platinol®-AQ).

[0216] The present invention also includes therapeutic combinations comprising any of the antigen-binding molecules mentioned herein and an inhibitor of one or more of VEGF, Ang2, DLL4,

EGFR, ErbB2, ErbB3, ErbB4, EGFRvIII, cMet, IGF1R, B-raf, PDGFR- α , PDGFR- β , FOLH1 (PSMA), PRLR, STEAP1, STEAP2, Tmprss2, MSLN, CA9, uroplakin, or any of the aforementioned cytokines, wherein the inhibitor is an aptamer, an antisense molecule, a ribozyme, an siRNA, a peptibody, a nanobody or an antibody fragment (e.g., Fab fragment; F(ab')₂ fragment; Fd fragment; Fv fragment; scFv; dAb fragment; or other engineered molecules, such as diabodies, triabodies, tetrabodies, minibodies and minimal recognition units). The antigen-binding molecules of the invention may also be administered and/or co-formulated in combination with antivirals, antibiotics, analgesics, corticosteroids and/or NSAIDs. The antigen-binding molecules of the invention may also be administered as part of a treatment regimen that also includes radiation treatment and/or conventional chemotherapy.

[0217] The additional therapeutically active component(s) may be administered just prior to, concurrent with, or shortly after the administration of an antigen-binding molecule of the present invention; (for purposes of the present disclosure, such administration regimens are considered the administration of an antigen-binding molecule “in combination with” an additional therapeutically active component).

[0218] The present invention includes pharmaceutical compositions in which an antigen-binding molecule of the present invention is co-formulated with one or more of the additional therapeutically active component(s) as described elsewhere herein.

Administration Regimens

[0219] According to certain embodiments of the present invention, multiple doses of an antigen-binding molecule (e.g., an anti-PSMA antibody or a bispecific antigen-binding molecule that specifically binds PSMA and CD3) may be administered to a subject over a defined time course. The methods according to this aspect of the invention comprise sequentially administering to a subject multiple doses of an antigen-binding molecule of the invention. As used herein, “sequentially administering” means that each dose of an antigen-binding molecule is administered to the subject at a different point in time, e.g., on different days separated by a predetermined interval (e.g., hours, days, weeks or months). The present invention includes methods which comprise sequentially administering to the patient a single initial dose of an antigen-binding molecule, followed by one or more secondary doses of the antigen-binding molecule, and optionally followed by one or more tertiary doses of the antigen-binding molecule.

[0220] The terms “initial dose,” “secondary doses,” and “tertiary doses,” refer to the temporal sequence of administration of the antigen-binding molecule of the invention. Thus, the “initial dose” is the dose which is administered at the beginning of the treatment regimen (also referred to as the “baseline dose”); the “secondary doses” are the doses which are administered after the initial dose; and the “tertiary doses” are the doses which are administered after the secondary doses. The initial, secondary, and tertiary doses may all contain the same amount of the antigen-binding molecule, but generally may differ from one another in terms of frequency of administration. In certain embodiments, however, the amount of an antigen-binding molecule contained in the initial, secondary and/or tertiary doses varies from one another (e.g., adjusted up or down as appropriate) during the course of treatment. In certain embodiments, two or more (e.g., 2, 3, 4, or 5) doses are administered at the beginning of the treatment regimen as “loading doses” followed by subsequent doses that are administered on a less frequent basis (e.g., “maintenance doses”).

[0221] In one exemplary embodiment of the present invention, each secondary and/or tertiary dose is administered 1 to 26 (e.g., 1, 1½, 2, 2½, 3, 3½, 4, 4½, 5, 5½, 6, 6½, 7, 7½, 8, 8½, 9, 9½, 10, 10½, 11, 11½, 12, 12½, 13, 13½, 14, 14½, 15, 15½, 16, 16½, 17, 17½, 18, 18½, 19, 19½, 20, 20½, 21, 21½, 22, 22½, 23, 23½, 24, 24½, 25, 25½, 26, 26½, or more) weeks after the immediately preceding dose. The phrase “the immediately preceding dose,” as used herein, means, in a sequence of multiple administrations, the dose of antigen-binding molecule which is administered to a patient prior to the administration of the very next dose in the sequence with no intervening doses.

[0222] The methods according to this aspect of the invention may comprise administering to a

patient any number of secondary and/or tertiary doses of an antigen-binding molecule (e.g., an anti-PSMA antibody or a bispecific antigen-binding molecule that specifically binds PSMA and CD3). For example, in certain embodiments, only a single secondary dose is administered to the patient. In other embodiments, two or more (e.g., 2, 3, 4, 5, 6, 7, 8, or more) secondary doses are administered to the patient. Likewise, in certain embodiments, only a single tertiary dose is administered to the patient. In other embodiments, two or more (e.g., 2, 3, 4, 5, 6, 7, 8, or more) tertiary doses are administered to the patient.

[0223] In embodiments involving multiple secondary doses, each secondary dose may be administered at the same frequency as the other secondary doses. For example, each secondary dose may be administered to the patient 1 to 2 weeks after the immediately preceding dose. Similarly, in embodiments involving multiple tertiary doses, each tertiary dose may be administered at the same frequency as the other tertiary doses. For example, each tertiary dose may be administered to the patient 2 to 4 weeks after the immediately preceding dose. Alternatively, the frequency at which the secondary and/or tertiary doses are administered to a patient can vary over the course of the treatment regimen. The frequency of administration may also be adjusted during the course of treatment by a physician depending on the needs of the individual patient following clinical examination.

Diagnostic Uses of the Antibodies

[0224] The anti-PSMA antibodies of the present invention may also be used to detect and/or measure PSMA, or PSMA-expressing cells in a sample, e.g., for diagnostic purposes. For example, an anti-PSMA antibody, or fragment thereof, may be used to diagnose a condition or disease characterized by aberrant expression (e.g., over-expression, under-expression, lack of expression, etc.) of PSMA. Exemplary diagnostic assays for PSMA may comprise, e.g., contacting a sample, obtained from a patient, with an anti-PSMA antibody of the invention, wherein the anti-PSMA antibody is labeled with a detectable label or reporter molecule. Alternatively, an unlabeled anti-PSMA antibody can be used in diagnostic applications in combination with a secondary antibody which is itself detectably labeled. The detectable label or reporter molecule can be a radioisotope, such as ^3H , ^{14}C , ^{32}P , ^{35}S , or ^{125}I ; a fluorescent or chemiluminescent moiety such as fluorescein isothiocyanate, or rhodamine; or an enzyme such as alkaline phosphatase, beta-galactosidase, horseradish peroxidase, or luciferase. Another exemplary diagnostic use of the anti-PSMA antibodies of the invention includes ^{89}Zr -labeled, such as ^{89}Zr -desferrioxamine-labeled, antibody for the purpose of noninvasive identification and tracking of tumor cells in a subject (e.g. positron emission tomography (PET) imaging). (See, e.g., Tavaré, R. et al. Cancer Res. 2016 Jan. 1; 76(1):73-82; and Azad, B B. et al. Oncotarget. 2016 Mar. 15; 7(11):12344-58.) Specific exemplary assays that can be used to detect or measure PSMA in a sample include enzyme-linked immunosorbent assay (ELISA), radioimmunoassay (RIA), and fluorescence-activated cell sorting (FACS).

[0225] Samples that can be used in PSMA diagnostic assays according to the present invention include any tissue or fluid sample obtainable from a patient which contains detectable quantities of PSMA protein, or fragments thereof, under normal or pathological conditions. Generally, levels of PSMA in a particular sample obtained from a healthy patient (e.g., a patient not afflicted with a disease or condition associated with abnormal PSMA levels or activity) will be measured to initially establish a baseline, or standard, level of PSMA. This baseline level of PSMA can then be compared against the levels of PSMA measured in samples obtained from individuals suspected of having a PSMA related disease (e.g., a tumor containing PSMA-expressing cells) or condition.

EXAMPLES

[0226] The following examples are put forth so as to provide those of ordinary skill in the art with a complete disclosure and description of how to make and use the methods and compositions of the invention, and are not intended to limit the scope of what the inventors regard as their invention. Efforts have been made to ensure accuracy with respect to numbers used (e.g., amounts,

temperature, etc.) but some experimental errors and deviations should be accounted for. Unless indicated otherwise, parts are parts by weight, molecular weight is average molecular weight, temperature is in degrees Centigrade, and pressure is at or near atmospheric.

Example 1: Generation of Anti-PSMA Antibodies

[0227] Anti-PSMA antibodies were obtained by immunizing a genetically modified mouse with a human PSMA antigen or by immunizing an engineered mouse comprising DNA encoding human immunoglobulin heavy and kappa light chain variable regions with a human PSMA antigen.

[0228] Mice were immunized with human prostate cancer cells (LNCaP, ATTC®, Manassas, Virginia, USA) expressing human PSMA (SEQ ID NO:1651; UniProtKB/Swiss-Prot. No. Q04609). Following immunization, splenocytes were harvested from each mouse and either (1) fused with mouse myeloma cells to preserve their viability and form hybridoma cells and screened for PSMA specificity, or (2) B-cell sorted (as described in US 2007/0280945A1) using a human PSMA with an N-terminal 6-His tag (R&D, Cat #4234-ZN) as the sorting reagent that binds and identifies reactive antibodies (antigen-positive B cells).

[0229] Chimeric antibodies to PSMA were initially isolated having a human variable region and a mouse constant region. The antibodies were characterized and selected for desirable characteristics, including affinity, selectivity, etc. If necessary, mouse constant regions were replaced with a desired human constant region, for example wild-type or modified IgG1 or IgG4, to generate a fully human anti-PSMA antibody. While the constant region selected may vary according to specific use, high affinity antigen-binding and target specificity characteristics reside in the variable region. The antibody name designations such as H1H11453N2 and H1M11900N denote fully human antibodies “H1H” or chimeric human variable/mouse constant region antibodies “H1M”. Antibodies identified by the hybridoma method are indicated with antibody ID numbers ending with “N” or “N2”; Antibodies identified by the B-cell sorting method are indicated with antibody ID numbers ending with “P” or “P2”.

[0230] Certain biological properties of the exemplary anti-PSMA antibodies generated in accordance with the methods of this Example are described in detail in the Examples set forth below.

Example 2: Heavy and Light Chain Variable Region Amino Acid and Nucleic Acid Sequences of Anti-PSMA Antibodies

[0231] Table 1 sets forth the amino acid sequence identifiers of the heavy and light chain variable regions and CDRs of selected anti-PSMA antibodies of the invention. The corresponding nucleic acid sequence identifiers are set forth in Table 2.

TABLE-US-00001 TABLE 1 Amino Acid Sequence Identifiers Antibody SEQ ID NOs:																																																																																																																																																																	
Designation	HCVR	HCDR1	HCDR2	HCDR3	LCVR	LCDR1	LCDR2	LCDR3	H1H11453N2	2	4	6	8	1642	1644	1646	1648	H1H11792P2	10	12	14	16	1642	1644	1646	1648	H1H11797P2	18	20	22	24	1642	1644	1646	1648	H1H11800P2	26	28	30	32	1642	1644	1646	1648	H1H11803P2	34	36	38	40	1642	1644	1646	1648	H1H11804P2	42	44	46	48	1642	1644	1646	1648	H1H11805P2	50	52	54	56	1642	1644	1646	1648	H1H11808P2	58	60	62	64	1642	1644	1646	1648	H1H11810P2	66	68	70	72	1642	1644	1646	1648	H1H11835P2	74	76	78	80	1642	1644	1646	1648	H1H11836P2	82	84	86	88	1642	1644	1646	1648	H1H11837P2	90	92	94	96	1642	1644	1646	1648	H1H11838P2	98	100	102	104	1642	1644	1646	1648	H1H11841P2	106	108	110	112	1642	1644	1646	1648	H1H11899N2	114	116	118	120	1642	1644	1646	1648	H1H3465P	122	124	126	128	130	132	134	136	H1M11900N	138	140	142	144	146	148	150	152

TABLE-US-00002 TABLE 2 Nucleic Acid Sequence Identifiers Antibody SEQ ID NOs:																																																																																					
Designation	HCVR	HCDR1	HCDR2	HCDR3	LCVR	LCDR1	LCDR2	LCDR3	H1H11453N2	1	3	5	7	1641	1643	1645	1647	H1H11792P2	9	11	13	15	1641	1643	1645	1647	H1H11797P2	17	19	21	23	1641	1643	1645	1647	H1H11800P2	25	27	29	31	1641	1643	1645	1647	H1H11803P2	33	35	37	39	1641	1643	1645	1647	H1H11804P2	41	43	45	47	1641	1643	1645	1647	H1H11805P2	49	51	53	55	1641	1643	1645	1647	H1H11808P2	57	59	61	63	1641	1643	1645	1647	H1H11810P2	65	67	69	71

1641 1643 1645 1647 H1H11835P2 73 75 77 79 1641 1643 1645 1647 H1H11836P2 81 83 85 87
1641 1643 1645 1647 H1H11837P2 89 91 93 95 1641 1643 1645 1647 H1H11838P2 97 99 101
103 1641 1643 1645 1647 H1H11841P2 105 107 109 111 1641 1643 1645 1647 H1H11899N2 113
115 117 119 1641 1643 1645 1647 H1H3465P 121 123 125 127 129 131 133 135 H1M11900N 137
139 141 143 145 147 149 151

Example 3: Surface Plasmon Resonance Derived Binding Affinities and Kinetic Constants of Human Monoclonal Anti-PSMA Antibodies

[0232] In this example, anti-PSMA antibodies were assessed for their ability to bind to human PSMA. Binding affinities and kinetic constants of anti-PSMA antibodies to soluble human PSMA protein were determined by surface plasmon resonance at 37° C. using an antibody-capture format. Results are shown in Tables 3 and 4. Measurements were conducted on a Biacore™ T-200 instrument (GE Healthcare).

[0233] Briefly, a CM5 Biacore™ sensor surface was derivatized via amine coupling with a monoclonal mouse anti-human Fc antibody (GE, #BR-1008-39) or monoclonal goat anti-mouse Fc antibody (GE, #BR-1008-38) to capture purified anti-PSMA antibodies. Binding studies were performed in HBSP++ buffer composed of 0.01M HEPES, 0.15M NaCl, 2 mM Ca.sup.2+, 2 mM Mg.sup.2+, 0.05% v/v Surfactant P20, pH7.4. Varying concentrations of human PSMA expressed with an N-terminal hexahistidine tag (6h.hPSMA, R&D) prepared in HBSP++ running buffer (ranging from 50 to 0.78 nM, 4-fold dilutions) were injected over the anti-PSMA antibody captured surface at a flow rate of 30 µL/minute. Antibody-reagent association was monitored for 2 minutes while dissociation in HBSP++ running buffer was monitored for 8 minutes.

[0234] Kinetic association (k.sub.a) and dissociation (k.sub.d) rate constants were determined by fitting the real-time sensorgrams to a 1:1 binding model using Scrubber 2.0c curve fitting software. Binding dissociation equilibrium constants (K.sub.D) and dissociative half-lives (t_{1/2}) were calculated from the kinetic rate constants as: K.sub.D (M)=k.sub.d/K.sub.a; and t.sub.1/2 (min)=(ln 2/(60*k.sub.d)).

[0235] As shown in Tables 3 and 4, the anti-PSMA antibodies of the invention bound to human PSMA in the antibody capture format with varying affinities and K.sub.D values ranging from 19.9 pM to 75.6 nM. Several exemplary anti-PSMA antibodies, such as H1H3465P and H1H11810P2, displayed strong affinity to human PSMA protein, with sub-nanomolar K.sub.D values.

TABLE-US-00003 TABLE 3 Affinities of anti-PSMA human IgG1 antibodies to soluble human PSMA at 37° C. Antibody ID ka (1/Ms) kd (1/s) KD (M) t.sub.1/2 (min)

H1H3465P	2.67E+05	6.06E-05	2.27E-10	190.6
H1H11792P2	4.73E+05	9.70E-05	2.05E-10	119.1
H1H11797P2	1.68E+05	6.30E-04	3.74E-09	18.3
H1H11800P2	2.51E+05	4.10E-05	1.63E-10	281.8
H1H11803P2	4.57E+05	6.08E-04	1.33E-09	19
H1H11804P2	2.03E+04	8.01E-04	3.94E-08	14.4
H1H11805P2	1.29E+05	9.74E-03	7.56E-08	1.2
H1H11808P2	1.78E+05	≤1E-05	≤5.63E-11	≤1.155
H1H11810P2	5.03E+05	≤1E-05	≤1.99E-11	≤1.155
H1H11835P2	1.75E+05	2.46E-03	1.40E-08	4.7
H1H11836P2	3.27E+05	2.80E-04	8.55E-10	41.3
H1H11837P2	4.41E+05	7.34E-04	1.66E-09	15.7
H1H11838P2	2.37E+05	4.71E-04	1.99E-09	24.5
H1H11841P2	IC	IC	IC	IC

IC: inconclusive

TABLE-US-00004 TABLE 4 Affinities of anti-PSMA mouse constant antibodies to soluble human PSMA at 37° C. Antibody ID ka (1/Ms) kd (1/s) KD (M) t.sub.1/2 (min)

H2M11899N	1.51E+05	2.65E-04	1.76E-09	43.6
H1M11453N	1.85E+05	3.56E-04	1.93E-09	32.4
H1M11900N	1.57E+05	4.06E-03	2.58E-08	2.8

Example 4: Generation of Bispecific Antibodies that Bind Prostate-Specific Membrane Antigen (PSMA) and CD3

[0236] The present invention provides bispecific antigen-binding molecules that bind CD3 and Prostate-Specific Membrane Antigen (PSMA); such bispecific antigen-binding molecules are also referred to herein as “anti-PSMA/anti-CD3 bispecific molecules.” The anti-PSMA portion of the anti-PSMA/anti-CD3 bispecific molecule is useful for targeting tumor cells that express PSMA,

and the anti-CD3 portion of the bispecific molecule is useful for activating T-cells. The simultaneous binding of PSMA on a tumor cell and CD3 on a T-cell facilitates directed killing (cell lysis) of the targeted tumor cell by the activated T-cell.

[0237] Bispecific antibodies comprising an anti-PSMA-specific binding domain and an anti-CD3-specific binding domain were constructed using standard methodologies, wherein the anti-PSMA antigen binding domain and the anti-CD3 antigen binding domain each comprise different, distinct HCVRs paired with a common LCVR. In some instances the bispecific antibodies were constructed utilizing a heavy chain from an anti-CD3 antibody, a heavy chain from an anti-PSMA antibody and a common light chain. In other instances, the bispecific antibodies were constructed utilizing a heavy chain from an anti-CD3 antibody, a heavy chain from an anti-PSMA antibody and a light chain from an anti-CD3 antibody.

[0238] The bispecific antibodies described in the following examples consist of binding arms known to bind to human soluble heterodimeric hCD3 ϵ/δ protein (as described in Examples 9-13 herein); and human PSMA (see Examples 1-3 above). Exemplified bispecific antibodies were manufactured having a modified (chimeric) IgG4 Fc domain as set forth in US Patent Application Publication No. US20140243504A1, published on Aug. 28, 2014.

[0239] A summary of the component parts of the antigen-binding domains of the various anti-PSMAxCD3 bispecific antibodies constructed is set forth in Table 5.

TABLE-US-00005 TABLE 5 Summary of Component Parts of PSMAxCD3 Bispecific Antibodies									
Anti-PSMA	Anti-CD3	Antigen-Binding	Antigen-Binding	Common Domain	Domain	Light Chain			
Bispecific	Heavy Chain	Heavy Chain	Variable	Antibody Identifier	Variable Region	Variable Region			
BSPSMA/CD3-001	PSMA-VH-3465	CD3-VH-A	CD3-VL-A	BSPSMA/CD3-002	PSMA-VH-3465	CD3-VH-B	CD3-VL-B	BSPSMA/CD3-003	PSMA-VH-11810
CD3-VH-G	VK 1-39 JK 5 (SEQ ID NO: 1386)	BSPSMA/CD3-200	PSMA-VH-11810	CD3-VH-G2	VK 1-39 JK 5 (SEQ ID NO: 1386)	BSPSMA/CD3-300	PSMA-VH-11810	CD3-VH-G3	VK 1-39 JK 5 (SEQ ID NO: 1386)
BSPSMA/CD3-400	PSMA-VH-11810	CD3-VH-G4	VK 1-39 JK 5 (SEQ ID NO: 1386)	BSPSMA/CD3-004	PSMA-VH-11810	CD3-VH-G5	VK 1-39 JK 5 (SEQ ID NO: 1386)	BSPSMA/CD3-800	PSMA-VH-11810
CD3-VH-G8	VK 1-39 JK 5 (SEQ ID NO: 1386)	BSPSMA/CD3-900	PSMA-VH-11810	CD3-VH-G9	VK 1-39 JK 5 (SEQ ID NO: 1386)	BSPSMA/CD3- PSMA-VH-11810	CD3-VH-G10	VK 1-39 JK 5 1000 (SEQ ID NO: 1386)	BSPSMA/CD3- PSMA-VH-11810
CD3-VH-G11	VK 1-39 JK 5 1100 (SEQ ID NO: 1386)	BSPSMA/CD3- PSMA-VH-11810	CD3-VH-G12	VK 1-39 JK 5 1200 (SEQ ID NO: 1386)	BSPSMA/CD3- PSMA-VH-11810	CD3-VH-G13	VK 1-39 JK 5 1300 (SEQ ID NO: 1386)	BSPSMA/CD3- PSMA-VH-11810	CD3-VH-G14
VK 1-39 JK 5 1400 (SEQ ID NO: 1386)	BSPSMA/CD3- PSMA-VH-11810	CD3-VH-G15	VK 1-39 JK 5 1500 (SEQ ID NO: 1386)	BSPSMA/CD3- PSMA-VH-11810	CD3-VH-G16	VK 1-39 JK 5 1600 (SEQ ID NO: 1386)	BSPSMA/CD3- PSMA-VH-11810	CD3-VH-G17	VK 1-39 JK 5 1700 (SEQ ID NO: 1386)
BSPSMA/CD3- PSMA-VH-11810	CD3-VH-G18	VK 1-39 JK 5 1800 (SEQ ID NO: 1386)	BSPSMA/CD3- PSMA-VH-11810	CD3-VH-G19	VK 1-39 JK 5 1900 (SEQ ID NO: 1386)	BSPSMA/CD3-005	PSMA-VH-11810	CD3-VH-G20	VK 1-39 JK 5 (SEQ ID NO: 1386)
BSPSMA/CD3- PSMA-VH-11810	CD3-VH-G21	VK 1-39 JK 5 2100 (SEQ ID NO: 1386)							

[0240] The anti-PSMA heavy chain variable region PSMA-VH-3465 is the HCVR of H1H3465P (SEQ ID NO:122) from Table 1. The anti-PSMA heavy chain variable region PSMA-VH-11810 is the HCVR of H1H11810P2 (SEQ ID NO:66) from Table 1.

[0241] The anti-CD3 heavy chain variable region CD3-VH-A is the HCVR of H1H5778P (SEQ ID NO:922) from Table 12. The anti-CD3 heavy chain variable region CD3-VH-B is the HCVR of H1H2712N (SEQ ID NO:154) from Table 12. The anti-CD3 heavy chain variable regions CD3-VH-G, CD3-VH-G2, CD3-VH-G3, CD3-VH-G4, CD3-VH-G5, CD3-VH-G8, CD3-VH-G9, CD3-VH-G10, CD3-VH-G11, CD3-VH-G12, CD3-VH-G13, CD3-VH-G14, CD3-VH-G15, CD3-VH-G16, CD3-VH-G17, CD3-VH-G18, CD3-VH-G19, CD3-VH-G20, and CD3-VH-G21 are

described in Table 18.

[0242] The light chains in Table 5 were common to both the CD3 and PSMA targeting arms of the bispecific antibodies. The anti-CD3 light chain variable region CD3-VL-A is the LCVR of H1H5778P (SEQ ID NO:930) from Table 12. The anti-CD3 light chain variable region CD3-VL-B is the LCVR of H1H2712N (SEQ ID NO:162) from Table 12. The light chain variable region VK 1-39 JK 5 is SEQ ID NO: 1386 from Table 20. Table 1 sets out amino acid sequence identifiers for the various heavy chain variable regions, and their corresponding CDRs, of the anti-PSMA arms of the bispecific antibodies of this Example. Table 2 sets out the sequence identifiers for the nucleotide sequences encoding the heavy chain variable regions, and their corresponding CDRs, of the anti-PSMA antigen-binding domains of the bispecific antibodies of this Example.

[0243] Tables 12, 14, 15, 18, and 20 describe amino acid sequences of the heavy chain variable regions, and their corresponding CDRs, for the anti-CD3 arms of the bispecific antibodies, as well as amino acid sequences for the light chain variable regions, and their corresponding CDRs, common to both arms of the bispecific antibodies of this Example. Tables 13, 16, 17, 19, and 21 describe the corresponding nucleotide sequences for these features of the bispecific antibodies of this Example.

Example 5: Binding Affinities of Exemplified Bispecific Antibodies as Measured by FACS Analysis

[0244] In this example, the ability of the anti-PSMA/anti-CD3 bispecific antibodies described in Example 4 to bind to human PSMA expressing cell lines and to human and cynomolgus CD3-expressing cell lines via FACS was determined. As described above, the bispecific antibodies of this invention utilized a PSMA-specific heavy chain (HC) binding arm paired with a panel of anti-CD3 HC binding arms and a common light chain. The PSMA-HC binding arms in the bispecific antibodies, below, demonstrated potent binding to human PSMA protein via surface plasmon resonance (Example 3). As described in Examples 6 and 13 herein, the CD3-binding HC arms also displayed a range of affinities to human soluble heterodimeric hCD3 ϵ/δ .mFc protein via surface plasmon resonance.

[0245] Briefly, 2 $\times$ 10⁵ cells/well of human CD3-expressing Jurkat, cynomolgus T, or human PSMA-specific expressing cells were incubated with a serial dilution of bispecific antibodies for 30 min at 4° C. After incubation, cells were washed and a goat F(ab')₂ anti-human Fc γ PE labeled secondary (Jackson Immunolabs) was added to the cells for an additional 30 min. Next, cells were washed, re-suspended in cold PBS+1% BSA and analyzed via flow cytometry on a BD FACS Canto II.

[0246] For FACS analysis, cells were gated by forward scatter height vs. forward scatter area for single events selection, followed by side and forward scatters. The EC₅₀ for cell binding titration was determined using Prism software. Values were calculated using 4-parameter non-linear regression analysis.

TABLE-US-00006 TABLE 6 FACS Binding on CD3 and PSMA-Specific Cell lines Bispecific Anti-CD3- Cyno T- B16F10.9/ Antibody Binding Jurkat cells PSMA 22RV1 Identifier Arm
EC₅₀ [M] EC₅₀ [M] EC₅₀ [M] EC₅₀ [M] BPSMA/ CD3-VH- 3.91E-08 NT
NT 7.85E-08 CD3-001 A BPSMA/ CD3-VH- NT NT NT NT CD3-002 B BPSMA/ CD3-VH-
1.65E-08 1.42E-08 2.26E-09 NT CD3-003 G BPSMA/ CD3-VH- NB NB 1.88E-09 NT CD3-
200 G2 BPSMA/ CD3-VH- NB NB 1.90E-09 NT CD3-300 G3 BPSMA/ CD3-VH- NB NB
1.72E-09 NT CD3-400 G4 BPSMA/ CD3-VH- ~1.0E-06 NB 1.31E-09 NT CD3-004 G5
BPSMA/ CD3-VH- 1.93E-08 1.96E-08 1.31E-09 NT CD3-800 G8 BPSMA/ CD3-VH-
2.74E-07 NB 1.43E-09 NT CD3-900 G9 BPSMA/ CD3-VH- 2.77E-07 NB 1.19E-09 NT CD3-
1000 G10 BPSMA/ CD3-VH- 1.83E-08 8.90E-07 1.03E-09 NT CD3-1100 G11 BPSMA/
CD3-VH- 4.72E-08 NB 1.16E-09 NT CD3-1200 G12 BPSMA/ CD3-VH- 1.02E-07 2.17E-06
1.25E-09 NT CD3-1300 G13 BPSMA/ CD3-VH- 3.19E-08 1.70E-07 1.30E-09 NT CD3-1400
G14 BPSMA/ CD3-VH- 9.30E-08 NB 1.21E-09 NT CD3-1500 G15 BPSMA/ CD3-VH-

5.68E-08 NB 1.03E-09 NT CD3-1600 G16 BSPSMA/ CD3-VH- 2.00E-07 3.35E-06 1.34E-09 NT CD3-1700 G17 BSPSMA/ CD3-VH- 1.26E-07 NB 2.16E-09 NT CD3-1800 G18 BSPSMA/ CD3-VH- 6.07E-08 NB 1.35E-09 NT CD3-1900 G19 BSPSMA/ CD3-VH- 2.10E-07 6.14E-06 2.09E-09 NT CD3-005 G20 BSPSMA/ CD3-VH- 1.06E-07 NB 1.14E-09 NT CD3-2100 G21 NB = no binding; NT = not tested

[0247] As shown in Table 6, the anti-PSMA/anti-CD3 bispecific antibodies tested demonstrated specificity of binding to human PSMA-expressing B16F10.9/hPSMA and 22RV1 cell lines via FACS. The detection limit for FACS binding is 1 μ M EC50.

[0248] As shown in Table 6, the CD3 binding arms of each CD3 $\times$ PSMA bispecific antibody displayed a range of cell binding affinity to human CD3 expressing Jurkat cells (15 to 300 nM EC50 range). Importantly, the CD3 arms that showed weak-to-no binding to human CD3 heterodimeric protein via surface plasmon resonance (see Table 7 hereinbelow) also correlated with weak to no observable binding on Jurkat cells (i.e. CD3-VH-G2, CD3-VH-G3, CD3-VH-G5). Several CD3-binding arms also displayed cross reactivity to cynomolgus T-cells. All tested bispecific antibodies displayed similar cell binding on respective PSMA-expressing cell lines, confirming that bispecific pairing with individual CD3 arms did not affect or diminish PSMA-specific binding (PSMA-specific binding was less than or equal to 5.6 nM (high affinity) in all examples tested).

[0249] Antibodies exhibiting weak-to-no detectable binding to human CD3, and also exhibiting weak-to-no binding to cynomolgus CD3, were considered advantageous for avidity-driven bispecific pairing in accordance with the present invention, and were further tested for cytotoxicity in in vitro and in vivo assays.

Example 6: Binding Affinities of Exemplified Antibodies as Measured by a Surface Plasmon Resonance Binding Assay

[0250] Binding affinities and kinetic constants of anti-PSMA $\times$ anti-CD3 bispecific antibodies to soluble heterodimeric hCD3 ϵ/δ .mFc protein (hCD3 ϵ =UniProtKB/Swiss-Prot: P07766.2; SEQ_ID NO: 1652; hCD3 δ =UniProtKB/Swiss-Prot: P04234.1, SEQ ID NO: 1653) were determined by surface plasmon resonance at 37° C. using an antigen-capture format (Table 7). Measurements were conducted on a Sierra Sensors MASS-1 instrument.

[0251] In the antigen-capture format, the MASS-1 high-density amine sensor surface was derivatized with a goat anti-mouse IgG2a polyclonal antibody (Southern Biotech). Soluble heterodimeric CD3 protein was captured and the respective antibodies were injected over the captured antigen.

[0252] Kinetic association (k.sub.a) and dissociation (k.sub.d) rate constants were determined by processing and fitting the data to a 1:1 binding model using MASS-1 AnalyserR2 curve fitting software. Binding dissociation equilibrium constants (K.sub.D) and dissociative half-lives (t.sub.1/2) were calculated from the kinetic rate constants as: K.sub.D (M)=k.sub.d/k.sub.a; and t.sub.1/2 (min)=(ln 2)/(60*k.sub.d).

TABLE-US-00007 TABLE 7 Affinities of anti-CD3 Bispecific Antibodies to Soluble Human CD3 Binding at 37° C./Antigen-Capture Format Corresponding anti- Bispecific CD3 Antigen- Antibody Binding HCVR T $\frac{1}{2}$ Identifier Identifier ka (Ms.sup.-1) kd (s.sup.-1) K.sub.D (M) (min)

BSPSMA/CD3- CD3-VH-G	1.32E+05	7.62E-04	5.78E-09	15.2	003	BSPSMA/CD3- CD3-VH-G2
NB NB NB NB	200	BSPSMA/CD3- CD3-VH-G3	NB NB NB NB	300	BSPSMA/CD3- CD3-VH-G4	
NB NB NB NB	400	BSPSMA/CD3- CD3-VH-G5	NB NB NB NB	004	BSPSMA/CD3- CD3-VH-G8	
5.95E+04	1.15E-03	1.94E-08	10.0	800	BSPSMA/CD3- CD3-VH-G9	
4.38E+04	4.95E-03	1.13E-07	2.3	900	BSPSMA/CD3- CD3-VH-G10	
3.44E+04	6.37E-03	1.85E-07	1.8	1000	BSPSMA/CD3- CD3-VH-G11	
9.21E+04	1.02E-03	1.11E-08	11.3	1100	BSPSMA/CD3- CD3-VH-G12	
3.85E+04	2.47E-03	6.42E-08	4.7	1200	BSPSMA/CD3- CD3-VH-G13	
2.03E+04	2.48E-03	1.22E-07	4.7	1300	BSPSMA/CD3- CD3-VH-G14	
6.21E+04	3.31E-03	5.33E-08	3.5	1400	BSPSMA/CD3- CD3-VH-G15	
7.36E+04	6.11E-03	8.29E-08	1.9	1500	BSPSMA/CD3-	

CD3-VH-G16 6.43E+04 2.43E-03 3.78E-08 4.7 1600 BSPSMA/CD3- CD3-VH-G17 4.70E+04 3.07E-03 6.52E-08 3.8 1700 BSPSMA/CD3- CD3-VH-G18 NB NB NB NB 1800 BSPSMA/CD3- CD3-VH-G19 4.43E+04 5.09E-03 1.15E-07 2.3 1900 BSPSMA/CD3- CD3-VH-G20 1.73E+04 5.77E-03 3.34E-07 2.0 005 BSPSMA/CD3- CD3-VH-G21 3.02E+04 2.34E-03 7.75E-08 4.9 2100 Control 1 CD3-L2K 3.68E+05 2.66E-03 7.22E-09 4.3

[0253] As shown in Table 7, the anti-CD3×anti-PSMA bispecific antibodies either maintained very weak binding to soluble CD3 in the surface plasmon resonance binding assay, e.g. having a K_{sub.D} value greater than 11 nM up to 334 nM which is weaker than that of the bispecific anti-CD3 arm derived from germline frameworks, CD3-VH-G, or did not exhibit any detectable binding.

[0254] As such, several bispecific antibodies exhibited greater than 50 nM KD values, and some were greater than 100 nM (i.e. BSPSMA/CD3-900, BSPSMA/CD3-1000, BSPSMA/CD3-1900, BSPSMA/CD3-005) and even beyond the detection limit of the assay, i.e. showed no detectable binding to soluble human CD3 (i.e. BSPSMA/CD3-200, BSPSMA/CD3-300, BSPSMA/CD3-400, BSPSMA/CD3-004 and BSPSMA/CD3-1800).

Example 7: T Cell Activation and Tumor-Specific Cytotoxicity Exhibited by Bispecific Antibodies of the Invention as Measured In Vitro

[0255] In this example, the specific killing of PSMA-expressing target cells in the presence of anti-PSMA×anti-CD3 bispecific antibodies was monitored via flow cytometry. As reported previously, the bispecific antibodies displayed a range of affinity to CD3 protein and CD3-expressing cell lines (i.e. weak, moderate and strong binding). This same panel of bispecific antibodies was tested for the ability to induce naïve human T-cells to re-direct killing toward target-expressing cells.

[0256] Briefly, PSMA-expressing (C4-2, 22Rv1 and TRAMPC2_PSMA) cell lines were labeled with 1 μM of the fluorescent tracking dye Violet Cell Tracker. After labeling, cells were plated overnight at 37° C. Separately, human PBMCs were plated in supplemented RPMI media at 1×10^{sup.6} cells/mL and incubated overnight at 37° C. in order to enrich for lymphocytes by depleting adherent macrophages, dendritic cells, and some monocytes. The next day, target cells were co-incubated with adherent cell-depleted naïve PBMC (Effector/Target cell 4:1 ratio) and a serial dilution of relevant bispecific antibodies or Isotype control (concentration range: 66.7 nM to 0.25 pM) for 48 hours at 37° C. Cells were removed from cell culture plates using an enzyme-free cell dissociation buffer, and analyzed by FACS.

[0257] For FACS analysis, cells were stained with a dead/live far red cell tracker (Invitrogen). 5×10^{sup.5} counting beads were added to each well immediately before FACS analysis. 1×10^{sup.4} beads were collected for each sample. For the assessment of specificity of killing, cells were gated on live Violet labeled populations. Percent of live population was recorded and used for the calculation of normalized survival.

[0258] T cell activation was assessed by incubating cells with directly conjugated antibodies to CD2 and CD69, and by reporting the percent of activated (CD69+) T cells out of total T cells (CD2+).

[0259] As the results in Table 8 show, depletion of PSMA-expressing cells was observed with anti-PSMA×anti-CD3 bispecific antibodies. Most of the tested bispecific antibodies activated and directed human T cells to deplete the target cells with EC_{sub.50s} in picomolar range. Additionally, the observed target-cell lysis was associated with an up-regulation of CD69 cells on CD2+ T cells, with pM EC_{sub.50S}.

[0260] Importantly, the results of this example demonstrate that several bispecifics which utilized a CD3 binding arm that displayed weak-to-non-observable binding to CD3 protein or CD3-expressing cells (i.e. CD3-VH-G5) still retained the ability to activate T-cells and exhibited potent cytotoxicity of tumor antigen-expressing cells.

TABLE-US-00008 TABLE 8 Cytotoxicity and T-cell activation properties of selected PSMA×CD3 Bispecific Antibodies 22RV1 TrampC2. Bispecific Anti-CD3 C4-2 Cell Cell PSMA T cell

Antibody Binding depletion depletion Cell depletion activation Identifier Arm EC.sub.50 [M]
 EC.sub.50 [M] EC.sub.50 [M] EC.sub.50 [M] BSPSMA/ CD3-VH- 1.03E-11 NT 6.43E-12
 1.23E-12 CD3-003 G BSPSMA/ CD3-VH- NT No NT No CD3-200 G2 activity activity
 BSPSMA/ CD3-VH- NT Very NT 1.85E-11 CD3-300 G3 weak BSPSMA/ CD3-VH- NT Very NT
 Very CD3-400 G4 weak weak BSPSMA/ CD3-VH- 2.15E-11 6.31E-12 1.15E-11 1.34E-11 CD3-
 004 G5 BSPSMA/ CD3-VH- NT NT 9.27E-12 1.76E-12 CD3-800 G8 BSPSMA/ CD3-VH- NT
 NT 3.50E-12 1.12E-12 CD3-900 G9 BSPSMA/ CD3-VH- NT NT 5.97E-12 1.28E-12 CD3-1000
 G10 BSPSMA/ CD3-VH- NT NT 3.86E-12 1.11E-12 CD3-1100 G11 BSPSMA/ CD3-VH-
 8.74E-12 NT NT 2.31E-12 CD3-1300 G13 BSPSMA/ CD3-VH- 7.37E-12 2.07E-12 NT
 3.89E-12 CD3-1700 G17 BSPSMA/ CD3-VH- 1.39E-11 8.32E-12 NT 6.11E-12 CD3-005 G20
 NT = not tested

Example 8: Anti-PSMA/Anti-CD3 Bispecific Antibodies Display Potent Anti-Tumor Efficacy In Vivo

[0261] To determine the in vivo efficacy of exemplary anti-PSMA/anti-CD3 bispecific antibodies, studies were performed in immunocompromised mice bearing human prostate cancer xenografts. Additional studies were also carried out in immunocompetent mice bearing mouse prostate cancer xenografts engineered to express human PSMA.

Efficacy of Anti-PSMA/Anti-CD3 Bispecific Antibodies in Human Tumor Xenograft Models

[0262] To assess the in vivo efficacy of the anti-PSMA/anti-CD3 bispecifics in human tumor xenograft studies, NOD scid gamma (NSG) mice (Jackson Laboratories, Bar Harbor, Maine) were co-implanted with human peripheral blood mononuclear cells (PBMCs) along with 22Rv1 or C4-2 human prostate tumor cells which endogenously express PSMA.

[0263] Briefly, 4×10^6 22Rv1 or 5×10^6 C4-2 cells (MD Anderson, TX) cells were co-implanted s.c. with 1×10^6 human PBMCs (ReachBio, LLC., Seattle, WA) in a 50:50 mix of matrigel matrix (BD Biosciences) into the right flank of male NSG mice. In the 22Rv1 study, mice were treated i.p. on days 0, 3 and 7 with 1 μ g of BSPSMA/CD3-001 or an isotype control (FIG. 1). In the C4-2 study, mice were treated i.p. on days 0, 4, and 7 post tumor implantation with 0.1 mg/kg BSPSMA/CD3-001, BSPSMA/CD3-003 or BSPSMA/CD3-005.

[0264] In an additional xenogenic model, anti-PSMA/anti-CD3 bispecifics were tested in mice engrafted with human hematopoietic CD34+ stem cells. Briefly, newborn SIRP α BALB/c-Rag2-IL2 γ -(BRG) pups were engrafted with hCD34+ fetal liver cells. 3-6 months later hCD34-engrafted SIRP α BRG mice were then implanted with C4-2 cells (5×10^6 s.c. in matrigel). 8 days later, mice were treated with 10 μ g of BSPSMA/CD3-004 or an isotype control antibody, followed by 2 $\times$ /week doses throughout the study.

[0265] In all studies, tumor size was measured 2 $\times$ /week using calipers and tumor volume calculated as $\text{Volume} = (\text{length} \times \text{width}^2) / 2$.

[0266] As the results in Table 9 show, the bispecific antibodies tested in the xenogenic models described above were all effective at inhibiting tumor growth compared to treatment with the isotype control.

Efficacy of Anti-PSMA/Anti-CD3 Bispecific Antibodies in Established Human Tumor Xenograft Model

[0267] Next, the efficacy of anti-PSMA/anti-CD3 bispecific antibodies in suppressing the growth of established tumors was assessed. NSG mice were first injected with 2.5×10^6 human PBMCs i.p. to allow for engraftment of human T cells. Fourteen days later, mice were co-implanted with C4-2 cells and PBMCs as above. 20 μ g of BSPSMA/CD3-002 or an isotype control were administered i.p. 18 days post tumor implantation and continued 2 $\times$ /week for the duration of the study. Additional PBMCs were given i.p. on days 20 and 40 post tumor implantation.

[0268] As the results in Table 9 show, BSPSMA/CD3-002 showed efficacy in suppressing the growth of established tumors, decreasing tumor growth by 95%.

Efficacy of Anti-PSMA/Anti-CD3 Bispecific Antibodies in Immune-Competent Tumor Model

[0269] Additionally, anti-PSMA/anti-CD3 bispecifics were assessed for anti-tumor activity in an immune-competent model. Mice humanized for the three chains ($\delta\gamma\epsilon$) of CD3 as well as for PSMA were implanted with a variant murine prostate cancer cell line TRAMP-C2 transfected with human PSMA.

[0270] Prior to study initiation, the tumorigenic cell line variant TRAMP-C2_hPSMAv #1 was generated. Briefly, 7.5×10^6 TRAMP-C2_hPSMA cells were implanted s.c. into the right flank of male mice humanized for CD3 and PSMA. A tumor was excised and cut into 3 mm fragments and subsequently implanted into the right flank of new male humanized mice. A tumor arising from the implanted tumor fragments was then harvested and disaggregated into a single cell suspension. These cells (TRAMP-C2_hPSMAv #1) were then cultured in vitro under G418 selection.

4×10^6 cells of this variant cell line were then implanted into the right flank of male PSMA/CD3 humanized mice for the bispecific antibody efficacy studies.

[0271] Humanized PSMA/CD3 mice implanted with TRAMP-C2_hPSMAv #1 were treated with 100 μ g or 10 μ g of anti-PSMA/anti-CD3 bispecific antibody BPSMA/CD3-001 or BPSMA/CD3-004 or an isotype control 2 $\times$ /week starting from the day of tumor implantation. Serum cytokine levels 4h post-injection were also examined, as well as spleen T-cell levels. Study was terminated at Day 27.

[0272] As the results in Table 10 show, both anti-PSMA/anti-CD3 bispecific molecules showed efficacy in significantly delaying tumor growth across treatment groups. Dose dependent cytokine release was observed after treatment with BPSMA/CD3-001. Minimal cytokine release was observed after administration of BPSMA/CD3-004, possibly due to the weak binding of the anti-CD3. BPSMA/CD3-001 showed anti-tumor efficacy without depleting T cells in the spleen. Efficacy of Anti-PSMA/Anti-CD3 Bispecific Antibodies on Established Tumors in Immune-Competent Model

[0273] Lastly, the efficacy of selected anti-PSMA/anti-CD3 bispecific molecules on reducing growth of established tumors in humanized PSMA/CD3 mice was assessed. TRAMP-C2_hPSMAv #1 cells were transplanted in humanized mice as described above, and 100 μ g BPSMA/CD3-001 or isotype control was administered i.p. 2 $\times$ /week 14 days after tumor implantation, when tumor sizes ranged from 50 mm³-100 mm³. As the results in Table 11 show, BPSMA/CD3-001 was efficacious in this established tumor model, displaying an 84% decrease in tumor growth compared to the control group.

[0274] In summary, the anti-PSMA/anti-CD3 bispecific antibodies of this invention display potent anti-tumor efficacy in both immune-compromised and immune-competent tumor models. Additionally, several of the tested bispecific antibodies (BPSMA/CD3-001 and 002) displayed potent ability to reduce the volume of established tumors.

[0275] Of note, in the absence of PSMA-expressing tumor cells, no T cell activation was seen.

[0276] Additionally, in mice bearing no tumors, blood samples were collected 4 hours following PSMAxCD3 bispecific antibody treatment, and serum cytokine levels were determined. Transient increases in levels of cytokines, namely interferon-gamma (IFN- γ), tumor necrosis factor (TNF), interleukin-2 (IL-2), and interleukin-6 (IL-6) were determined and the transient increases were dose-dependent (FIGS. 2A-2D).

[0277] In order to validate the specificity of the bispecific antibodies, mice either humanized for both PSMA and CD3 or mice humanized for CD3 alone were dosed with 100 μ g of PSMAxCD3 and examined for serum cytokines (4 hrs post dose) and transient T cell loss from the blood (24 hrs post dose). Treatment with PSMAxCD3 bispecific antibody in a humanized T cell mouse (100 μ g/mouse) induces acute increase in cytokines (e.g. IFN γ) as well as transient decrease in circulating T cells (FIGS. 3A-3B). This finding reproduces cytokine and T cell changes that have been observed in human patients treated with tumor antigen x CD3 bispecific antibodies.

[0278] PSMAxCD3 bispecific antibodies are efficacious without depleting effector T cells in the spleen of the immunocompetent mice, as shown in FIGS. 4A-4C. Briefly, humanized PSMA and

CD3 Velocigene® mice were implanted with hPSMA-expressing tumors and treated with PSMA×CD3 twice weekly. T cells were present at normal numbers at final harvest. Spleens were examined for CD4+ T cells, CD8+ T cells, and Tregs at the end of the experiment after treatment with PSMA×CD3 bispecific antibody twice per week throughout study. Mice humanized for PSMA and CD3 were implanted with TRAMP-C2_hPSMA tumors and dosed from day 0 with 100 ug or 10 ug of PSMA×CD3. Cell populations in the spleen were analyzed by flow cytometry. Data was analyzed using analysis of variance (ANOVA) for any significant effects compared to the isotype control group but no significant differences were found (FIGS. 4A-4C).

TABLE-US-00009 TABLE 9 Efficacy of anti-PSMA/anti-CD3 Bispecific Antibodies in Immune-Compromised Xenograft Models

Xenogenic model:	suppression of tumor growth	Tumor N	Final Tumor Model/ #mice/ Volume	Mouse treatment Bispecific Antibody (mm.sup.3)	Strain group
Identifier	Dose	Mean ± SD	22Rv1/ 5	BSPSMA/CD3-001 1.0	370 ± 270 NSG 5
Isotype Control	ug/mouse	1260 ± 730	on day 0, 3 & 7	C4-/2 5	BSPSMA/CD3-001 0.1 mg/kg 0 ± 0 NSG 5
BSPSMA/CD3-003	on day 0, 4	0 ± 0	5	BSPSMA/CD3-005 & 7	0 ± 0 5
Isotype Control	960 ± 660				
C4-2/ 5	BSPSMA/CD3-004 1.0	70 ± 60	SIRPα	ug/mouse Balb/c- 2×/week	Rag2- IL2ry- BRG
engrafted with hCD34+ HSC	5	Isotype Control 260 ± 180			
Xenogenic model:	inhibition of established tumor growth	Tumor Growth	Tumor N (mm.sup.3)	from % Decrease	Model/ # mice/ Bispecific Antibody start of Tumor Mouse treatment Identifier treatment Growth vs. Strain group
Dose: 20 ug/ mouse (mean ± SD)	Control C4-2/ 5	BSPSMA/CD3-002 60 ± 100	95%	NSG 4	Isotype Control 1170 ± 600 (-)

TABLE-US-00010 TABLE 10 Efficacy of anti-PSMA/anti-CD3 Bispecific antibodies in immune-competent syngeneic models

N Tumor Spleen T-cell Dose # mice/ Volume (mm.sup.3)	Mean Serum Cytokine level %, Tumor Model/ Bispecific Antibody (ug/mouse) treatment at Day 27 Concentrations, (pg/mL) (mean ± SD).sup.#	Mouse Strain Identifier 2×/week* group (Mean ± SD)
IFNγ TNFα IL-2 IL-12p70 IL-6 CD4+ CD8+ TRAMP-C2/ BSPSMA/CD3-001	100 4 260 ± 250	130 180 150 70 20 6.0 ± 1.0 13.0 ± 3.0
PSMA.sup.Hum/hum 10 6 600 ± 330	80 140 140 30 1130	7.0 ± 2.0 14.0 ± 3.0
CD3.sup.Hum/Hum BSPSMA/CD3-004	100 4 50 ± 60	30 60 60 40 370 8.0 ± 1.0 12.0 ± 2.0
10 5 380 ± 650	10 50 50 10 330 8.0 ± 3.0 14.0 ± 4.0	Isotype Control 100 5 1740 ± 560
4 30 30 10 230 5.0 ± 1.0 8.0 ± 2.0		*Mice were dosed with antibody or isotype control 2×/week starting on the day of tumor implantation .sup.#Measured as the percentage of CD4+ or CD8+ cells in spleen out of live mCD45+ cells

TABLE-US-00011 TABLE 11 Efficacy of anti-PSMA/anti-CD3 Bispecific antibodies in suppression of established tumor growth in immune competent syngeneic model

Tumor efficacy in immune-competent model, established tumors	Bispecific Tumor Growth % Decrease Antibody Identifier (mm.sup.3) from start Tumor Tumor Model/ Dose: 100 of treatment Growth vs. Mouse Strain ug/mouse (mean ± SD)
Control TRAMP-C2/ BSPSMA/CD3-001	170 ± 170 84
psmA.sup.Hum/hum Isotype Control	740 ± 570 (-) CD3.sup.Hum/Hum

Example 9: Generation of Anti-CD3 Antibodies

[0279] Anti-CD3 antibodies were obtained by immunizing an engineered mouse comprising DNA encoding human Immunoglobulin heavy and kappa light chain variable regions with cells expressing CD3 or with DNA encoding CD3. The antibody immune response was monitored by a CD3-specific immunoassay. When a desired immune response was achieved, splenocytes were harvested and fused with mouse myeloma cells to preserve their viability and form hybridoma cell lines. The hybridoma cell lines were screened and selected to identify cell lines that produce CD3-specific antibodies. Using this technique several anti-CD3 chimeric antibodies (i.e., antibodies possessing human variable domains and mouse constant domains) were obtained. In addition, several fully human anti-CD3 antibodies were isolated directly from antigen-positive B cells without fusion to myeloma cells, as described in US 2007/0280945A1.

[0280] Certain biological properties of the exemplary anti-CD3 antibodies generated in accordance with the methods of this Example are described in detail in the Examples herein.

Example 10: Heavy and Light Chain Variable Region Amino Acid and Nucleic Acid Sequences [0281] Table 12 sets forth the amino acid sequence identifiers of the heavy and light chain variable regions and CDRs of selected anti-CD3 antibodies of the invention. The corresponding nucleic acid sequence identifiers are set forth in Table 13. Methods of making the anti-CD3 antibodies disclosed herein can also be found in US publication 2014/0088295.

TABLE-US-00012 TABLE 12 Amino Acid Sequence Identifiers Antibody SEQ ID NOs:

Designation	HCVR	HCDR1	HCDR3	LCVR	LCDR11	LCDR2	LCDR3	H1H2712N	154	156	158
160	162	164	166	168	H1M2692N	170	172	174	176	178	180
182	184	H1M3542N	186	188	190	192	194	196	198	200	H1M3544N
202	204	206	208	210	212	214	216	H1M3549N	218	220	222
224	226	228	230	232	H1M3613N	234	236	238	240	242	244
246	248	H2M2689N	250	252	254	256	258	260	262	264	H2M2690N
266	268	270	272	274	276	278	280	H2M2691N	282	284	286
288	290	292	294	296	H2M2704N	298	300	302	304	306	308
310	312	H2M2705N	314	316	318	320	322	324	326	328	H2M2706N
330	332	334	336	338	340	342	344	H2M2707N	346	348	350
352	354	356	358	360	H2M2708N	362	364	366	368	370	372
374	376	H2M2709N	378	380	382	384	386	388	390	392	H2M2710N
394	396	398	400	402	404	406	408	H2M2711N	410	412	414
416	418	420	422	424	H2M2774N	426	428	430	432	434	436
438	440	H2M2775N	442	444	446	448	450	452	454	456	H2M2776N
458	460	462	464	466	468	470	472	H2M2777N	474	476	478
480	482	484	486	488	H2M2778N	490	492	494	496	498	500
502	504	H2M2779N	506	508	510	512	514	516	518	520	H2M2789N
522	524	526	528	530	532	534	536	H2M2862N	538	540	542
544	546	548	550	552	H2M2885N	554	556	558	560	562	564
566	568	H2M2886N	570	572	574	576	578	580	582	584	H2M3540N
586	588	590	592	594	596	598	600	H2M3541N	602	604	606
608	610	612	614	616	H2M3543N	618	620	622	624	626	628
630	632	H2M3547N	634	636	638	640	642	644	646	648	H2M3548N
650	652	654	656	658	660	662	664	H2M3563N	666	668	670
672	674	676	678	680	H1H5751P	682	684	686	688	690	692
694	696	H1H5752P	698	700	702	704	706	708	710	712	H1H5753B
714	716	718	720	722	724	726	728	H1H5754B	730	732	734
736	738	740	742	744	H1H5755B	746	748	750	752	754	756
758	760	H1H5756B	762	764	766	768	770	772	774	776	H1H5757B
778	780	782	784	786	788	790	792	H1H5758B	794	796	798
800	802	804	806	808	H1H5761P	810	812	814	816	818	820
822	824	H1H5763P	826	828	830	832	834	836	838	840	H1H5764P
842	844	846	848	850	852	854	856	H1H5769P	858	860	862
864	866	868	870	872	H1H5771P	874	876	878	880	882	884
886	888	H1H5772P	890	892	894	896	898	900	902	904	H1H5777P
906	908	910	912	914	916	918	920	H1H5778P	922	924	926
928	930	932	934	936	H1H5780P	938	940	942	944	946	948
950	952	H1H5781P	954	956	958	960	962	964	966	968	H1H5782P
970	972	974	976	978	980	982	984	H1H5785B	986	988	990
992	994	996	998	1000	H1H5786B	1002	1004	1006	1008	1010	1012
1014	1016	H1H5788P	1018	1020	1022	1024	1026	1028	1030	1032	H1H5790B
1034	1036	1038	1040	1042	1044	1046	1048	H1H5791B	1050	1052	1054
1056	1058	1060	1062	1064	H1H5792B	1066	1068	1070	1072	1074	1076
1078	1080	H1H5793B	1082	1084	1086	1088	1090	1092	1094	1096	H1H5795B
1098	1100	1102	1104	1106	1108	1110	1112	H1H5796B	1114	1116	1118
1120	1122	1124	1126	1128	H1H5797B	1130	1132	1134	1136	1138	1140
1142	1144	H1H5798B	1146	1148	1150	1152	1154	1156	1158	1160	H1H5799P
1162	1164	1166	1168	1170	1172	1174	1176	H1H5801B	1178	1180	1182
1184	1186	1188	1190	1192	H1H7194B	1194	1196	1198	1200	1386	1388
1390	1392	H1H7195B	1202	1204	1206	1208	1386	1388	1390	1392	H1H7196B
1210	1212	1214	1216	1386	1388	1390	1392	H1H7198B	1218	1220	1222
1224	1386	1388	1390	1392	H1H7203B	1226	1228	1230	1232	1386	1388
1390	1392	H1H7204B	1234	1236	1238	1240	1386	1388	1390	1392	H1H7208B
1242	1244	1246	1248	1386	1388	1390	1392	H1H7211B	1250	1252	1254
1256	1386	1388	1390	1392	H1H7221B	1258	1260	1262	1264	1386	1388
1390	1392	H1H7223B	1266	1268	1270	1272	1386	1388	1390	1392	H1H7226B
1274	1276	1278	1280	1386	1388	1390	1392	H1H7232B	1282	1284	1286
1288	1386	1388	1390	1392	H1H7233B	1290	1292	1294	1296	1386	1388
1390	1392	H1H7241B	1298	1300	1302	1304	1386	1388	1390	1392	H1H7242B
1306	1308	1310	1312	1386	1388	1390	1392	H1H7250B	1314	1316	1318
1320	1386	1388	1390	1392	H1H7251B	1322	1324				

1326 1386 1386 1386 1390 1392 H1H7254B 1330 1332 1334 1336 1386 1386 1390 1392
H1H7258B 1338 1340 1342 1344 1386 1388 1390 1392 H1H7269B 1346 1348 1350 1352 1386
1388 1390 1392 H1H7279B 1354 1356 1358 1360 1386 1388 1390 1392 H1xH7221G 1362 1364
1366 1368 1386 1388 1390 1392 H1xH7221G3 1370 1372 1374 1376 1386 1388 1390 1392
H1xH7221G5 1378 1380 1382 1384 1386 1388 1390 1392
TABLE-US-00013 TABLE 13 Nucleic Acid Sequence Identifiers Antibody SEQ ID NOs:
Designation HCVR HCDR1 HCDR2 HCDR3 LCVR LCDR1 LCDR2 LCDR3 H1H2712N 153
155 157 159 161 163 165 167 H1M2692N 169 171 173 175 177 179 181 183 H1M3542N 185 187
189 191 193 195 197 199 H1M3544N 201 203 205 207 209 211 213 215 H1M3549N 217 219 221
223 225 227 229 231 H1M3613N 233 235 237 239 241 243 245 247 H2M2689N 249 251 253 255
257 259 261 263 H2M2690N 265 267 269 271 273 275 277 279 H2M2691N 281 283 285 287 289
291 293 295 H2M2704N 297 299 301 303 305 307 309 311 H2M2705N 313 315 317 319 321 323
325 327 H2M2706N 329 331 333 335 337 339 341 343 H2M2707N 345 347 349 351 353 355 357
359 H2M2708N 361 363 365 367 369 371 373 375 H2M2709N 377 379 381 383 385 387 389 391
H2M2710N 393 395 397 399 401 403 405 407 H2M2711N 409 411 413 415 417 419 421 423
H2M2774N 425 427 429 431 433 435 437 439 H2M2775N 441 443 445 447 449 451 453 455
H2M2776N 457 459 461 463 465 467 469 471 H2M2777N 473 475 477 479 481 483 485 487
H2M2778N 489 491 493 495 497 499 501 503 H2M2779N 505 507 509 511 513 515 517 519
H2M2789N 521 523 525 527 529 531 533 535 H2M2862N 537 539 541 543 545 547 549 551
H2M2885N 553 555 557 559 561 563 565 567 H2M2886N 569 571 573 575 577 579 581 583
H2M3540N 585 587 589 591 593 595 597 599 H2M3541N 601 603 605 607 609 611 613 615
H2M3543N 617 619 621 623 625 627 629 631 H2M3547N 633 635 637 639 641 643 645 647
H2M3548N 649 651 653 655 657 659 661 663 H2M3563N 665 667 669 671 673 675 677 679
H1H5751P 681 683 685 687 689 691 693 695 H1H5752P 697 699 701 703 705 707 709 711
H1H5753B 713 715 717 719 721 723 725 727 H1H5754B 729 731 733 735 737 739 741 743
H1H5755B 745 747 749 751 753 755 757 759 H1H5756B 761 763 765 767 769 771 773 775
H1H5757B 777 779 781 783 785 787 789 791 H1H5758B 793 795 797 799 801 803 805 807
H1H5761P 809 811 813 815 817 819 821 823 H1H5763P 825 827 829 831 833 835 837 839
H1H5764P 841 843 845 847 849 851 853 855 H1H5769P 857 859 861 863 865 867 869 871
H1H5771P 873 875 877 879 881 883 885 887 H1H5772P 889 891 893 895 897 899 901 903
H1H5777P 905 907 909 911 913 915 917 919 H1H5778P 921 923 925 927 929 931 933 935
H1H5780P 937 939 941 943 945 947 949 951 H1H5781P 953 955 957 959 961 963 965 967
H1H5782P 969 971 973 975 977 979 981 983 H1H5785B 985 987 989 991 993 995 997 999
H1H5786B 1001 1003 1005 1007 1009 1011 1013 1015 H1H5788P 1017 1019 1021 1023 1025
1027 1029 1031 H1H5790B 1033 1035 1037 1039 1041 1043 1045 1047 H1H5791B 1049 1051
1053 1055 1057 1059 1061 1063 H1H5792B 1065 1067 1069 1071 1073 1075 1077 1079
H1H5793B 1081 1083 1085 1087 1089 1091 1093 1095 H1H5795B 1097 1099 1101 1103 1105
1107 1109 1111 H1H5796B 1113 1115 1117 1119 1121 1123 1125 1127 H1H5797B 1129 1131
1133 1135 1137 1139 1141 1143 H1H5798B 1145 1147 1149 1151 1153 1155 1157 1159
H1H5799P 1161 1163 1165 1167 1169 1171 1173 1175 H1H5801B 1177 1179 1181 1183 1185
1187 1189 1191 H1H7194B 1193 1195 1197 1199 1385 1387 1389 1391 H1H7195B 1201 1203
1205 1207 1385 1387 1389 1391 H1H7196B 1209 1211 1213 1215 1385 1387 1389 1391
H1H7198B 1217 1219 1221 1223 1385 1387 1389 1391 H1H7203B 1225 1227 1229 1231 1385
1387 1389 1391 H1H7204B 1233 1235 1237 1239 1385 1387 1389 1391 H1H7208B 1241 1243
1245 1247 1385 1387 1389 1391 H1H7211B 1249 1251 1253 1255 1385 1387 1389 1391
H1H7221B 1257 1259 1261 1263 1385 1387 1389 1391 H1H7223B 1265 1267 1269 1271 1385
1387 1389 1391 H1H7226B 1273 1275 1277 1279 1385 1387 1389 1391 H1H7232B 1281 1283
1285 1287 1385 1387 1389 1391 H1H7233B 1289 1291 1293 1295 1385 1387 1389 1391
H1H7241B 1297 1299 1301 1303 1385 1387 1389 1391 H1H7242B 1305 1307 1309 1311 1385
1387 1389 1391 H1H7250B 1313 1315 1317 1319 1385 1387 1389 1391 H1H7251B 1321 1323

1325 1327 1385 1387 1389 1391 H1H7254B 1329 1331 1333 1335 1385 1387 1389 1391
H1H7258B 1337 1339 1341 1343 1385 1387 1389 1391 H1H7269B 1345 1347 1349 1351 1385
1387 1389 1391 H1H7279B 1353 1355 1357 1359 1385 1387 1389 1391 H1xH7221G 1361 1363
1365 1367 1385 1387 1389 1391 H1xH7221G3 1369 1371 1373 1375 1385 1387 1389 1391
H1xH7221G5 1377 1379 1381 1383 1385 1387 1389 1391

[0282] Antibodies are typically referred to herein according to the following nomenclature: Fc prefix (e.g. “H1H,” “H1M,” “H2M,” etc.), followed by a numerical identifier (e.g. “2712,” “2692,” etc., as shown in Table 1), followed by a “P,” “N,” or “B” suffix. Thus, according to this nomenclature, an antibody may be referred to herein as, e.g., “H1H2712N,” “H1M2692N,” “H2M2689N,” etc. The H1H, H1M and H2M prefixes on the antibody designations used herein indicate the particular Fc region isotype of the antibody. For example, an “H1H” antibody has a human IgG1 Fc, an “H1M” antibody has a mouse IgG1 Fc, and an “H2M” antibody has a mouse IgG2 Fc, (all variable regions are fully human as denoted by the first ‘H’ in the antibody designation). As will be appreciated by a person of ordinary skill in the art, an antibody having a particular Fc isotype can be converted to an antibody with a different Fc isotype (e.g., an antibody with a mouse IgG1 Fc can be converted to an antibody with a human IgG4, etc.), but in any event, the variable domains (including the CDRs)—which are indicated by the numerical identifiers shown in Table 1—will remain the same, and the binding properties are expected to be identical or substantially similar regardless of the nature of the Fc domain.

[0283] Tables 14 and 15 set out the amino acid sequence identifiers for heavy chain variable regions (Table 14) and light chain variable regions (Table 15), and their corresponding CDRs, of additional anti-CD3 HCVRs and LCVRs useful in anti-PSMA×anti-CD3 bispecific antibodies of the invention.

TABLE-US-00014 TABLE 14 (Heavy Chain Variable Region Amino Acid Sequences) SEQ ID
NOs Heavy Chain Identifier HCVR HCDR1 HCDR2 HCDR3 CD3-VH-AA 1394 1396 1398 1400
CD3-VH-B 1410 1412 1414 1416 CD3-VH-C 1426 1428 1430 1432 CD3-VH-D 1442 1444 1446
1448 CD3-VH-E 1458 1460 1462 1464 CD3-VH-F.sup.# 1473 1474 1475 1476

TABLE-US-00015 TABLE 15 (Light Chain Variable Region Amino Acid Sequences) SEQ ID NOs
Light Chain Identifier LCVR LCDR1 LCDR2 LCDR3 CD3-VL-AA 1402 1404 1406 1408 CD3-
VL-B 1418 1420 1422 1424 CD3-VL-C 1434 1436 1438 1440 CD3-VL-D 1450 1452 1454 1456
CD3-VL-E 1466 1468 1470 1472 CD3-VL-F.sup.# 1477 1478 1479 1480

[0284] The heavy and light chain variable regions of CD3-VH-F and CD3-VL-F were derived from the anti-CD3 antibody designated “L2K” as set forth in WO2004/106380.

[0285] In addition, Tables 16 and 17 set out the sequence identifiers for the nucleotide sequences encoding the heavy chain variable regions (Table 16) and light chain variable regions (Table 17), and their corresponding CDRs, of additional anti-CD3 HCVRs and LCVRs useful in anti-PSMA×anti-CD3 bispecific antibodies of the invention.

TABLE-US-00016 TABLE 16 (Nucleotide Sequences Encoding Heavy Chain Variable Region
Sequences) SEQ ID NOs Heavy Chain Identifier HCVR HCDR1 HCDR2 HCDR3 CD3-VH-AA
1393 1395 1397 1399 CD3-VH-B 1409 1411 1413 1415 CD3-VH-C 1425 1427 1429 1431 CD3-
VH-D 1441 1443 1445 1447 CD3-VH-E 1457 1459 1461 1463

TABLE-US-00017 TABLE 17 (Nucleotide Sequences Encoding Light Chain Variable Region
Sequences) SEQ ID NOs Light Chain Identifier LCVR LCDR1 LCDR2 LCDR3 CD3-VL-AA
1401 1403 1405 1407 CD3-VL-B 1417 1419 1421 1423 CD3-VL-C 1433 1435 1437 1439 CD3-
VL-D 1449 1451 1453 1455 CD3-VL-E 1465 1467 1469 1471

Control Constructs Used in the Following Examples

[0286] Various control constructs (anti-CD3 antibodies) were included in the following experiments for comparative purposes: “OKT-3,” a mouse monoclonal antibody against human T-cell surface antigens available from the American Type Culture Collection (ATCC) under catalog no. CRL-8001; and “SP34,” a commercially available mouse monoclonal antibody obtained, e.g.,

from Biolegend, San Diego, CA (Cat. No. 302914) or BD Pharmagen, Cat. 55052, reactive against the epsilon chain of the T3 complex on human T lymphocyte cells.

Example 11: Generation of Additional Anti-CD3 Antibodies

[0287] The following procedures were aimed at identifying antibodies that specifically recognized CD3 (T cell co-receptor) as an antigen.

[0288] A pool of anti-CD3 antibodies were derived from a genetically modified mouse. Briefly, mice were immunized with a CD3 antigen and generated B cells that comprised a diversity of human VH rearrangements in order to express a diverse repertoire of high-affinity antigen-specific antibodies. Antibodies described in Tables 18-21 have the same light chain sequence of V κ 1-39JK5 (LCVR set forth in SEQ ID NO: 1386).

[0289] Generated antibodies were tested for affinity to human and cynomolgus monkey CD3 antigen in an in vitro binding assay, and e.g. one CD3 antibody: designated CD3-VH-P (HCVR set forth in SEQ ID NO: 1634) was identified, amongst a few others, that were found to bind to both human and cynomolgus CD3 having an EC₅₀ between 1 and 40 nM affinity, as determined in a FACS titration of Jurkat cells and cynomolgus T cells, respectively. See, e.g. FACS binding experiments outlined in Example 5.

[0290] The germline amino acid residues of CD3-VH-P were subsequently identified and an antibody designated "CD3-VH-G" was engineered to contain only germline frameworks. Other antibody derivatives were engineered by well-known molecular cloning techniques to replace amino acid residues in a stepwise manner based on differences between the germline sequence and the CD3-VH-P sequence. Each antibody derivative is given a "CD3-VH-G" number designation. See Table 18.

[0291] While CD3-VH-G and some other engineered antibodies retained their binding affinity as seen in the FACS assays, several anti-CD3 antibodies bound to human or cyno CD3 in vitro with weak to no measurable binding affinity, such as 40 nM EC₅₀. Binding affinities, binding kinetics, and other biological properties to elucidate toxicity and pharmacokinetic (pK) profiles were subsequently investigated for bispecific antibodies comprising the exemplary anti-CD3 antibodies generated in accordance with the methods of this Example, are described in detail in the Examples herein.

Example 12: Heavy and Light Chain Variable Regions (Amino Acid and Nucleic Acid Sequences of the CDRs)

[0292] Table 18 sets forth the amino acid sequence identifiers of the heavy chain variable regions and CDRs of selected anti-CD3 antibodies of the invention. The corresponding nucleic acid sequence identifiers are set forth in Table 19.

[0293] Amino acid and nucleic acid sequences were determined for each antibody heavy chain sequence. Each antibody heavy chain derived from the germline sequence (SEQ ID NO: 1662) was assigned a "G" number designation for consistent nomenclature. Table 2 sets forth the amino acid sequence identifiers of the heavy chain variable regions and CDRs of engineered anti-CD3 antibodies of the invention. The corresponding nucleic acid sequence identifiers are set forth in Table 19. The amino acid and nucleic acid sequence identifiers of the light chain variable region and CDRs are also identified below in Tables 20 and 21, respectively.

TABLE-US-00018 TABLE 18 Heavy Chain Amino Acid Sequence Identifiers Antibody CD3-VH
SEQ ID NOs: Designation HCVR CDR1 CDR2 CDR3 CD3-VH-G 1482 1484 1486 1488 CD3-
VH-G2 1490 1492 1494 1496 CD3-VH-G3 1498 1500 1502 1504 CD3-VH-G4 1506 1508 1510
1512 CD3-VH-G5 1514 1516 1518 1520 CD3-VH-G8 1522 1524 1526 1528 CD3-VH-G9 1530
1532 1534 1536 CD3-VH-G10 1538 1540 1542 1544 CD3-VH-G11 1546 1548 1550 1552 CD3-
VH-G12 1554 1556 1558 1560 CD3-VH-G13 1562 1564 1566 1568 CD3-VH-G14 1570 1572
1574 1576 CD3-VH-G15 1578 1580 1582 1584 CD3-VH-G16 1586 1588 1590 1592 CD3-VH-
G17 1594 1596 1598 1600 CD3-VH-G18 1602 1604 1606 1608 CD3-VH-G19 1610 1612 1614
1616 CD3-VH-G20 1618 1620 1622 1624 CD3-VH-G21 1626 1628 1630 1632 CD3-VH-P 1634

1636 1638 1640

TABLE-US-00019 TABLE 19 Heavy Chain Nucleic Acid Sequence Identifiers Antibody CD3-VH
SEQ ID NOs: Designation HCVR CDR1 CDR2 CDR3 CD3-VH-G 1481 1483 1485 1487 CD3-
VH-G2 1489 1491 1493 1495 CD3-VH-G3 1497 1499 1501 1503 CD3-VH-G4 1505 1507 1509
1511 CD3-VH-G5 1513 1515 1517 1519 CD3-VH-G8 1521 1523 1525 1527 CD3-VH-G9 1529
1531 1533 1535 CD3-VH-G10 1537 1539 1541 1543 CD3-VH-G11 1545 1547 1549 1551 CD3-
VH-G12 1553 1555 1557 1559 CD3-VH-G13 1561 1563 1565 1567 CD3-VH-G14 1569 1571
1573 1575 CD3-VH-G15 1577 1579 1581 1583 CD3-VH-G16 1585 1587 1589 1591 CD3-VH-
G17 1593 1595 1597 1599 CD3-VH-G18 1601 1603 1605 1607 CD3-VH-G19 1609 1611 1613
1615 CD3-VH-G20 1617 1619 1621 1623 CD3-VH-G21 1625 1627 1629 1631 CD3-VH-P 1633
1635 1637 1639

TABLE-US-00020 TABLE 20 Light Chain Amino Acid Sequence Identifiers Antibody SEQ ID
NOs: Designation LCVR CDR1 CDR2 CDR3 VK1-39JK5 1386 1388 1390 1392

TABLE-US-00021 TABLE 21 Light Chain Nucleic Acid Sequence Identifiers Antibody SEQ ID
NOs: Designation LCVR CDR1 CDR2 CDR3 VK1-39JK5 1385 1387 1389 1391

[0294] Control 1 antibody designated “CD3-L2K” was constructed based on a known anti-CD3
antibody (i.e., the anti-CD3 antibody “L2K” as set forth in WO2004/106380).

[0295] Isotype Control Antibody, referred to in the Examples herein, is an isotype matched
(modified IgG4) antibody that interacts with an irrelevant antigen, i.e. FelD1 antigen.

Example 13: Surface Plasmon Resonance Derived Binding Affinities and Kinetic Constants of
Human Monoclonal Anti-CD3 Antibodies

[0296] Binding affinities and kinetic constants of human monoclonal anti-CD3 antibodies were
determined by surface plasmon resonance at 25° C. using either an antibody-capture format (Tables
22, 24, and 26) or an antigen-capture format (Tables 23, 25, and 27). Measurements were
conducted on a T200 Biacore™ instrument.

[0297] In the antibody-capture format, the Biacore™ sensor surface was derivatized with a rabbit
anti-mouse Fc for hybridoma capture (antibody prefix H1M or H2M) or a mouse anti-human Fc
surface for human IgG formatted antibodies (antibody prefix H1H). Soluble heterodimeric CD3
protein (hCD3-epsilon/hCD3-delta; SEQ ID NOs: 1652/1653) with either a human Fc tag
(hFcΔAdp/hFc; SEQ ID NOs: 1683/1684) or a mouse Fc tag (mFcΔAdp/mFc; SEQ ID NOs:
1685/1686) was injected over the antibody captured surface and the binding response was recorded.
Heterodimeric CD3 protein was purified using the method described in Davis et al.
(US2010/0331527).

[0298] In the antigen-capture format, heterodimeric CD3 protein was captured using a rabbit anti-
mouse Fc or mouse anti-human Fc and the respective antibodies were injected over the captured
antigen.

[0299] Antibodies were analyzed in their conventional divalent format (Tables 22-25) or in a
monovalent 1-arm configuration (Tables 26-27) in which the second Fab from the antibody was
removed and only the Fc portion (CH2-CH3) was expressed.

[0300] Kinetic association (k.sub.a) and dissociation (k.sub.d) rate constants were determined by
processing and fitting the data to a 1:1 binding model using Scrubber 2.0 curve fitting software.
Binding dissociation equilibrium constants (K.sub.D) and dissociative half-lives (t.sub.1/2) were
calculated from the kinetic rate constants as: K.sub.D (M)=k.sub.d/k.sub.a; and t.sub.1/2 (min)=(ln
2/(60*k.sub.d). NT=not tested; NB=no binding observed.

TABLE-US-00022 TABLE 22 Biacore™ Binding Affinities of Hybridoma mAbs (H1M and H2M)
Binding at 25° C./Antibody-Capture Format Antibody ka (Ms.sup.-1) kd (s.sup.-1) K.sub.D
(Molar) T½ (min) H2M2689N 7.73E+05 3.23E-03 4.18E-09 4 H2M2690N 9.70E+03 2.02E-04
2.09E-08 57 H2M2691N 1.03E+04 2.07E-04 2.01E-08 56 H1M2692N 8.05E+03 4.34E-04
5.39E-08 27 H2M2704N 3.46E+04 6.92E-04 2.00E-08 17 H2M2705N 6.62E+04 9.10E-04
1.37E-08 13 H2M2706N 3.29E+04 4.44E-03 1.35E-07 3 H2M2707N 2.95E+04 1.87E-03

6.35E-08 6 H2M2708N 6.94E+04 6.12E-04 8.82E-09 19 H2M2709N NT NT NT NT
H2M2710N 6.72E+04 7.53E-04 1.12E-08 15 H2M2711N 6.72E+04 7.67E-04 1.14E-08 15
H1M2712N 9.32E+03 2.19E-04 2.35E-08 53 H2M2774N 7.79E+04 9.18E-04 1.18E-08 13
H2M2775N 6.97E+04 6.26E-04 8.98E-09 18 H2M2776N 6.29E+04 6.39E-04 1.02E-08 18
H2M2777N 3.70E+04 1.63E-03 4.39E-08 7 H2M2778N 2.13E+04 1.89E-04 8.90E-09 61
H2M2779N 2.18E+04 2.28E-04 1.05E-08 51 H2M2789N NT NT NT NT H2M2862N 3.72E+04
3.00E-03 8.07E-08 4 H2M2885N 6.82E+04 6.51E-04 9.54E-09 18 H2M2886N 7.29E+04
6.53E-04 8.96E-09 18 H2M3540N 3.77E+04 6.11E-04 1.62E-08 19 H2M3541N 7.10E+03
1.35E-03 1.89E-07 9 H1M3542N 2.37E+04 5.08E-04 2.14E-08 23 H2M3543N 7.53E+03
2.26E-04 3.00E-08 51 H1M3544N 9.69E+03 1.42E-04 1.46E-08 82 H2M3547N 2.18E+04
3.47E-04 1.59E-08 33 H2M3548N 3.87E+04 5.04E-03 1.30E-07 2 H1M3549N 1.18E+04
9.19E-04 7.76E-08 13 H2M3563N 3.24E+04 1.19E-04 3.66E-09 97 H1M3613N 1.93E+04
3.04E-04 1.57E-08 38

TABLE-US-00023 TABLE 23 Biacore™ Binding Affinities of Hybridoma mAbs (H1M and H2M)
Binding at 25° C./Antigen-Capture Format Antibody ka (Ms.sup.-1) kd (s.sup.-1) K.sub.D (Molar)
T½ (min) H2M2689N 1.71E+06 9.97E-05 5.83E-11 116 H2M2690N 7.51E+04 6.35E-06
7.99E-11 1820 H2M2691N 3.94E+04 9.98E-06 2.54E-10 1158 H1M2692N 4.19E+04 9.90E-06
2.38E-10 1167 H2M2704N 1.32E+06 2.48E-04 1.87E-10 47 H2M2705N 2.43E+06 3.41E-04
1.40E-10 34 H2M2706N 5.63E+05 3.06E-04 5.44E-10 38 H2M2707N 3.99E+05 2.85E-04
7.15E-10 41 H2M2708N 1.73E+06 2.27E-04 1.31E-10 51 H2M2709N NT NT NT NT
H2M2710N 1.59E+06 2.43E-04 1.53E-10 48 H2M2711N 1.59E+06 2.40E-04 1.51E-10 48
H1M2712N 4.75E+04 1.37E-05 2.95E-10 846 H2M2774N 2.49E+06 3.36E-04 1.35E-10 34
H2M2775N 1.56E+06 2.16E-04 1.38E-10 53 H2M2776N 1.58E+06 2.22E-04 1.40E-10 52
H2M2777N 5.80E+05 3.21E-04 5.54E-10 36 H2M2778N 1.50E+05 6.57E-06 4.68E-11 1758
H2M2779N 1.28E+05 1.23E-05 9.38E-11 941 H2M2789N NT NT NT NT H2M2862N 5.91E+05
3.21E-04 5.41E-10 36 H2M2885N 1.37E+06 1.52E-04 1.11E-10 76 H2M2886N 1.42E+06
1.36E-04 9.56E-11 85 H2M3540N 2.55E+06 5.87E-04 2.31E-10 20 H2M3541N 8.40E+04
1.16E-03 1.38E-08 10 H1M3542N 4.37E+05 2.00E-04 4.57E-10 58 H2M3543N 1.22E+05
7.96E-05 6.53E-10 145 H1M3544N 5.74E+04 5.98E-05 1.04E-09 193 H2M3547N 4.70E-05
1.00E-05 2.15E-11 1155 H2M3548N NT NT NT NT H1M3549N 2.81E+05 2.89E-04 1.03E-09
40 H2M3563N 6.16E+05 4.77E-05 7.73E-11 242 H1M3613N 2.20E+05 9.60E-05 4.35E-10 120

TABLE-US-00024 TABLE 24 Biacore™ Binding Affinities of Human Fc mAbs (H1H) Binding at
25° C./Antibody-Capture Format Antibody ka (Ms.sup.-1) kd (s.sup.-1) K.sub.D (Molar) T½
(min) H1H2690N NT NT NT NT H1H2712N 3.06E+03 2.70E-04 8.82E-08 43 H1H5751P
4.01E+03 5.18E-04 1.29E-07 22 H1H5752P NB NB NB NB H1H5753B NT NT NT NT
H1H5755B 8.21E+03 4.72E-04 5.75E-08 24 H1H5756B 8.15E+03 2.66E-04 3.26E-08 43
H1H5757B 6.63E+03 7.85E-04 1.18E-07 15 H1H5758B 5.02E+03 1.17E-03 2.33E-07 10
H1H5761P 4.72E+03 2.44E-02 5.16E-06 0 H1H5763P 1.85E+04 5.40E-02 2.92E-06 0
H1H5764P 4.16E+03 1.59E-02 3.82E-06 1 H1H5769P 7.80E+03 9.41E-04 1.21E-07 12
H1H5771P 3.00E+04 6.26E-04 2.09E-08 18 H1H5772S 1.56E+04 1.55E-03 9.96E-08 7
H1H5777P 1.35E+04 3.02E-03 2.24E-07 4 H1H5778P 5.52E+03 1.54E-04 2.78E-08 75
H1H5780P 1.31E+04 3.99E-04 3.04E-08 29 H1H5781P 8.61E+03 4.97E-04 5.77E-08 23
H1H5782P NB NB NB NB H1H5785B NT NT NT NT H1H5786B 1.26E+04 1.08E-03 8.54E-08
11 H1H5788P 2.88E+03 2.91E-04 1.01E-07 40 H1H5790B 1.82E+04 5.17E-04 2.83E-08 22
H1H5791B 1.09E+04 7.90E-04 7.25E-08 15 H1H5792B NT NT NT NT H1H5793B 8.54E+03
3.82E-04 4.47E-08 30 H1H5795B 1.73E+04 5.76E-04 3.33E-08 20 H1H5796B 1.47E+04
8.91E-04 6.05E-08 13 H1H5797B NT NT NT NT H1H5798B NT NT NT NT H1H5799P
1.36E+04 7.88E-03 5.79E-07 1 H1H5801B 6.57E+03 1.62E-03 2.46E-07 7 OKT3 2.10E+06
2.00E+00 1.00E-06 0.35 sec

TABLE-US-00025 TABLE 25 Biacore™ Binding Affinities of Human Fc mAbs (H1H) Binding at

25° C./Antigen-Capture Format Antibody ka (Ms.sup.-1) kd (s.sup.-1) K.sub.D (Molar) T½ (min)

H1H2690N	NT	NT	NT	NT	H1H2712N	8.93E+04	8.68E-05	9.71E-10	133	H1H5751P	7.24E+04
2.47E-04	3.42E-09	47	H1H5752P	NB	NB	NB	NB	H1H5753B	NT	NT	NT
NT	NT	NT	H1H5755B	2.15E+05	2.01E-04	9.36E-10	57	H1H5756B	1.44E+05	1.11E-04	7.67E-10
105	H1H5757B	1.80E+05	2.95E-04	1.64E-09	39	H1H5758B	1.42E+05	5.62E-04	3.97E-09	21	H1H5761P
2.11E+05	1.13E-02	5.34E-08	1	H1H5763P	1.84E+05	1.70E-02	9.24E-08	1	H1H5764P	3.50E+05	
7.36E-03	2.10E-08	2	H1H5769P	1.19E+05	5.23E-04	4.41E-09	22	H1H5771P	9.23E+05		
3.42E-04	3.71E-10	34	H1H5772S	5.19E+05	8.69E-04	1.67E-09	13	H1H5777P	4.83E+05		
1.70E-03	3.52E-09	7	H1H5778P	3.99E+05	3.42E-05	8.56E-11	338	H1H5780P	4.78E+05		
1.71E-04	3.58E-10	68	H1H5781P	1.40E+05	2.68E-04	1.92E-09	43	H1H5782P	NB	NB	NB
NB	NB	NB	H1H5785B	NT	NT	NT	NT	H1H5786B	3.00E+06	4.24E-04	1.41E-10
27	H1H5788P	7.06E+04	1.64E-04	2.33E-09	70	H1H5790B	9.25E+05	2.36E-04	2.54E-10	49	H1H5791B
7.86E+05	3.40E-04	4.33E-10	34	H1H5792B	NT	NT	NT	NT	H1H5793B	4.78E+05	1.59E-04
3.33E-10	73	H1H5795B	1.58E+06	2.29E-04	1.45E-10	50	H1H5796B	1.05E+05	2.44E-04	2.32E-09	47
H1H5797B	NT	NT	NT	NT	H1H5798B	NT	NT	NT	NT	H1H5799P	7.18E+05
5.64E-03	7.85E-09	2	H1H5801B	3.31E+05	1.12E-03	3.38E-09	10	OKT3	3.94E+06	2.18E-02	5.53E-09
0.5											

TABLE-US-00026 TABLE 26 Biacore™ Binding Affinities of monovalent 1-arm mAbs Binding at 25° C./Antibody-Capture Format Antibody ka (Ms.sup.-1) kd (s.sup.-1) K.sub.D (Molar) T½ (min)

H1H7194P	1.16E+04	1.51E-04	1.30E-08	76	H1H7195P	3.13E+04	9.89E-05	3.16E-09	117
H1H7196P	1.07E+04	4.43E-04	4.13E-08	26	H1H7198P	2.63E+04	1.58E-04	6.02E-09	73
H1H7203P	1.46E+04	2.67E-04	1.83E-08	43	H1H7204P	1.43E+04	3.62E-04	2.53E-08	32
H1H7208P	NT	NT	NT	NT	H1H7211P	1.41E+04	1.59E-04	1.13E-08	73
H1H7221P	1.07E+04	2.92E-04	2.75E-08	40	H1H7223P	1.60E+04	3.07E-04	1.92E-08	38
H1H7226P	1.30E+04	3.55E-04	2.72E-08	33	H1H7232P	8.03E+03	1.77E-03	2.20E-07	7
H1H7233P	1.11E+04	2.69E-04	2.42E-08	43	H1H7241P	1.34E+04	2.95E-04	2.20E-08	39
H1H7242P	2.15E+04	6.64E-04	3.09E-08	17	H1H7250P	2.34E+04	2.47E-04	1.05E-08	47
H1H7251P	2.56E+04	1.07E-03	4.17E-08	11	H1H7254P	2.60E+04	3.88E-04	1.49E-08	30
H1H7258P	1.26E+04	3.02E-04	2.40E-08	38	H1H7269P	2.57E+04	6.24E-03	2.43E-07	2
H1H7279P	NB	NB	NB	NB	H1xH7221G	NT	NT	NT	NT
H1xH7221G3	NB	NB	NB	NB	H1xH7221G5	NB	NB	NB	NB

TABLE-US-00027 TABLE 27 Biacore™ Binding Affinities of monovalent 1-arm mAbs Binding at 25° C./Antigen-Capture Format Antibody ka (Ms.sup.-1) kd (s.sup.-1) K.sub.D (Molar) T½ (min)

H1H7194P	3.50E+05	8.43E-05	2.41E-10	137	H1H7195P	5.66E+05	7.14E-05	1.26E-10	162
H1H7196P	1.85E+05	4.61E-04	2.49E-09	25	H1H7198P	6.28E+05	7.07E-05	1.12E-10	163
H1H7203P	4.79E+05	2.38E-04	4.98E-10	48	H1H7204P	1.73E+05	3.65E-04	2.12E-09	32
H1H7208P	NT	NT	NT	NT	H1H7211P	3.45E+05	9.61E-05	2.79E-10	120
H1H7221P	1.36E+05	2.39E-04	1.75E-09	48	H1H7223P	1.87E+05	2.86E-04	1.53E-09	40
H1H7226P	4.18E+05	2.36E-04	5.65E-10	49	H1H7232P	1.49E+05	1.49E-03	1.00E-08	8
H1H7233P	1.61E+05	2.04E-04	1.27E-09	57	H1H7241P	1.87E+05	2.36E-04	1.26E-09	49
H1H7242P	3.83E+05	1.01E-03	2.63E-09	11	H1H7250P	2.31E+05	1.89E-04	8.20E-10	61
H1H7251P	4.47E+05	1.19E-03	2.67E-09	10	H1H7254P	4.33E+05	3.30E-04	7.62E-10	35
H1H7258P	1.33E+05	2.90E-04	2.18E-09	40	H1H7269P	2.77E+05	6.89E-03	2.49E-08	2
H1H7279P	NB	NB	NB	NB	H1xH7221G	NT	NT	NT	NT
H1xH7221G3	NB	NB	NB	NB	H1xH7221G5	NB	NB	NB	NB

[0301] As shown in Tables 22-27, Several anti-CD3 antibodies of the present invention bind CD3, in either antibody-capture or antigen-capture formats, with high affinity.

Example 14: Anti-CD3 Antibodies Bind and Proliferate Human T-Cells

[0302] Anti-CD3 antibodies of the present invention were tested for their ability to bind to human T-cells and induce their proliferation. Binding was assessed using Jurkat cells (a CD3+human T-cell line), while proliferation of Peripheral Blood Mononuclear Cells (PBMC) was assessed using ATP catalyzed quantification (CellTiter Glo®). Anti-CD3 antibody OKT3 acted as a positive control and irrelevant isotype matched antibodies served as negative controls.

[0303] FACS data was acquired using the following protocol: Cells at 2×10^5 per well were incubated with serially diluted antibodies for 30 min on ice. Post incubation, cells were washed and secondary antibody was added and incubated for an additional 30 minutes. After incubation, cells were washed, re-suspended in cold PBS containing 1% BSA and analyzed by flow cytometry with viable Jurkat cells gated by side and forward scatters. The EC₅₀s for cell binding titration were determined using Prism software with values calculated using a 4-parameter non-linear regression analysis.

[0304] Proliferation data was acquired using the following protocol: Human PBMC (5×10^4 /well) were incubated with a 3-fold serial dilution of anti-CD3 and a fixed concentration of a commercial anti-CD28 antibody (200 ng/ml) in 96 well plates for 72 h at 37° C. Following incubation, CellTiter Glo® was added and luminescence was measured using a VICTOR X5 multi-label plate reader (PerkinElmer). The EC₅₀ of cell viability (ATP catalyzed quantification) was calculated using a 4-parameter non-linear regression analysis in GraphPad Prism.

[0305] Results of the binding and proliferation experiments are summarized in Tables 28-30.

TABLE-US-00028 TABLE 28 Hybridoma Anti-CD3 mAbs Bind & Proliferate Human T-Cells

EC ₅₀ [M]	EC ₅₀ [M]	hPBMC Antibody	FACS JURKAT Proliferation
H2M2689N	NB	0.00E+00	H2M2690N 4.37E-09 5.37E-12 H2M2691N 6.77E-09 3.43E-11 H1M2692N 5.99E-09 1.42E-10 H2M2704N 8.45E-10 2.93E-12 H2M2705N 2.96E-10 1.76E-11 H2M2706N 2.37E-09 3.86E-12 H2M2707N 1.24E-07 1.92E-12 H2M2708N 6.58E-10 2.69E-08 H2M2709N 7.11E-10 2.48E-11 H2M2710N 7.10E-10 2.11E-10 H2M2711N 1.16E-09 6.48E-10 H1M2712N 2.19E-08 1.28E-10 H2M2774N 3.52E-10 4.92E-10 H2M2775N 1.32E-09 1.09E-09 H2M2776N 4.91E-10 2.84E-11 H2M2777N 2.16E-09 2.51E-11 H2M2778N 3.62E-09 0.00E+00 H2M2779N NT 0.00E+00 H2M2789N NT 2.85E-08 H2M2862N 7.68E-09 6.72E-13 H2M2885N 2.09E-09 2.49E-12 H2M2886N 3.97E-09 2.69E-12 H2M3540N 3.99E-09 3.16E-12 H2M3541N 3.70E-09 6.40E-12 H1M3542N 2.01E-09 0.00E+00 H2M3543N 5.63E-09 6.12E-12 H1M3544N 2.32E-08 0.00E+00 H2M3547N 2.71E-09 5.02E-12 H2M3548N 1.10E-09 1.89E-12 H1M3549N 2.30E-09 0.00E+00 H2M3563N 1.07E-09 7.74E-12 H1M3613N 1.03E-08 0.00E+00

Isotype Ctrl NB 0.00E+00 NB: No Binding; NT: Not Tested

TABLE-US-00029 TABLE 29 Human Fc Anti-CD3 mAbs Bind & Proliferate Human T-Cells

EC ₅₀ [M]	EC ₅₀ [M]	hPBMC Antibody	FACS JURKAT Proliferation
H1H5751P	2.12E-09 9.29E-12	H1H5752P 3.43E-10 1.09E-12	H1H5753B NB 9.14E-11 H1H5755B 1.23E-09 4.24E-12 H1H5756B NB 0.00E+00 H1H5757B 3.38E-09 4.86E-12 H1H5758B 1.90E-09 2.13E-12 H1H5761P 2.10E-09 3.62E-13 H1H5763P 2.76E-09 3.11E-13 H1H5764P 8.80E-10 3.27E-13 H1H5769P 4.10E-09 6.17E-12 H1H5771P NT 6.35E-12 H1H5772S 6.64E-10 4.42E-12 H1H5777P 5.71E-10 3.04E-12 H1H5778P 6.85E-10 5.04E-12 H1H5780P 7.62E-10 3.44E-12 H1H5781P 1.23E-09 6.08E-12 H1H5782P NB 5.17E-12 H1H5785B NB 0.00E+00 H1H5786B 1.10E-09 1.79E-12 H1H5788P 3.53E-09 4.62E-12 H1H5790B 3.55E-09 2.71E-12 H1H5791B 3.77E-09 1.75E-12 H1H5792B 5.87E-09 6.47E-12 H1H5793B 4.62E-09 3.28E-12 H1H5795B 2.04E-09 3.09E-12 H1H5796B 9.82E-09 4.37E-12 H1H5797B 3.96E-08 1.07E-11 H1H5798B 5.57E-09 2.59E-12 H1H5799P NT 1.63E-13 H1H5801B 1.55E-08 1.09E-12 OKT3 1.96E-10 3.30E-13

Isotype Ctrl NB 0.00E+00 NB: No Binding; NT: Not Tested

TABLE-US-00030 TABLE 30 Monovalent 1-arm Anti-CD3 mAbs Bind & Proliferate Human T-Cells

EC ₅₀ [M]	EC ₅₀ [M]	hPBMC Antibody	FACS JURKAT Proliferation
H1H7194P	1.50E-09 2.37E-12	H1H7195P 3.42E-10 2.42E-12	H1H7196P 3.44E-08 1.27E-12 H1H7198P 7.26E-10 2.55E-12 H1H7203P 3.24E-09 1.64E-12 H1H7204P 2.29E-09 1.51E-12 H1H7208P 5.19E-08 1.46E-12 H1H7211P 7.01E-10 2.75E-12 H1H7221P 1.40E-09 2.60E-12 H1H7223P 9.37E-10 1.07E-12 H1H7226P 7.95E-10 9.52E-13 H1H7232P 1.50E-09 1.03E-12 H1H7233P 7.15E-10 7.34E-13 H1H7241P 1.01E-09 1.05E-12 H1H7242P 1.83E-09 2.13E-12 H1H7250P

1.37E-09 2.43E-12 H1H7251P 1.45E-09 1.30E-12 H1H7254P 1.09E-09 2.80E-12 H1H7258P
1.07E-09 2.17E-12 H1H7269P 1.95E-09 1.15E-12 H1H7279P NB 0.00E+00 Isotype Ctrl NB
0.00E+00 NB: No Binding; NT: Not Tested

[0306] As shown in Tables 7-9, the vast majority of anti-CD3 antibodies of the invention bound human T-cells and induced T-cell proliferation.

Example 15: Anti-CD3 Antibodies Bind and Proliferate Monkey T-Cells

[0307] A subset of anti-CD3 antibodies of the invention was tested for the ability to bind to and induce proliferation of monkey T-cells.

[0308] FACS data was acquired using the following protocol: Cells at 2×10^5 per well were incubated with serially diluted antibodies for 30 min on ice. Post incubation, cells were washed and secondary antibodies were added and incubated for an additional 30 minutes. After incubation, cells were washed, re-suspended in cold PBS containing 1% BSA and analyzed by flow cytometry. CD4⁺ monkey T cells were gated by side and forward scatters, and on the CD2⁺CD4⁺CD20⁻ population. The EC₅₀s for cell binding titration were calculated using a 4-parameter non-linear regression analysis in GraphPad Prism.

[0309] Proliferation data was acquired using the following protocol: Freshly isolated cynomolgus monkey derived PBMC (5×10^4 /well) were incubated with a 3-fold serial dilution of anti-CD3 antibody and a fixed concentration of a commercial anti-CD28 antibody (500 ng/ml) antibody in 96 well plates for 72 h at 37° C. Following incubation, CellTiter Glo® was added and luminescence was measured using a VICTOR X5 multi-label plate reader (PerkinElmer). The EC₅₀ of cell viability (ATP catalyzed quantification) was calculated using a 4-parameter non-linear regression analysis in GraphPad Prism.

[0310] Results of the binding and proliferation experiments are summarized in Tables 31 and 32.

TABLE-US-00031 TABLE 31 Anti-CD3 mAbs Bind & Proliferate monkey PBMCs EC₅₀ [M]

EC ₅₀ [M]	mfPBMC Antibody	FACS PBMCs Proliferation
5.66E-09	H1H2690N	2.71E-12
2.29E-09	H1H2712N	2.72E-12
1.12E-10	H2M3547N	NT
1.65E-10	H2M3563N	NT
2.81E-09	H1H5761P	NT
0.00E+00	H1H5763P	NT
4.06E-10	H1H5764P	NT
8.33E-13	H1H5769P	NT
2.74E-12	H1H5771P	NT
1.47E-12	H1H5772S	NT
5.93E-13	H1H5778P	NT
3.13E-13	H1H5780P	NT
7.92E-13	H1H5781P	NT
2.01E-12	H1H5788P	NT
7.03E-11	OKT3 NB	NT
1.71E-12	SP34	7.03E-11

NB: No Binding; NT: not tested

TABLE-US-00032 TABLE 32 Monovalent 1-arm Anti-CD3 mAbs Bind & Proliferate Monkey

EC ₅₀ [M]	EC ₅₀ [M]	mfPBMC Antibody	FACS PBMCs Proliferation
4.84E-12	NT	H1H7194P	NT
1.36E-12	NT	H1H7195P	NT
1.40E-08	NT	H1H7196P	NT
2.29E-12	NT	H1H7198P	NT
4.97E-13	NT	H1H7203P	NT
1.26E-11	NT	H1H7204P	NT
7.02E-12	NT	H1H7208P	NT
2.81E-13	NT	H1H7211P	NT
1.72E-12	NT	H1H7221P	NT
6.75E-11	NT	H1H7223P	NT
2.26E-11	NT	H1H7226P	NT
4.90E-11	NT	H1H7232P	NT
4.35E-12	NT	H1H7233P	NT
2.05E-11	NT	H1H7241P	NT
1.38E-11	NT	H1H7242P	NT
7.27E-11	NT	H1H7250P	NT
1.83E-11	NT	H1H7251P	NT
8.88E-11	NT	H1H7254P	NT
1.11E-11	NT	H1H7258P	NT

NB: No Binding; NT: not tested

[0311] As shown in Tables 31 and 32, several anti-CD3 antibodies of the invention bound CD2⁺CD4⁺ monkey T-cells and induced their proliferation. OKT3 did not drive monkey PBMC proliferation, while SP34 was active against monkey PBMCs.

Example 16: Anti-CD3 mAbs Support T-Cell-Mediated Killing of Tumor Cells

[0312] The ability of anti-CD3 antibodies to redirect T-cell mediated killing via Fc/FcR interactions was studied using a calcein based U937 killing assay. Briefly, human PBMC were isolated over FICOLL-PAQUE™ and activated over a course of several days with media containing human IL-2 (30 U/ml) and T-cell activation beads (anti-CD3/CD28). U937 cells were labeled with calcein, and then incubated with activated T-cells at a 10:1 effector: target ratio using 3-fold serial dilutions of antibodies over a course of 3 hours at 37° C. Following incubation, the plates were centrifuged and supernatants were transferred to a translucent black clear bottom plate for fluorescence analysis. EC₅₀ values, defined as the molar concentration of CD3 antibody that induces 50%

cytotoxicity, were calculated using a 4-parameter non-linear regression analysis in GraphPad Prism. Results using hybridoma antibodies, human Fc antibodies, and monovalent one-arm antibodies are shown in Tables 33, 34 and 35, respectively.

TABLE-US-00033 TABLE 33 Hybridoma Anti-CD3 mAbs Redirect T-Cell Killing to U937 Cells

U937 Cytotoxicity Antibody Human T-cells [M] H2M2689N 0.00E+00 H2M2690N 2.79E-11
H2M2691N 2.34E-11 H1M2692N 3.59E-10 H2M2704N 2.49E-12 H2M2705N 1.73E-12
H2M2706N 7.91E-12 H2M2707N 7.21E-12 H2M2708N 3.27E-12 H2M2709N 3.47E-12
H2M2710N 3.97E-12 H2M2711N 3.66E-12 H1M2712N 3.14E-10 H2M2774N 2.46E-12
H2M2775N 3.38E-12 H2M2776N 4.06E-12 H2M2777N 4.86E-12 H2M2778N 0.00E+00
H2M2779N 6.75E-10 H2M2789N NT H2M2862N 7.66E-12 H2M2885N 3.71E-12 H2M2886N
8.06E-12 H2M3540N 1.25E-11 H2M3541N 5.39E-11 H1M3542N 2.92E-11 H2M3543N
1.31E-11 H1M3544N 1.72E-10 H2M3547N 3.17E-11 H2M3548N 5.50E-12 H1M3549N
1.07E-10 H2M3563N 4.05E-11 H1M3613N 8.66E-10 Isotype Ctrl 0.00E+00 NT: Not Tested

TABLE-US-00034 TABLE 34 Human Fc formatted Anti-CD3 mAbs Redirect T-Cell Killing to

U937 Cells U937 Cytotoxicity Antibody Human T-cells [M] H1H5751P 1.30E-10 H1H5752P
1.85E-11 H1H5753B 3.79E-10 H1H5755B 5.16E-11 H1H5756B 7.69E-11 H1H5757B 9.65E-11
H1H5758B 8.86E-08 H1H5761P 2.00E-12 H1H5763P NT H1H5764P NT H1H5769P 5.65E-11
H1H5771P NT H1H5772S 6.89E-13 H1H5777P 4.87E-13 H1H5778P 3.41E-13 H1H5780P
4.03E-12 H1H5781P 1.83E-12 H1H5782P 5.18E-12 H1H5785B 4.43E-11 H1H5786B 6.10E-11
H1H5788P 1.54E-11 H1H5790B 8.71E-11 H1H5791B 8.01E-11 H1H5792B 1.40E-10
H1H5793B 8.85E-11 H1H5795B 6.74E-11 H1H5796B 5.03E-10 H1H5797B 5.76E-10
H1H5798B 1.81E-10 H1H5799P NT H1H5801B 9.23E-11 OKT3 2.35E-12 Isotype Ctrl
0.00E+00 NT: Not Tested

TABLE-US-00035 TABLE 35 Monovalent 1-arm Anti-CD3 mAb Redirect T-Cell Killing to U937

Cells U937 Cytotoxicity Antibody Human T-cells [M] H1H7194P 4.71E-12 H1H7195P 6.10E-12
H1H7196P 1.96E-11 H1H7198P 5.21E-12 H1H7203P 5.47E-12 H1H7204P 1.08E-11
H1H7208P 4.59E-11 H1H7211P 7.89E-12 H1H7221P 9.21E-12 H1H7223P 5.30E-12
H1H7226P 1.04E-11 H1H7232P 9.96E-12 H1H7233P 1.19E-11 H1H7241P 1.23E-11 H1H7242P
7.50E-12 H1H7250P 5.91E-12 H1H7251P 1.81E-12 H1H7254P 4.18E-12 H1H7258P 1.53E-11
H1H7269P 1.08E-11 H1H7279P 0.00E+00 Isotype Ctrl 0.00E+00 NT: Not Tested

[0313] As shown in Tables 33-35, most anti-CD3 antibodies, as well as OKT3, supported redirected T-cell mediated killing in this assay system. The observed killing, believed to be dependent on the antibody's Fc engagement with the Fc Receptor on U937 cells leading to clustering of CD3 on adjacent T-cells, was squelched by addition of non-specific human IgG (data not shown).

[0314] The present invention is not to be limited in scope by the specific embodiments described herein. Indeed, various modifications of the invention in addition to those described herein will become apparent to those skilled in the art from the foregoing description. Such modifications are intended to fall within the scope of the appended claims.

TABLE-US-00036 TABLE 36 Sequences Excluded from ST.26-Formatted Sequence

Listing SEQ ID NO: Sequence 133 ggtgcatcc 134 GAS 149 actgcatcc 150 TAS 165 ggtgcatcc
166 GAS 181 ggtgcatcc 182 GAS 197 gctgcatcc 198 AAS 213 ggtgcatcc 214 GAS 229 gctgcatcc
230 AAS 245 gctgcatcc 246 AAS 261 gatgcatcc 262 DAS 277 ggtgcatcc 278 GAS 293 ggtgcatcc
294 GAS 309 ggtgcatcc 310 GAS 325 ggtgcatcc 326 GAS 341 ggtgcatcc 342 GAS 357 ggtgcatcc
358 GAS 373 ggtgcatcc 374 GAS 389 ggtgcatcc 390 GAS 405 ggtgcatcc 406 GAS 421 ggtgcatcc
422 GAS 437 ggtgcatcc 438 GAS 453 ggtgcaacc 454 GAT 469 ggtgcaacc 470 GAT 485 ggtgcatcc
486 GAS 501 ggtgcatcc 502 GAS 517 tgggcatct 518 WAS 533 gctgcatcc 534 AAS 549 ggtgcatcc
550 GAS 565 ggtgcaacc 566 GAT 581 ggtgcaacc 582 GAT 597 gctgcatcc 598 AAS 613 gctgcatcc
614 AAS 629 gctgcatcc 630 AAS 645 gctgcatcc 646 AAS 661 gctgcatcc 662 AAS 677 gctgcatcc
678 AAS 693 gctgcttcc 694 AAS 709 gaagcttct 710 EAS 725 gctgtatcc 726 AVS 741 gatgcatcc 742
DAS 757 gctgcatcc 758 AAS 773 gctgcatcc 774 AAS 789 gctgcatcc 790 AAS 805 gctgcatcc 806

AAS 821 gctgcatcc 822 AAS 837 gctgcatcc 838 AAS 853 gctgcatcc 854 AAS 869 ggtgcatcc 870
GAS 885 ggtgcatcc 886 GAS 901 ggtgcatcc 902 GAS 917 ggtgcatcc 918 GAS 933 ggtgcatcc 934
GAS 949 ggtgcatcc 950 GAS 965 ggtgcatcc 966 GAS 981 aaggcatct 982 KAS 997 tgggcatct 998
WAS 1013 actgcatcc 1014 TAS 1029 ggtgcatcc 1030 GAS 1045 actgcatcc 1046 TAS 1061
actgcatcc 1062 TAS 1077 actgcatcc 1078 TAS 1093 gttgcatcc 1094 VAS 1109 actgcatcc 1110 TAS
1125 gctgcatcc 1126 AAS 1141 gttgcatcc 1142 VAS 1157 actgcatcc 1158 TAS 1173 actgcatcc 1174
TAS 1189 gttgcatcc 1190 VAS 1389 gctgcatcc 1390 AAS 1405 ggtgcatcc 1406 GAS 1421
ggtgcatcc 1422 GAS 1437 ggtgcatcc 1438 GAS 1453 ggtgcatcc 1454 GAS 1469 ggtgcatcc 1470
GAS 1479 DTS 1645 actgcatcc 1646 TAS

Claims

1. An isolated antibody or antigen-binding fragment thereof that binds human prostate-specific membrane antigen (PSMA) with a binding dissociation equilibrium constant ($K_{sub.D}$) value of less than about 80 nM as measured in a surface plasmon resonance assay at 37° C.
2. An isolated antibody or antigen-binding fragment thereof that binds human PSMA with a dissociative half-life ($t_{1/2}$) of greater than about 10 minutes as measured in a surface plasmon resonance assay at 37° C.
3. The antibody or antigen-binding fragment of any one of claims 1 to 2, wherein the antibody or antigen-binding fragment thereof competes for binding to human PSMA with a reference antibody comprising an HCVR/LCVR amino acid sequence pair as set forth in Table 1.
4. The antibody or antigen-binding fragment of claim 3, wherein the reference antibody comprises an HCVR/LCVR amino acid sequence pair selected from the group consisting of SEQ ID NOs: 2/1642; 10/1642; 18/1642; 26/1642; 34/1642; 42/1642; 50/1642; 58/1642; 66/1642; 74/1642; 82/1642; 90/1642; 98/1642; 106/1642; 114/1642; 122/130; and 138/146.
5. The antibody or antigen-binding fragment of any one of claims 1 to 4, wherein the antibody or antigen-binding fragment thereof binds to the same epitope on human PSMA as a reference antibody comprising an HCVR/LCVR amino acid sequence pair as set forth in Table 1.
6. The antibody or antigen-binding fragment of claim 5, wherein the antibody or antigen-binding fragment thereof binds to the same epitope on human PSMA as a reference antibody comprising an HCVR/LCVR amino acid sequence pair selected from the group consisting of SEQ ID NOs: 2/1642; 10/1642; 18/1642; 26/1642; 34/1642; 42/1642; 50/1642; 58/1642; 66/1642; 74/1642; 82/1642; 90/1642; 98/1642; 106/1642; 114/1642; 122/130; and 138/146.
7. An isolated antibody or antigen-binding fragment thereof that binds human PSMA, wherein the antibody or antigen-binding fragment comprises: (a) the complementarity determining regions (CDRs) of a heavy chain variable region (HCVR) having an amino acid sequence as set forth in Table 1; and (b) the CDRs of a light chain variable region (LCVR) having an amino acid sequence as set forth in Table 1.
8. The isolated antibody or antigen-binding fragment of claim 7, wherein the antibody or antigen-binding fragment comprises the heavy and light chain CDRs of a HCVR/LCVR amino acid sequence pair selected from the group consisting of: SEQ ID NOs: 2/1642; 10/1642; 18/1642; 26/1642; 34/1642; 42/1642; 50/1642; 58/1642; 66/1642; 74/1642; 82/1642; 90/1642; 98/1642; 106/1642; 114/1642; 122/130; and 138/146.
9. The isolated antibody or antigen-binding fragment of claim 8, wherein the antibody or antigen-binding fragment comprises HCDR1-HCDR2-HCDR3-LCDR1-LCDR2-LCDR3 domains, respectively, selected from the group consisting of: SEQ ID NOs: 4-6-8-1644-1646-1648; 12-14-16-1644-1646-1648; 20-22-24-1644-1646-1648; 28-30-32-1644-1646-1648; 36-38-40-1644-1646-1648; 44-46-48-1644-1646-1648; 52-54-56-1644-1646-1648; 60-62-64-1644-1646-1648; 68-70-72-1644-1646-1648; 76-78-80-1644-1646-1648; 84-86-88-1644-1646-1648; 92-94-96-1644-1646-1648; 100-102-104-1644-1646-1648; 108-110-112-1644-1646-1648; 116-118-120-1644-1646-

1648; 124-126-128-132-134-136; and 140-142-144-148-150-152.

10. An isolated antibody or antigen-binding fragment thereof that binds human PSMA, wherein the antibody or antigen-binding fragment comprises: (a) a heavy chain variable region (HCVR) having an amino acid sequence selected from the group consisting of SEQ ID NOs: 2, 10, 18, 26, 34, 42, 50, 58, 66, 74, 82, 90, 98, 106, 114, 122, and 138; and (b) a light chain variable region (LCVR) having an amino acid sequence selected from the group consisting of SEQ ID NOs: 130 and 146.

11. The isolated antibody or antigen-binding fragment of claim 10, wherein the antibody or antigen-binding fragment comprises a HCVR/LCVR amino acid sequence pair selected from the group consisting of: SEQ ID NOs: 2/1642; 10/1642; 18/1642; 26/1642; 34/1642; 42/1642; 50/1642; 58/1642; 66/1642; 74/1642; 82/1642; 90/1642; 98/1642; 106/1642; 114/1642; 122/130; and 138/146.

12. A bispecific antigen-binding molecule comprising a first antigen-binding domain that binds human CD3 and a second antigen-binding domain that binds human PSMA, wherein the second antigen-binding domain is derived from the antibody or antigen-binding fragment of any one of claims 1-11.

13. A bispecific antigen-binding molecule comprising a first antigen-binding domain that specifically binds human CD3, and a second antigen-binding domain that specifically binds human PSMA.

14. The bispecific antigen-binding molecule of claim 12 or claim 13, wherein the first antigen-binding domain binds human cells expressing human CD3 and cynomolgus monkey cells expressing cynomolgus CD3.

15. The bispecific antigen-binding molecule of claim 12 or claim 13, wherein the second antigen-binding domain binds human cells expressing human PSMA and cynomolgus monkey cells expressing cynomolgus PSMA.

16. The bispecific antigen-binding molecule of claim 12 or claim 13, wherein the antigen-binding molecule binds both human CD3 and human PSMA and induces T cell-mediated cell killing of PSMA-expressing cells.

17. The bispecific antigen-binding molecule of claim 12 or claim 13 wherein the antigen-binding molecule inhibits tumor growth in immunocompromised mice bearing human prostate cancer xenografts.

18. The bispecific antigen-binding molecule of claim 12 or claim 13, wherein the antigen-binding molecule inhibits tumor growth in immunocompetent mice bearing human prostate cancer xenografts.

19. The bispecific antigen-binding molecule of claim 12 or claim 13, wherein the antigen-binding molecule suppresses tumor growth of established tumors in immunocompromised mice bearing human prostate cancer xenografts.

20. The bispecific antigen-binding molecule of claim 12 or claim 13, wherein the antigen-binding molecule reduces tumor growth of established tumors in immunocompetent mice bearing human prostate cancer xenografts.

21. The bispecific antigen-binding molecule of any one of claims 12-20, wherein the antigen-binding molecule induces T cell-mediated tumor cell killing with an EC₅₀ value of less than about 1.3 nM, as measured in an in vitro T cell-mediated tumor cell killing assay.

22. A bispecific antigen-binding molecule of any one of claims 12-21, wherein the second antigen-binding domain specifically binds human PSMA with an K_D value of less than about 80 nM, as measured in an in vitro surface plasmon resonance binding assay.

23. The bispecific antigen-binding molecule of any one of claims 12-22, wherein the second antigen-binding domain specifically binds each of human PSMA with an K_D value of less than about 5 nM, less than about 2 nM, less than about 1 nM, less than about 800 pM, or less than about 600 pM, as measured in an in vitro surface plasmon resonance binding assay.

24. The bispecific antigen-binding molecule of any one of claims 12-23 that is a bispecific antibody

or bispecific antigen-binding fragment thereof.

25. The bispecific antigen-binding molecule of any one of claims 12-24, wherein the second antigen-binding domain that specifically binds human PSMA comprises the heavy chain complementarity determining regions (HCDR1, HCDR2 and HCDR3) from a heavy chain variable region (HCVR) selected from the group consisting of SEQ ID NOs: 2, 10, 18, 26, 34, 42, 50, 58, 66, 74, 82, 90, 98, 106, 114, 122, and 138; and the light chain complementarity determining regions (LCDR1, LCDR2 and LCDR3) from a light chain variable region (LCVR) comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 162, 930 and 1642.

26. The bispecific antigen-binding molecule of any one of claims 12-24, wherein the second antigen-binding domain that specifically binds human PSMA comprises three heavy chain complementarity determining regions (A2-HCDR1, A2-HCDR2 and A2-HCDR3) and three light chain complementarity determining regions (A2-LCDR1, A2-LCDR2 and A2-LCDR3), wherein A2-HCDR1 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 4, 12, 20, 28, 36, 44, 52, 60, 68, 76, 84, 92, 100, 108, 116, 124, and 140; A2-HCDR2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 6, 14, 22, 30, 38, 46, 54, 62, 70, 78, 86, 94, 102, 110, 118, 126, and 142; A2-HCDR3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 8, 16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104, 112, 120, 128, and 144; A2-LCDR1 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 164, 932 and 1644; A2-LCDR2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 166, 934 and 1646; and A2-LCDR3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 168, 936 and 1648.

27. The bispecific antigen-binding molecule of any one of claims 12-24, wherein the second antigen-binding domain that specifically binds human PSMA comprises the heavy and light chain CDRs of a HCVR/LCVR amino acid sequence pair selected from the group consisting of: SEQ ID NOs: 122/162, 122/930 and 66/1642.

28. The bispecific antigen-binding molecule of any one of claims 12-24, wherein the first antigen-binding domain that specifically binds human CD3 comprises heavy chain complementarity determining regions (HCDR1, HCDR2 and HCDR3) from a heavy chain variable region (HCVR) comprising an amino acid sequence as set forth in Table 12, Table 14, or Table 18 and light chain complementarity determining regions (LCDR1, LCDR2 and LCDR3) from a light chain variable region (LCVR) comprising an amino acid sequence as set forth in Table 12, Table 15, or Table 20.

29. The bispecific antigen-binding molecule of any one of claims 12-24, wherein the first antigen-binding domain that specifically binds human CD3 comprises heavy chain complementarity determining regions (HCDR1, HCDR2 and HCDR3) from a heavy chain variable region (HCVR) selected from the group consisting of SEQ ID NOs: 922, 154, 1482, 1490, 1498, 1506, 1514, 1522, 1530, 1538, 1546, 1554, 1562, 1570, 1578, 1586, 1594, 1602, 1610, 1618, 1626, and 1634, and light chain complementarity determining regions (LCDR1, LCDR2 and LCDR3) from a light chain variable region (LCVR) comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 162, 930 and 1642.

30. The bispecific antigen-binding molecule of any one of claims 12-24, wherein the first antigen-binding domain that specifically binds human CD3 comprises three heavy chain complementarity determining regions (A1-HCDR1, A1-HCDR2 and A1-HCDR3) and three light chain complementarity determining regions (A1-LCDR1, A1-LCDR2 and A1-LCDR3), wherein A1-HCDR1 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 924, 156; 1484, 1492, 1500, 1508, 1516, 1524, 1532, 1540, 1548, 1556, 1564, 1572, 1580, 1588, 1596, 1604, 1612, 1620, 1628, and 1636; A1-HCDR2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 926, 158, 1486, 1494, 1502, 1510, 1518, 1526, 1534, 1542, 1550, 1558, 1566, 1574, 1582, 1590, 1598, 1606, 1614, 1622, 1630, and 1638; A1-HCDR3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 928, 160,

1488, 1496, 1504, 1512, 1520, 1528, 1536, 1544, 1552, 1560, 1568, 1576, 1584, 1592, 1600, 1608, 1616, 1624, 1632, and 1640; A1-LCDR1 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 932, 164, and 1644; A1-LCDR2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 166, 934 and 1646; and A1-LCDR3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 168, 936 and 1648.

31. The bispecific antigen-binding molecule of any one of claims 12-24, wherein the first antigen-binding domain that specifically binds human CD3 comprises the heavy and light chain CDRs of a HCVR/LCVR amino acid sequence pair selected from the group consisting of: SEQ ID NOs: 922/930, 154/162, 1482/1642, 1490/1642, 1498/1642, 1506/1642, 1514/1642, 1522/1642, 1530/1642, 1538/1642, 1546/1642, 1554/1642, 1562/1642, 1570/1642, 1578/1642, 1586/1642, 1594/1642, 1602/1642, 1610/1642, 1618/1642, 1626/1642, and 1634/1642.

32. The bispecific antigen-binding molecule of any one of claims 12-24, wherein the first antigen-binding domain that specifically binds human CD3 comprises three heavy chain complementarity determining regions (A1-HCDR1, A1-HCDR2 and A1-HCDR3) and three light chain complementarity determining regions (A1-LCDR1, A1-LCDR2 and A1-LCDR3), and wherein the second antigen-binding domain that specifically binds human PSMA comprises three heavy chain complementarity determining regions (A2-HCDR1, A2-HCDR2 and A2-HCDR3) and three light chain complementarity determining regions (A2-LCDR1, A2-LCDR2 and A2-LCDR3); wherein A1-HCDR1 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 924, 156, 1484, 1492, 1500, 1508, 1516, 1524, 1532, 1540, 1548, 1556, 1564, 1572, 1580, 1588, 1596, 1604, 1612, 1620, 1628, and 1636; A1-HCDR2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 926, 158, 1486, 1494, 1502, 1510, 1518, 1526, 1534, 1542, 1550, 1558, 1566, 1574, 1582, 1590, 1598, 1606, 1614, 1622, 1630, and 1638; A1-HCDR3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 928, 160, 1488, 1496, 1504, 1512, 1520, 1528, 1536, 1544, 1552, 1560, 1568, 1576, 1584, 1592, 1600, 1608, 1616, 1624, 1632, and 1640; A1-LCDR1 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 164, 932, and 1644; A1-LCDR2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 166, 934, and 1646; and A1-LCDR3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 168, 936, and 1648; and wherein A2-HCDR1 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 124 and 68; A2-HCDR2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 126 and 70; A2-HCDR3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 128 and 72; A2-LCDR1 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 932, 164, and 1644; A2-LCDR2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 934, 166, and 1646; and A2-LCDR3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 936, 168, and 1648.

33. The bispecific antigen-binding molecule of any one of claims 12-24, wherein the first antigen-binding domain that specifically binds human CD3 comprises a heavy chain comprising variable domain framework regions having an amino acid sequence selected from FR1 (SEQ ID NO: 1655), FR2 (SEQ ID NO: 1656), FR3 (SEQ ID NO: 1657), and FR4 (SEQ ID NO: 1658).

34. The bispecific antigen-binding molecule of any one of claims 12-24, wherein the first antigen-binding domain that specifically binds human CD3 comprises a HCVR comprising HCDR1-HCDR2-HCDR3 having the amino acid sequences of SEQ ID NOs: 1659-1660-1661.

35. The bispecific antigen-binding molecule of any one of claims 12-24, wherein the second antigen-binding domain competes for binding to human PSMA with a reference antigen-binding protein comprising three heavy chain complementarity determining regions (A2-HCDR1, A2-HCDR2 and A2-HCDR3) and three light chain complementarity determining regions (A2-LCDR1, A2-LCDR2 and A2-LCDR3), wherein A2-HCDR1 comprises an amino acid sequence selected

from the group consisting of SEQ ID NOs: 124 and 68; A2-HCDR2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 126 and 70; A2-HCDR3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 128 and 72; A2-LCDR1 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 164, 932 and 1644; A2-LCDR2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 166, 934 and 1646; and A2-LCDR3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 168, 936, and 1648.

36. The bispecific antigen-binding molecule of any one of claims 12-24, wherein the second antigen-binding domain competes for binding to human PSMA with a reference antigen-binding protein comprising a heavy chain variable region (HCVR) comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 122 and 66, and a light chain variable region (LCVR) comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 162, 930 and 1642.

37. The bispecific antigen-binding molecule of any one of claims 12-24 wherein the first antigen-binding domain competes for binding to human CD3 with a reference antigen-binding protein comprising three heavy chain complementarity determining regions (A1-HCDR1, A1-HCDR2 and A1-HCDR3) and three light chain complementarity determining regions (A1-LCDR1, A1-LCDR2 and A1-LCDR3), wherein A1-HCDR1 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 924, 156, 1484, 1492, 1500, 1508, 1516, 1524, 1532, 1540, 1548, 1556, 1564, 1572, 1580, 1588, 1596, 1604, 1612, 1620, 1628, and 1636; A1-HCDR2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 926, 158, 1486, 1494, 1502, 1510, 1518, 1526, 1534, 1542, 1550, 1558, 1566, 1574, 1582, 1590, 1598, 1606, 1614, 1622, 1630, and 1638; A1-HCDR3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 928, 160, 1488, 1496, 1504, 1512, 1520, 1528, 1536, 1544, 1552, 1560, 1568, 1576, 1584, 1592, 1600, 1608, 1616, 1624, 1632, and 1640; A1-LCDR1 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 164, 932, and 1644; A1-LCDR2 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 166, 934, and 1646; and A1-LCDR3 comprises an amino acid sequence selected from the group consisting of SEQ ID NOs: 168, 936, and 1648.

38. The bispecific antigen-binding molecule of any one of claims 12-24, wherein the first antigen-binding domain competes for binding to human CD3 with a reference antigen-binding protein comprising a heavy chain variable region (HCVR) comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 922, 154, 1482, 1490, 1498, 1506, 1514, 1522, 1530, 1538, 1546, 1554, 1562, 1570, 1578, 1586, 1594, 1602, 1610, 1618, 1626, and 1634, and a light chain variable region (LCVR) comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 930, 162, and 1642.

39. The bispecific antigen-binding molecule of any one of claims 12-24, wherein the first antigen-binding domain competes for binding to human CD3 with a reference antigen-binding protein comprising a heavy chain variable region (HCVR) comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 922, 154, 1482, 1490, 1498, 1506, 1514, 1522, 1530, 1538, 1546, 1554, 1562, 1570, 1578, 1586, 1594, 1602, 1610, 1618, 1626, and 1634, and a light chain variable region (LCVR) comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 930, 162, and 1642; and wherein the second antigen-binding domain competes for binding to human PSMA with a reference antigen-binding protein comprising a heavy chain variable region (HCVR) comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 122 and 66, and a light chain variable region (LCVR) comprising an amino acid sequence selected from the group consisting of SEQ ID NOs: 930, 162, and 1642.

40. A pharmaceutical composition comprising the antibody or antigen-binding fragment thereof of any one of claims 1-11 or the bispecific antigen-binding molecule of any one of claims 12-39 and a pharmaceutically acceptable carrier or diluent.

- 41.** A method for treating a cancer in a subject, the method comprising administering to the subject the pharmaceutical composition of claim 40.
- 42.** The method of claim 41, wherein the cancer is selected from the group consisting of prostate cancer, kidney cancer, bladder cancer, colorectal cancer, and gastric cancer.
- 43.** The method of claim 42, wherein the cancer is prostate cancer.
- 44.** The method of claim 43, wherein the prostate cancer is castrate-resistant prostate cancer.
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